Sensitivity Is Not Fragility: Shared Visual Responses and Task Outcomes in Vision-Language-Action Policies Supplementary Material

Anonymous CVPR submission

Reading Guide and Measurement Definitions

Part 1 maps the main paper's claims C1–C4 to their evidence, Part 2 its stated limits, Part 3 the studies and terms. Recording sites and interval conventions are in Sec. S1; a reusable protocol and recomputation checks in Sec. S16. Results are for $\pi_{0.5}$ unless stated.

1. How this supplement supports the main paper. C1: perturbations change the features along 16 shared directions per layer (main Sec. 4.1, Fig. 3a).

Sec. S3, Table S5. Held-out tasks and families, all 536 pairs: 36.7–84.2% of the energy around the fitting-data mean, against 0.5–10.6% (clean-feature) and 1.3–1.9% (random); 16.26–75.00% of the raw energy.

Sec. S3.4. The 404-pair subset of main Fig. 3: 36.5–83.6%, against 0.5–10.2% and 1.3–1.9%.

Table S4. Not the mean: uncentered shared directions retain 32.4–75.3% of the raw energy, the span of the position means 13.5–30.4%.

Sec. S3.3, Figs. S1, S2. Same ordering for 1 to 32 directions.

Secs. S19, S15, S8.1. Energy-weighted tasks: 38.4–84.0%. π_0: 39–77%, against 1–28% and about 1.5%. A later observation (original LIBERO; tasks held out, conditions seen in fitting): 32.7–64.2%, against 0.8–11.0% and 1.4–1.8%.

Sec. S2. A reconstruction uses 160 coefficients from the response it scores.

C1, linear readout (main Fig. 3b).

Sec. S3.4, Table S6. The 16 coordinates predict the command change better than no change at L18 in all families, worse at L1 and L9 in 7 of 8 cells. At L18 they match clean-feature coordinates (intervals span zero); first flow predicts best in every row.

Sec. S14. Nonlinear readouts change 102 and 76 of 216 interval categories.

C2: restoration at L9 reduces command error (main Sec. 4.2, Figs. 1b, 4).

Sec. S13, Tables S31–S33. 216 held-out cases: error 0.409 to 0.312 (Camera), 0.209 to 0.199 (Sensor Noise), lower in 25 of 27 and 24 of 30 tasks. Success $+1.85$ and $+3.33$ points, intervals include zero. The 16 directions reach 86.9% and 70.7% of replacing the whole L9 state, which gains 7.41 and 5.83 points (post hoc).

Sec. S13.5. At matched edit size, restoration beats clean-feature directions under Camera only.

Sec. S12. No clean observation (6,048 planned rollouts): L18 response lower by 8.8% and 7.6%; random directions at most 0.3%.

C3: command changes reach the robot (main Sec. 5.1).

Sec. S5.1, Fig. S21. Rank correlation with end-effector separation 0.95/0.97 after 5 controls, 0.47/0.41 after 60.

Secs. S7, S10.3. Agreement across noise draws 0.88–0.92 and 0.80–0.89; lower for the gripper.

C4: the size of a change is not its cost (main Sec. 5.2, Fig. 5).

Sec. S5, Table S10. π_0, Sensor Noise: 0.58 times a clean rerun's motion, 14.3 points lost; every lost case listed.

Sec. S6. Lost and gained successes against the clean rerun; bounds for 39 of 2,680 missing outcomes. Lost successes exceed the rerun level under Camera and Sensor Noise, gained ones do not. AUROC 0.72/0.64 pooled, 0.62/0.55 within a family (post hoc).

Secs. S10.2, S10.6. Rank correlation -0.90 across ten condition means, -0.075 across 400 cells. Headlight: 5.1 times the command change of matched dimming, both 77 of 80 (Fig. S17).

Sec. S18. Complete main-paper panels.

Additional studies, not relied on by the main paper: internal associations (Sec. S4, Table S7), exposure timing (Sec. S9), original LIBERO (Sec. S10), LIBERO-Plus outcomes (Sec. S11), attention (Sec. S20, Table S39, Fig. S23; details in Secs. S10.4, S10.5, S8.2, Figs. S19, S13).

2. Limits stated in the main paper and where they are documented (main Sec. 6).

Sensitivity is measured at the episode start: Sec. S1; the command–motion link weakens by 60 controls (Sec. S5.1); one later observation (Sec. S8.1).

Restoration: one layer, 16 directions, two families, 216 cases, needs a clean observation: Sec. S13; the test without one: Sec. S12.

The shared directions carry the Background response, which costs $\pi_{0.5}$ no net success: Table S5; Sec. S6.

Simulation, two related policies, moderate settings: Table S2; Sec. S11.

3. Study index.

Internal recordings ($\pi_{0.5}$, LIBERO-Plus): 536 pairs on 40 tasks fit the shared directions (Secs. S3, S11); 404 pairs on 28 common

tasks link features, commands, motion and outcomes (Sec. S4).

Restoration ($\pi_{0.5}$): 31 held-out tasks, 108 settings, two draws, six runs, 1,296 rollouts (Sec. S13).

Paired executions ($\pi_{0.5}$, π_0): five runs per setting; 268 settings per policy on 40 tasks, 202 on 28 for outcomes (Sec. S6). Same settings as the recordings; π_0 internals regenerated on 369 of 404 pairs (Sec. S15).

Original LIBERO ($\pi_{0.5}$): later observation, 3,360 episodes (Sec. S8.1; its L9 directions are the ones removed in Sec. S12); removal without a clean observation, 288 cases $\times$ 3 conditions (clean, Camera $+15^\circ$, headlight off) $\times$ 7 runs; matched lighting, 80 initial states (Sec. S10.6); 480-pair camera study (Sec. S10); fixed-state camera $0^\circ \rightarrow 15^\circ$, 36 tasks (Table S26); ten conditions, 2,344 of 2,400 pairs on 40 tasks.

Glossary.

Setting, pair, case. Setting: one task under one visual variant. Pair: one setting under one policy-noise draw (label 0 or 1); its clean and perturbed runs share task, start state and noise. Case: a pair in the restoration study. Start state: task, initial state and, where used, noise label.

Runs. Clean R_0, R_1; perturbed T_0, T_1 (noise of R_0, R_1); R'_0 repeats R_0; clean rerun: R_1 vs. R_0. Restoration: N_C clean, N_T perturbed unedited, R_{resp} restoration, R_{clean}, R_{rand} controls, R_{full} whole L9 state.

Directions. Sixteen channel directions fitted to responses (shared; "Response" in tables), to clean features ("Clean") or random ("Random"); position-centered fits subtract the ten position means of the fitting data.

Energy shares. One minus reconstruction error over the energy around the fitting-data mean (main Eq. 3), or over the raw energy.

Computation. Query: one observation and plan; generation step $j = 0, \ldots, 9$; plan position: one planned action; control: one executed action. First five commands: their translation part, 15 numbers. First flow: clean–perturbed difference of the step-0 update.

Internal quantities. Prefix H, normalized Z, cached values V; action-expert H after a block; attention probability P; attention output W before gating (Table S1). Conditional TV: redistribution within a region; mass-weighted TV rescales it by the region's mass. Participation: effective number of channels (tokens) carrying a response. Raw output response: a difference RMS; relative response divides it by the matched-clean norm.

Command change, footprint. Perturbed minus clean first five commands; an edit's footprint is its edited-minus-unedited change as a fraction of it.

Outcomes. Clean then perturbed (S success, F failure): SF lost, FS gained success. Success change (RD in some tables): perturbed minus clean success rate. Command–cost correlation: across the ten conditions.

Intervals resample whole tasks. **Bounds over missing outcomes** cover all feasible values of unknown outcomes.

Difficulty level: LIBERO-Plus level 1–5, higher is harder. **Camera:** Camera Viewpoints family; on original LIBERO, $+15^\circ$ unless stated.

Abbreviations. RMS root mean square; MSE mean squared error; EEF end effector; TV total variation; AUROC (AUC in some tables) area under the ROC curve; pp percentage points.

S1. Policy Computation, Recorded Quantities, and Evaluation Design

This section states where each quantity is recorded (Table S1), which study compares it (Table S2) and with what sampling unit and intervals. For C1, the shared directions are fitted to the action-expert state H after L1, L9 and L18 at the first query and first generation step of the LIBERO-Plus recordings (main Figs. 1a and 2).

Comparisons along the control loop. At every block, the action-expert tokens attend to that block's prefix keys and values, which are computed once per query; the image therefore enters the action expert at every layer and every generation step. The robot executes the first five commands of each plan and then replans from a new observation. Reconstruction asks how much of an observed response a fitted span retains; its information, mean and capacity controls are in Secs. S2 and S3. Command readouts compare predictive information (Sec. S3.4), while associations describe covariation on matched cases (Secs. S11.2 and S4). Paired executions measure repeatability, motion and outcomes (Secs. S7, S5 and S6). The later-observation, exposure-timing and original-LIBERO studies extend these questions to other states and schedules (Secs. S8, S9 and S10).

Policies and evaluation design. The two policies generate plans of different length, 10×32 for $\pi_{0.5}$ and 50×32 for π_0, but both use ten generation steps and execute five controls before replanning. The π_0 paired executions and the separate π_0 720-episode study use the same checkpoint; each has its own executions and interventions. The paired executions reuse the 268 settings of the LIBERO-Plus recordings. Their assignments were fixed before their outcomes were read, but after the results of the LIBERO-Plus recordings were known.

Hidden states come from the policy's own run; attention comes from a replay. Action-expert H is recorded along the policy's own generation; P and W describe attention recomputed at each branch's saved generation state x_j and time. The policy's forward pass applies all 18 blocks through a rematerialized scan. The diagnostic pass applies the same frozen block parameters through a separate full-depth scan, retaining blocks 0, 8 and 17 (L1, L9 and L18). Saved generation states, times and visibility masks define its inputs. This distinction applies throughout the LIBERO-Plus analyses (Table S1). Final actions of the policy's own run are checked against the reference baseline before H is adopted; diagnostic readouts and velocities are checked exactly against in-graph controls. Section S16 compares diagnostic and policy hidden states and velocities; no such comparison exists for P or W.

The replay shows how the allocation P becomes the projected output W. It recomputes the prefix; normalized suffix

states produce queries, keys and values. Rotary embeddings transform queries and keys, queries are scaled by the head dimension, and cached prefix keys and values are concatenated with the suffix's. Softmax spans all visible keys before value aggregation. The output projection produces W before the adaptive attention gate and residual addition. The feed-forward sublayer and its gated residual then produce post-block H. Prefix V stays fixed within a query, while suffix V changes with the generation state.

Sampling units and intervals by study. Layer responses compare response size with each layer's clean scale. The comparison uses 1,199 of 1,200 internal pairs on 40 tasks; one headlight pair is missing. It reports all 200 task–condition means and five equal-task condition means at the first query, first generation step and first five plan positions, each response RMS divided by its own layer's clean RMS.

Camera calibration predicts a larger-angle response on the same task domain, contrasting 234 original cases with 240 separate start-state/run cases. Its full-minus-compact loss carries 99.375% conditional intervals from 10,000 suite-stratified task draws, with an eight-effect adjustment and fixed predictions. Training tasks include the target angle.

The exploratory joint task/family reconstruction tests transfer when both the task and the family are held out, and retains all 12 family/layer cells. Background, Camera, Light and Sensor Noise contribute 34/32/40/39 task components and 118/112/152/154 cases; Light includes 18 records without a LIBERO-Plus difficulty level. Its 95% task-component intervals use 2,000 draws, condition on the fitted directions and carry no multiplicity adjustment; each query keeps 160 coefficients across ten positions.

The common-task comparison places internal summaries and behavior on the same 28 tasks and 404 pairs. Hidden-state and attention summaries are descriptive means; commands and motion have pointwise 95% task-block intervals; outcomes keep the four-family Bonferroni adjustment for simultaneous 95% coverage. Motion ratios of task-equal means have no interval here; Sec. S11.4 gives the paired motion differences.

The paired executions estimate the paired success change against two clean runs on the prespecified primary support of 28 tasks and 202 settings per policy, with fixed noise weights. All 536 policy–setting cases are archived; 39 run outcomes are unknown (25 technical failures and 14 that fail the analysis validation criteria) and enter bounds over missing outcomes. The 2,000 task-bootstrap draws give four-family-adjusted 98.75% intervals for the success changes (Sec. S6).

The exposure-window study separately compares four timing schedules. Its complete readout (138/138 start states, 552/552 episodes) uses 23 whole tasks, three noise draws per task and two fixed initial states. The prespecified end-point for initial state 24 is the terminal world-XY distance from the target to the current goal center, $R_{30} - N_{30}$; initial state 25 repeats the fixed rule. Its 10,000 shared whole-task bootstrap draws and pointwise 95% intervals describe this timing comparison (Sec. S9).

Implementation notes. Masked attention logits are accumulated at higher precision and cast to the model precision before value aggregation. The exposure-window bootstrap uses seed 20261008.

S2. Information Used for Reconstruction, Prediction, and Readout

Each test of a response receives different information and scores a different target (Table S3). Reconstruction, the C1 test, observes the response it scores: the 16 directions are fixed in advance, but their 160 coefficients per query come from the test response. C1 therefore shows that fixed directions, fitted without the test tasks and family, capture new responses, not that a response can be predicted unobserved. Templates, camera calibration and command readout are listed for contrast.

Reconstruction observes the response it scores. For one layer, $D \in \mathbb{R}^{10\times1024}$ is the paired response over ten plan positions, M a broadcast global training mean or a ten-position training template, and $B \in \mathbb{R}^{1024\times16}$ an orthonormal channel basis. Fixed-template prediction is $\widehat{D} = M$; reconstruction projects the residual of the observed response onto the basis:

$$\widehat{D} = M + [(D - M)B]B^{\top}. \tag{S1}$$

The 10×16 coefficients are computed from the test response. Sixteen directions therefore give 16 coefficients at each position, 160 per query, in addition to the stored directions and template. The test evaluates reuse of the directions, not of the coefficients.

The clean-feature control learns directions from training clean features but projects the same test residual around the same template, so both sets of directions have the same target and test information. Mean-span controls also observe the target through their projection coefficients, unlike mean-only prediction: a global mean direction uses ten coefficients, a rank-ten span of the position means 100.

Calibration predicts a larger-angle response from a measured probe. Camera calibration keeps the measured D_{15} and adds the training shift $\mu_{30} - \mu_{15}$; its full correction also receives the additional observations in the table, detailed in Sec. S10.1. The calibration gain measures the value of a smaller-angle probe; template transfer tests a prediction from training data only.

Table S1. **Each internal quantity is read at one site through one recording path.** Definitions for measurements at the first query in the LIBERO-Plus recordings; widths are channel counts; P and W exist only in the diagnostic replay. The paired executions use the records in Sec. S6.

Quantity	Computation site	Recording support
Prefix H / Z	Pre-block state / pre-attention RMSNorm output; width 2048	Recomputed prefix; layers 1, 9, 18; image/wrist/text masks
Prefix V	Projected cache values; one KV head of width 256	Recomputed once per query; no generation-step axis
P	Full-key probability	Diagnostic suffix; eight heads, ten action queries; steps 0, 5, 9
W	Projected attention output before the adaptive gate	Diagnostic suffix; ten positions of width 1024; steps 0, 5, 9
Action-expert H	Post-block state of width 1024	Policy's own run; layers 1, 9, 18
Commands	First five commands returned after the output adapter	Fifteen translation coordinates; command units

Table S2. **Each study has its own sampling unit and support.** Rows are distinct studies of the frozen policies ($\pi_{0.5}$ unless stated); support gives planned and, where records are missing, available counts. Studies may share tasks or clean episodes but keep their own executions.

Study	Recorded support	Main role
Layer responses	40 tasks; 5 conditions; 1,199/1,200 internal pairs	First-query layer ordering; paired task outcomes
Response calibration	234 cases and 240 separate start-state/run cases	Predict a larger-angle response on the same 40-task domain
Matched-lighting study	80 initial states; 240 episodes	Headlight versus matched dimming
480-pair camera study	480 pairs; 720 episodes; 40 tasks	Commands, end-effector motion, paired outcomes
Ten-condition study	2,344/2,400 persistent pairs; 40 tasks	Response–command association and aggregation effects
LIBERO-Plus recordings	536 pairs; 40 tasks; 4 families	Shared directions and readout (C1); matched internal, command, motion and outcome measurements on 404 pairs / 28 tasks
Restoration	31 held-out tasks; 108 Camera/Sensor Noise settings; 1,296 episodes	Restoration at L9 at every generation step (C2); first five commands and complete paired task outcomes
Paired executions	268 planned settings per policy; 536/536 policy–setting cases archived	Paired behavior of both policies (C3, C4); 2,641/2,680 outcomes valid
Exposure windows	23 tasks; 138 four-run start states; 552 episodes	A separate $\pi_{0.5}$ comparison of exposure timing with complete terminal geometry and task success (Sec. S9)
Second-policy cameras	π_0: 240 triplets; 720 episodes	Confirms the success cost of camera changes

Command readout compares response coordinates with the first flow output. The first-flow baseline is the clean–perturbed difference Δv_0 in the initial predicted velocity, with 320 coordinates. Response coefficients contribute 160 coordinates; their concatenation with Δv_0 contributes 480. Comparing the concatenation with first flow alone tests the additional command information available to the fitted readout. Only sets of directions of equal size match the coefficient budget. Final commands, measured motion and task success remain separate endpoints.

S3. Shared Directions Beyond Training Means

This section is the main evidence for C1 (main Sec. 4.1). On held-out tasks and families of all 536 $\pi_{0.5}$ pairs, 16 shared directions retain 36.7–84.2% of the response energy around the fitting-data mean, against 0.5–10.6% for clean-feature and 1.3–1.9% for random directions (Table S5), for 1 to 32 directions alike (Sec. S3.3). The result does not come from the mean: uncentered shared directions retain more raw energy than training-mean templates and their spans (Table S4). Of the raw energy the centered fit retains 16.26–75.00%, lower in some cells for a reason given below. Section S3.4 gives the 404-pair subset quoted in the main paper.

S3.1. Shared directions against mean and capacity controls

Fits use fixed task/family splits. The analysis uses the first query, first generation step, ten ordered plan positions and

Table S3. **Four tests, four information conditions.** Columns give what each test observes, its scored target and its question. M is a training mean or position template, B a learned channel basis (for example the shared directions). Camera calibration trains on tasks including the target angle and withholds only the test case's larger-angle target; reconstruction and its clean-feature and random controls project the same test response. Task holdouts concern analysis fits, not policy pretraining.

Comparison	Information used to produce a test output	Scored target	Question
Fixed template	Training response mean or position template; no test response	Unobserved response D	Can a population template transfer?
Conditional calibration	Measured D_{15}; full correction also receives smaller-angle translation commands, generation noise, and response norms	Unobserved D_{30} of that case	Does an observed probe response help predict a known target angle?
Response reconstruction	Observed D; coefficients DB or $(D-M)B$	The same observed D	Do learned channel directions represent new responses?
Command readout	Paired first-query response features, paired first-flow difference, or both	Paired difference of the commands (before or after the output adapter)	What command information remains beyond strong baselines?

1,024 channels at each of L1, L9 and L18, fitted separately. Task components (tasks sharing a clean scene) are formed before folds are assigned; the original 60 joint task/family splits and 12 known-task, held-family splits are kept. Training weights are equal at each nested level (task component, family, setting, noise, position); test folds are joined before a family result, so each held-out component appears once. The families contribute 118/112/152/154 pairs and 34/32/40/39 task components; the full Light set includes 18 cases without a LIBERO-Plus difficulty level. Inputs, numbers of directions and fits do not use outcomes.

Every set of directions reconstructs the same residual about a training template. A global mean μ and ten position means μ_p use training data only, broadcast across positions or applied per position. Centered shared directions are fitted to $D - M$; centered clean-feature directions to clean features after subtracting their own training clean means; both, and the random directions, reconstruct the same residual $D - M$. Mean-only prediction has no test coefficients; the mean-direction and position-mean-span controls project the observed target (Sec. S2), the latter with rank ten in every fold.

Scores are means of task ratios. For each held-out task component t, the score is one minus the reconstruction error SSE_t over the response energy, raw (E_t) or around the training template ($E_{M,t}$):

$$S_{\mathrm{raw}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left(1 - \frac{\mathrm{SSE}_t}{E_t}\right), \tag{S2}$$

$$S_M = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left(1 - \frac{\mathrm{SSE}_t}{E_{M,t}}\right). \tag{S3}$$

Global and position templates give different residual energies. Zero denominators and too few directions are undefined. All primary comparisons have valid inputs and enough directions.

The tables report the primary comparisons with 16 directions on the original 72 training splits. Uncertainty uses 2,000 whole-task-component bootstrap draws shared across methods and numbers of directions, conditional on the fitted directions. All 95% intervals are pointwise: paired difference intervals for the 16-direction contrasts, marginal intervals for the curves. Resampling preserves the dependence among family/layer comparisons, which share data and training support.

S3.2. What remains after centering

The shared directions do more than the mean. The global and position mean-only predictors each have negative raw scores in seven of twelve cells (Table S4), yet the span of the position means reconstructs 13.5–30.4% of the raw response energy. Uncentered shared directions reconstruct 32.4–75.3%, with paired advantages of 18.2–57.5 percentage points over that span and positive pointwise intervals in all twelve cells. The span has rank ten, the shared directions sixteen; equal-size comparisons follow.

Centered shared directions keep a wide margin over clean-feature and random directions. With the training position template removed, the shared directions retain 36.7–84.2% of the energy around the mean, against 0.5–10.6% for centered clean-feature and 1.3–1.9% for random directions (Table S5). Around the global template they retain 35.8–84.3%. Every paired pointwise interval favors the shared directions under both raw and around-mean scoring. Projection reduces or preserves error relative to mean-only prediction.

Raw and around-mean shares. With the position template, the shared directions retain 36.73–84.24% of the energy around the mean and 16.26–75.00% of the raw energy

Table S4. **Uncentered shared directions retain more raw energy than training-mean templates and their spans.** Raw-energy scores S_{raw} (%) under joint task/family holdout in the LIBERO-Plus recordings ($\pi_{0.5}$), by family (B/C/L/S: Background Textures, Camera Viewpoints, Light Conditions, Sensor Noise incl. blur and fog-like conditions) and layer. Shared (PCA16): 16 uncentered shared directions. Mean-only columns use no test response; the others project it, with 160, ten and 100 coefficients. Negative scores mean greater error than predicting zero. Point values; pointwise 95% task-component intervals are in the text.

Family/layer	Shared (PCA16)	Global mean only	Position mean only	Global mean span	Position-mean span
B / L1	55.63	-47.41	-48.76	11.68	29.46
B / L9	33.86	-90.08	-112.16	2.52	14.60
B / L18	32.51	-94.23	-112.92	3.56	13.55
C / L1	62.33	-80.70	-81.72	5.78	28.96
C / L9	64.66	0.92	1.64	6.42	22.36
C / L18	42.96	4.20	3.28	9.05	20.64
L / L1	49.27	-272.46	-277.68	3.06	30.43
L / L9	37.86	-68.91	-81.52	3.30	14.84
L / L18	32.37	-47.67	-56.72	3.53	13.63
S / L1	75.27	4.00	4.03	11.03	17.75
S / L9	47.16	4.45	4.80	12.07	24.82
S / L18	38.59	3.48	1.69	11.34	20.37

Table S5. **Shared directions retain far more than clean-feature and random directions after centering; raw and around-mean shares.** Percent; same panel and holdouts. First two columns: raw shares S_{raw} of the 16 shared directions after global or position centering. Other columns: shares S_M around the global or position training template for 16 shared (Response), clean-feature (Clean) and random (Random) directions, each with 160 coefficients from the test response. Means of task-component ratios; pointwise 95% intervals are in the text. Both shares have the same error and different denominators: Light L1 retains 84.24% around the position template and 45.48% raw.

	Raw share, shared directions		Share around global template			Share around position template		
Family/layer	Global	Position	Response	Clean	Random	Response	Clean	Random
B / L1	54.87	54.63	65.19	2.53	1.33	65.28	2.53	1.33
B / L9	24.95	20.91	43.18	0.57	1.45	44.39	0.53	1.50
B / L18	24.65	16.26	38.55	9.65	1.60	39.30	8.12	1.63
C / L1	60.17	59.96	77.07	1.73	1.39	77.08	1.75	1.39
C / L9	64.92	64.82	64.07	0.53	1.65	63.70	0.50	1.63
C / L18	42.73	42.49	40.14	13.22	1.70	40.43	10.55	1.72
L / L1	46.42	45.48	84.28	1.91	1.85	84.24	1.90	1.84
L / L9	35.05	31.03	60.64	0.85	1.57	60.87	0.82	1.67
L / L18	29.16	24.19	50.85	7.72	1.92	50.55	6.90	1.90
S / L1	75.00	75.00	73.97	1.35	1.62	73.98	1.35	1.62
S / L9	46.81	46.96	43.73	0.61	1.39	43.46	0.61	1.39
S / L18	38.11	37.89	35.84	9.60	1.86	36.73	8.73	1.88

on this full 40-task panel (Table S5); the main paper quotes the first on the 404-pair subset (36.5–83.6%, Sec. S3.4) and the second from this table. The two differ only in the denominator. They are close where the fitting-data mean, learned without the test family, matches the held-out responses (Sensor Noise; Camera at L9 and L18). Where it does not, mean-only prediction has negative raw scores (Table S4), so the energy around the mean exceeds the raw energy and the raw share is lower (Background, Light, Camera at L1; largest for Light L1). Centering removes only the training template; an offset shared within the held-out family stays in the scored response.

The effect of position centering depends on the family and layer. On the common raw denominator, six cells have negative intervals for position-minus-global reconstruction, one has a small positive interval and five span zero. The positive cell is S/L9, $+0.151$ percentage points $[0.020, 0.281]$. B/L9 and B/L18 have position-centered raw-share intervals that cross zero, 20.91% $[-4.45, 35.50]$ and 16.26% $[-17.46, 34.83]$: for Background at L9 and L18 the raw share of the centered fit is not distinguishable from zero.

Section S11.3 reads commands from uncentered directions; Sec. S3.4 from the position-centered ones, fitted with the same splits.

S3.3. Other numbers of directions

The shared directions retain more than clean-feature and random directions for every number tested, $k \in \{1, 4, 8, 16, 32\}$ (Figs. S1 and S2), which supports C1's statement for other numbers of directions. It uses the 536-pair panel and all 72 training splits (60 joint task/family, 12 known-task, held-family). The 404 cases of Sec. S3.4 give the main paper's paired reconstruction–readout comparison.

Matched targets and nested directions. For every k, shared, clean-feature and random directions reconstruct the same observed residual $D_p - \mu_p$ about the training position mean, with the same denominator. The shared and clean-feature curves use the leading k directions of their training decompositions. Random directions are the first k columns of one fixed 32-column orthonormal basis, whose first 16 also give Table S5; the uncentered replication uses its own fixed random basis. Each uses k coefficients of the observed response per position.

The ordering holds for all five numbers of directions. Shared directions retain more energy around the mean than clean-feature and random directions of equal number in every family/layer cell under both holdout settings. Under joint task/family holdout, the shared directions retain 7.0–66.0% with one direction and 49.3–88.9% with 32; with 32, clean-feature and random directions retain 1.0–15.0% and 2.8–3.8%. Paired intervals against the controls are reported for the primary $k = 16$, where every interval favors the shared directions in both settings.

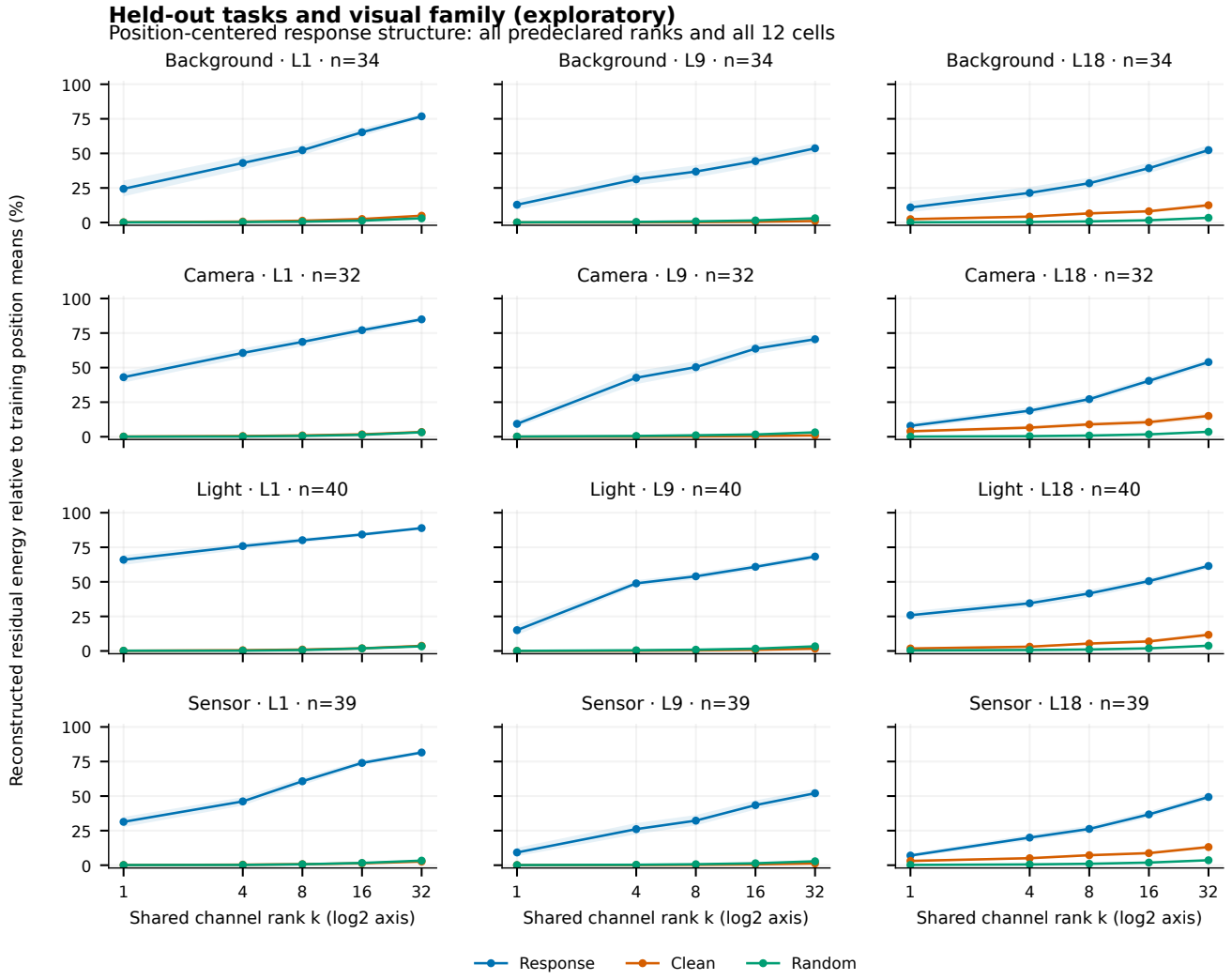


Figure S1. **Shared directions retain more than clean-feature and random directions for every number of directions, under joint task/family holdout.** Shares of the energy around the training position means on the original 536-pair LIBERO-Plus panel, for $k = 1, 4, 8, 16, 32$ directions. Twelve panels: four families by three layers; $34/32/40/39$ task components for Background/Camera/Light/Sensor Noise. Curves average task-component scores. Shading shows pointwise marginal 95% intervals from 2,000 whole-component bootstrap draws conditional on the fits. Each set uses k coefficients of the observed response per position, $10k$ in total.

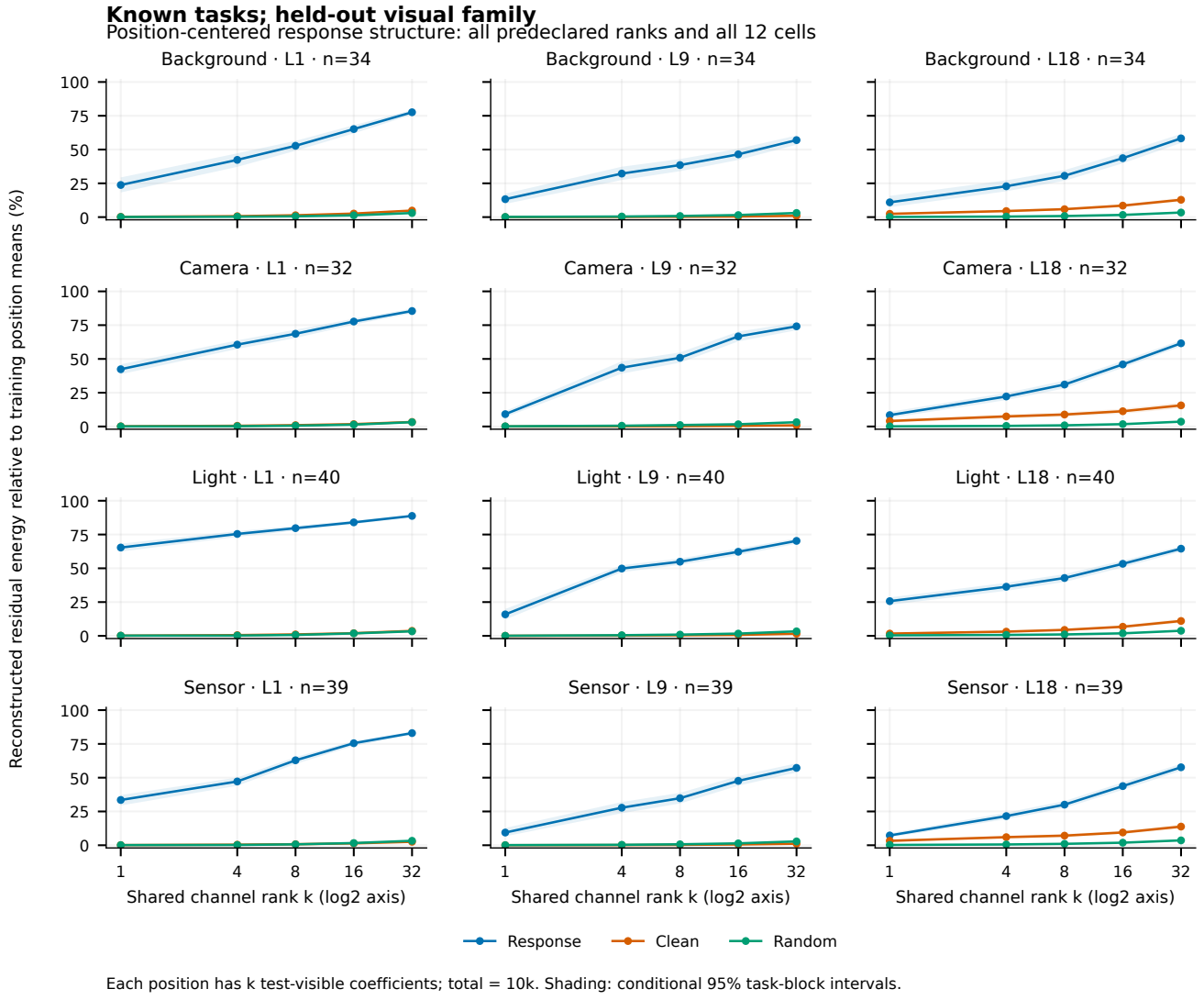


Figure S2. **The same ordering holds for known tasks and a held-out family.** The 536-pair panel under the twelve known-task, held-family splits; same target, denominator and $34/32/40/39$ task components as Fig. S1. Shading shows pointwise marginal 95% whole-task-component intervals conditional on the fits, using 2,000 shared draws. Each set uses k coefficients of the observed response per position, $10k$ in total.

S3.4. The 404-pair subset and the linear command readout

On the 404 pairs of main Fig. 3, the position-centered shared directions retain 36.5–83.6% of the response energy around the fitting-data mean, against 0.5–10.2% for clean-feature and 1.3–1.9% for random directions: the C1 numbers of the main paper. An exploratory linear readout of the same 16 coordinates predicts the change in the first five commands better than predicting no change at L18 in all four families, but not at L1 and L9 (Table S6; main Fig. 3b). This supports main Sec. 4.1 with two qualifications: at L18 the shared coordinates predict no better than clean-feature coordinates, and first flow predicts better than any set of coordinates.

What the readout is compared with. Affine readouts are compared on the same held-out cases with predicting no change (absolute accuracy), clean-feature and random coordinates (choice of directions), the full hidden response (prediction before compression) and first flow (information beyond a downstream output; it beats no change in every family). Table S6 puts all layers on one translation-error scale; the figures keep all command components.

Retention and prediction peak at different layers on the same cases. At L1 and L9, shared-direction coordinates reduce affine translation error by 20–75% and 27–71% relative to centered clean-feature coordinates. On the absolute scale, however, the shared-direction readout has higher error than predicting no change in 7 of these 8 cells, and the clean-feature readout in all 8 (point estimates). The first compares coordinates within a layer; the second asks whether a readout beats keeping the clean commands. Against the full-hidden readout, 8 of 12 translation intervals include zero (3 favor the shared directions, 1 the full readout).

Within most tasks, L1 retains most and L18 predicts best. We compare retained response energy and affine translation-prediction MSE at L1, L9 and L18 within each task, on the same pairs. In the table below, *best-layer separation* counts tasks in which L1 retains strictly more response energy than both other layers and L18 has strictly lower prediction MSE than both. This holds in 24–27 of the 28 tasks in each family. The *full ordering* column also requires both retention and prediction MSE to decrease strictly from L1 through L9 to L18, and so also checks the position of L9; it holds in fewer tasks. The same 28 tasks appear in all four families. These are descriptive counts, reported without an inferential test.

Family	Best-layer separation	Full ordering
Background	24/28	11/28
Camera	26/28	18/28
Light	27/28	15/28
Sensor Noise	27/28	17/28

At L9 the advantage over clean-feature coordinates depends on centering. At L9, shared-direction coordinates improve translation readout relative to centered clean-feature coordinates in each family. Uncentered clean-feature coordinates, however, give lower prediction error than the centered ones (Table S6), while the intervals for position-centered versus uncentered shared directions span zero in every family. The relative gain at L9 thus depends on how the clean-feature coordinates are built.

At L18 the shared directions do not beat clean-feature coordinates. At L18, translation and rotation readout differences against centered clean-feature coordinates have intervals spanning zero in every family. Continuous-gripper readout errors exceed those of the magnitude baseline in every family. The rotation and gripper figures show this component-specific pattern: reconstruction is scored once, prediction per component.

Adding shared coordinates to first flow rarely helps. Of the 36 first-flow-plus-shared-direction comparisons, one pointwise interval is positive, fifteen are negative and twenty span zero. The positive Camera/L1 translation contrast reduces error by 8.66%; the uncentered shared directions already give an 8.81% point reduction on the same support. Since first flow is already a downstream prediction, this measures additional command information available to the readout.

The directions are reused without refitting. The cached first-five adapter differences match all 35 returned command coordinates in each of the 404 cases. Translation, rotation and continuous gripper are scored separately; the gripper target precedes sign formatting. The saved position-centered shared, clean-feature and random directions are used without refitting or choosing their number from command results; each has sixteen channel directions with coefficients at ten positions. Comparators are the uncentered shared, clean-feature and random directions, ten positionwise response magnitudes, the full 10×1024 hidden difference, the first flow difference of the policy's own generation, zero prediction and the training target mean. The sixty joint task/family splits are kept, five held-out folds joined per family and layer. Training uses the LIBERO-Plus recordings; evaluation uses the matched subset on 28 tasks, with 96/102/96/110 cases for Background/Camera/Light/Sensor Noise.

Training the command readout. A weighted affine ridge readout predicts the target vector before or after the output adapter. Each feature block is centered on training data and divided by its training scalar RMS; the intercept is unpenalized and $\lambda = 0.01$ is fixed. Features are never pooled across layers or plan positions; adding the shared coordinates to first flow uses one training scale per block. The full-hidden readout has more features, so this compares prediction under fixed estimator and feature budgets.

Readout and reconstruction share cases and draws. Command and reconstruction comparisons share the 404 test cases; within tasks, noise and setting weights follow the original protocol. Reconstruction divides task-weighted error by task-weighted residual energy about the saved training position mean and averages task scores. Readout contrasts compare task MSEs on the same 2,000 whole-task-component draws per family/layer and report the relative error reduction $1 - \mathrm{MSE}_{\mathrm{candidate}}/\mathrm{MSE}_{\mathrm{baseline}}$. Intervals are pointwise 95%, conditional on the fitted directions and readouts; the repeated family/layer/component comparisons share data.

Table S6. **The linear readout beats predicting no change only at L18, and there it matches clean-feature coordinates.** Task-equal translation MSE of the first five commands $\times 10^3$, in squared adapter-command units; lower is better. All columns use the same 404 held-out cases on 28 tasks. Response, Clean and Random are 16 position-centered shared, clean-feature and random directions; the last two columns use uncentered fits. Full H uses the complete hidden response, and First flow the clean–perturbed difference in the initial predicted velocity. First flow and No-change are the same at every layer. Main Fig. 3b shows the errors of the 16-direction sets relative to No-change; First flow is kept here as a downstream comparator. Paired readout intervals appear in Table S36 and Fig. S3.

Layer	Family	Response	Clean	Random	Full H	First flow	No-change	Uncentered response	Uncentered clean
L1	Background	4.297	10.213	7.298	6.877	0.453	2.084	4.282	9.030
L1	Camera	16.280	20.428	18.038	15.072	1.258	15.414	16.263	19.629
L1	Light	1.875	7.400	3.937	3.578	0.227	1.063	1.881	6.898
L1	Sensor Noise	18.564	27.219	29.106	30.301	0.655	4.609	18.978	20.971
L9	Background	3.825	7.597	5.069	2.771	0.453	2.084	3.935	3.423
L9	Camera	14.407	19.831	16.383	13.418	1.258	15.414	14.442	17.308
L9	Light	1.365	4.713	2.478	1.693	0.227	1.063	1.351	2.278
L9	Sensor Noise	6.331	14.484	8.915	6.484	0.655	4.609	6.193	7.433
L18	Background	1.035	0.964	4.194	0.686	0.453	2.084	1.040	0.872
L18	Camera	3.737	3.457	11.073	4.349	1.258	15.414	3.940	3.568
L18	Light	0.433	0.496	1.482	0.389	0.227	1.063	0.416	0.435
L18	Sensor Noise	1.538	1.480	4.028	1.510	0.655	4.609	1.564	1.268

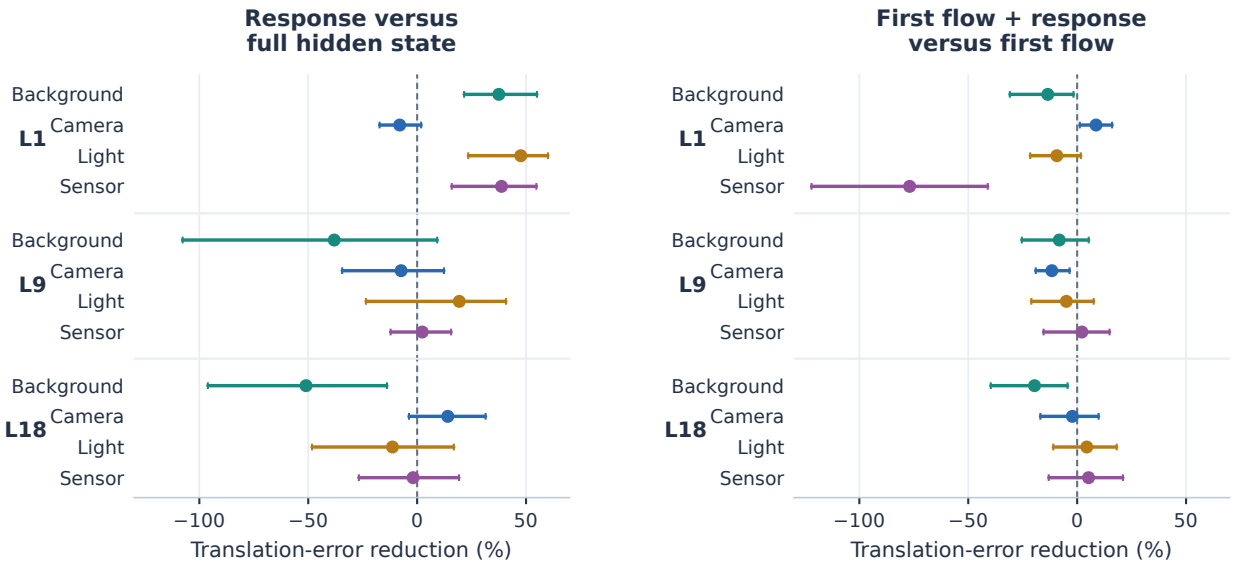


Figure S3. **Shared coordinates mostly match the full hidden response and rarely add to first flow.** All 24 translation contrasts across three layers and four families, on the 404 matched LIBERO-Plus pairs and original task/family holdouts. Left: shared-direction coordinates versus the full hidden response. Right: first flow plus shared coordinates versus first flow. Error reduction is $100(1 - \mathrm{MSE}_{\mathrm{candidate}}/\mathrm{MSE}_{\mathrm{baseline}})$; positive favors the candidate. Bars are pointwise 95% task-component intervals under fixed fitted readouts.

S4. Associations between Internal Responses, Commands, and Outcomes

Do larger internal responses in $\pi_{0.5}$ go with larger command changes and with lost success? Under Camera and Sensor Noise, the L18 attention and action-expert responses rank with the change in the first five commands (Table S7). Their links to outcomes are weak, except that the attention output under Camera separates lost from kept successes. This is an additional, post hoc study; the main paper does not rely on it.

Pairs. The analysis uses 404 of the 536 planned $\pi_{0.5}$ pairs of the internal recordings on LIBERO-Plus: the 202 settings on the 28 common tasks, each with two policy-noise draws. All 404 have the four L18 signals, verified first five commands (translation part), motion after controls 5 and 60, and a paired outcome; the other 132 pairs are not used. Pairs are matched on case, task, setting, noise draw, clean run and outcome before any signal is selected. Section S11.4 reports motion on these pairs; the paired executions of Sec. S6 are separate episodes.

Signals and endpoints. The four signals are fixed, all at L18 and the first query: the block-input prefix state for the main camera (prefix H); the total-variation change of the attention probabilities over all keys (P); the attention output after projection and before gating (W), both recomputed; and the action-expert state from the policy's own generation (AE). P, W and AE are taken at the first generation step. Wrist and text prefix states and the cached values V (main camera, wrist, text; no generation-step axis) are companion signals; Table S1 gives all recording sites. The command endpoint is the RMS of the fifteen translation differences of the first five commands whose environment call returned, in command units. The signed outcome is $dY = Y_T - Y_R$: a lost success (SF) counts -1, a gained success (FS) $+1$.

Rank correlation within tasks. Within a task with n_t pairs, each variable becomes $u_i = (\mathrm{rank}_{\mathrm{avg}}(x_i) - \frac{1}{2})/n_t$ (average ranks for ties), and pair i has weight $w_i = 1/(v_t n_{v(i)})$, where v_t counts the task's settings and $n_{v(i)}$ the observed noise draws in pair i's setting. After subtracting weighted task means, the task covariance is $C_t = \sum_i w_i \widetilde{u}_i \widetilde{z}_i$ and the variances $A_t = \sum_i w_i \widetilde{u}_i^2$, $B_t = \sum_i w_i \widetilde{z}_i^2$; the pooled coefficient is $\overline{C}/\sqrt{\overline{A}\,\overline{B}}$, averaging tasks equally. A variable constant within a task contributes zero variance and covariance, while the other variance still enters the denominator.

Lost-success AUROC and intervals. Among pairs whose clean run succeeds, the AUROC asks whether a larger response separates lost success (SF) from joint success (SS). Weighted SF–SS comparisons are averaged within each task, with half credit for ties, and eligible tasks are averaged equally. One shared set of 2,000 whole-task draws over the 28 tasks gives pointwise 95% intervals; an interval is suppressed with fewer than two eligible tasks, an undefined point, or fewer than 1,900 defined draws. Signals and layer were fixed; the estimates are post hoc.

What the table shows. Under Sensor Noise all four signals rank with the command change, and under Camera P, W and AE do; under Background and Light every interval includes zero. Only two correlations with dY exclude zero, Background AE and Camera W, both negative. Camera W, with AUROC $0.842[0.683, 0.967]$ on 10 tasks, is the largest of the 16 AUROC estimates (Sec. S4.1); the other Camera and Sensor Noise intervals include one half. Camera W thus has a positive command association and a negative signed-outcome association on the same pairs. It was not singled out in advance and rests on few tasks, so the main paper does not use it.

Attention output versus attention probabilities. The one contrast listed in advance, $\rho(W, \mathrm{command}) - \rho(P, \mathrm{command})$, does not show either association to be stronger: all four intervals include or touch zero (B $0.295[0.000, 0.592]$, C $0.119[-0.177, 0.430]$, L $-0.198[-0.648, 0.214]$, S $0.101[-0.124, 0.336]$). These estimates use task-centered ranks on LIBERO-Plus; the ten-condition study uses partial ranks with its own endpoint and support (Sec. S10.2).

Table S7. **Under Camera and Sensor Noise, L18 attention and action-expert responses rank with the command change; links to outcomes are weak.** All four fixed signals on the matched $\pi_{0.5}$ LIBERO-Plus pairs; B/C/L/S denote Background, Camera, Light and Sensor Noise, with 96/102/96/110 pairs. ρ: task-centered weighted rank correlation, tasks weighted equally, between each L18 signal at the first generation step and either the RMS of the first five commands (translation) or the signed outcome dY (SF, lost success, $= -1$); brackets are pointwise 95% whole-task intervals. AUROC: larger signal in SF than in SS among pairs whose clean run succeeds, half credit for ties, averaged over the eligible tasks in the last column; it conditions on clean success and the observed tasks. Background and Light have only three eligible tasks; under Background, prefix H has the same score in all eligible tasks, hence the degenerate interval $[0.833, 0.833]$.

Family	L18 signal	ρ with first five commands	ρ with signed dY	AUROC: SF versus SS	AUROC tasks
B	Prefix H	$-0.097[-0.397, 0.186]$	$-0.125[-0.243, 0.000]$	$0.833[0.833, 0.833]$	3
B	P TV	$-0.083[-0.357, 0.177]$	$-0.081[-0.352, 0.219]$	$0.333[0.000, 1.000]$	3
B	W	$0.212[-0.078, 0.492]$	$-0.178[-0.408, 0.101]$	$0.556[0.333, 1.000]$	3
B	AE	$0.015[-0.282, 0.296]$	$-0.275[-0.448, -0.067]$	$0.556[0.333, 1.000]$	3
C	Prefix H	$-0.072[-0.377, 0.245]$	$-0.185[-0.443, 0.107]$	$0.667[0.417, 0.896]$	10
C	P TV	$0.348[0.075, 0.596]$	$-0.076[-0.298, 0.164]$	$0.567[0.333, 0.788]$	10
C	W	$0.467[0.225, 0.679]$	$-0.284[-0.484, -0.044]$	$0.842[0.683, 0.967]$	10
C	AE	$0.400[0.116, 0.657]$	$-0.257[-0.475, 0.000]$	$0.700[0.455, 0.905]$	10
L	Prefix H	$-0.012[-0.262, 0.229]$	$-0.095[-0.233, 0.000]$	$0.611[0.500, 0.833]$	3
L	P TV	$0.175[-0.117, 0.468]$	$0.143[-0.047, 0.332]$	$0.222[0.000, 0.667]$	3
L	W	$-0.024[-0.344, 0.294]$	$-0.102[-0.280, 0.088]$	$0.722[0.500, 1.000]$	3
L	AE	$0.000[-0.302, 0.299]$	$-0.102[-0.291, 0.101]$	$0.528[0.250, 1.000]$	3
S	Prefix H	$0.620[0.359, 0.831]$	$-0.016[-0.297, 0.262]$	$0.463[0.194, 0.733]$	9
S	P TV	$0.453[0.174, 0.703]$	$-0.130[-0.360, 0.099]$	$0.593[0.333, 0.867]$	9
S	W	$0.554[0.312, 0.743]$	$-0.166[-0.393, 0.074]$	$0.630[0.394, 0.867]$	9
S	AE	$0.568[0.343, 0.757]$	$-0.036[-0.273, 0.193]$	$0.444[0.185, 0.692]$	9

S4.1. Influence of individual Camera tasks

The Camera attention-output AUROC is not carried by a single task: it stays above one half when any one task is removed (Table S8). Before deletion it is 0.842 for the L18 attention output at the first generation step, on ten eligible tasks with 15 lost successes and 23 joint successes.

Method. Each same-task SF–SS comparison keeps its original setting and noise-draw weights, and eligible tasks are weighted equally. Each eligible task is removed once, and the table gives the average over the remaining tasks. The deletion rule was fixed before these calculations but after the association had been seen, so this is a post hoc influence analysis.

S5. Command and Motion Changes Relative to a Clean Rerun

How large is a perturbation's effect on commands and motion, measured against a clean rerun? Under Sensor Noise, π_0 changes its first five commands and its end-effector position less than a clean rerun does, yet loses 14.29 points of success; under Camera, both policies end up further from their clean runs after 60 controls than a rerun does (Table S9).

Table S8. **The Camera attention-output AUROC stays above one half after each single-task deletion.** All ten deletions are shown; task indices follow the released table. Each entry is the exact mean after removing the indicated task.

Task	SF	SS	Task AUROC	After deletion
1	1	1	1.000	0.824
2	1	3	1.000	0.824
3	2	2	0.750	0.852
4	1	3	0.667	0.861
5	1	3	0.667	0.861
6	1	3	1.000	0.824
7	2	2	1.000	0.824
8	1	3	1.000	0.824
9	2	2	1.000	0.824
10	3	1	0.333	0.898

This supports Sec. 5.2 of the main paper: the size of a change is not its cost. Section S5.1 shows that command changes reach the robot.

Missing outcomes. The paired executions (Sec. S6) plan 268 settings for each of $\pi_{0.5}$ and π_0. All 536 policy–setting records exist, but 25 runs failed technically and 14 further outcomes failed validation; these 39 stay unknown in the fixed denominator of 2,680 runs. The primary scope is the 202 difficulty-labelled settings on the 28 common tasks per policy; all 268 settings form a sensitivity scope. For $\pi_{0.5}$ Light, the primary success change lies between 0 and $+8.04$ percentage points (-2.92 to $+5.21$ over all 268 settings); for π_0 Light it lies between -2.68 and $+5.36$ on the primary scope. These *bounds over missing outcomes* cover every possible assignment of the missing outcomes; sampling uncertainty is reported separately.

Perturbed and rerun distances on the same settings. Let R_0, R_1 be the clean runs with policy-noise draws 0 and 1, T_0, T_1 the perturbed runs with the same draws, and R_0' a repeat of R_0. For each policy, family, metric and control step, a setting enters only if the distances T_0–R_0, T_1–R_1 and R_0–R_1 are all measured in the same record. We write TR for the mean of the two perturbed–clean distances and RR for the clean-rerun distance, average settings within a task and tasks equally, and report $TR - RR$ and the ratio of task means $\overline{TR}/\overline{RR}$ on exactly this support; a zero RR leaves the ratio undefined. The repeat (R_0 vs. R_0') and the comparison of the two perturbed runs (T_0 vs. T_1) are separate panels, each with its own support.

Measurements. The 15 metrics listed in advance are: the RMS differences of the first five commands (translation, rotation, gripper); end-effector position distance and orientation

angle at controls 0/5/15/30/60; and two quantities at each run's own last step. Commands are in command units, positions in metres and orientations in radians. A fixed-time comparison uses only the pose stored at that control step; a comparison of final poses requires the two runs to end at the same control step.

π_0 under Sensor Noise moves less than a clean rerun and still loses success. For π_0 under Sensor Noise, the ratio to the clean rerun is $0.545[0.395, 0.718]$ for the first five commands and $0.580[0.409, 0.791]$ for end-effector position after control 5, while its primary success change is -14.29 percentage points. A family mean below the rerun level thus accompanies a net loss of success on the same settings. The position ratios after control 5 for all families, as listed in the main paper's Table 3, are 0.93 (Background), 2.68 (Camera), 0.57 (Light) and 1.32 (Sensor Noise) for $\pi_{0.5}$, and 0.62, 1.03, 0.24 and 0.58 for π_0. After control 60, Camera ratios exceed one in both policies, which also lose net success under Camera (a distance at a fixed step against an outcome at the end of the episode). For $\pi_{0.5}$ Background the ratio after control 60 is $1.273[0.9997, 1.6505]$, and the interval of its paired difference spans zero. Light's motion support is 43/48 settings on 26/28 tasks for $\pi_{0.5}$ and 44/48 on 27/28 for π_0.

Individual pairs against their own clean rerun. We also compare each perturbed pair of π_0 under Sensor Noise (primary scope) with the clean-rerun distance of its own setting. After control 5, let d_{vn} be the end-effector position distance between T_n and R_n for setting v and noise draw n, and let r_v be the distance between that setting's R_0 and R_1. Each recorded pair is classed by $d_{vn} < r_v$, $d_{vn} = r_v$ or $d_{vn} > r_v$, and all four outcome groups are kept (Table S10).

Most pairs that lose a success moved less than a clean rerun: of the 18 lost-success pairs, 12 have $d_{vn} < r_v$. All 55 settings on 28 tasks have both perturbed pairs, measured distances after control 5, a positive clean-rerun distance and known outcomes. The twelve pairs cover ten settings and eight tasks; all lost-success pairs cover fifteen settings and ten tasks. In every setting with a lost-success pair, both clean runs succeed, so the clean rerun changes no outcome there. The two pairs of a setting share one rerun distance, so the counts are nested within settings; they are a descriptive split, while the task-weighted estimates above keep the hierarchy.

Motion size is a poor predictor of lost success (post hoc AUROC). The main paper summarizes these pairs by a post hoc AUROC. The score is the ratio d_{vn}/r_v; the 18 lost-success pairs are the positive class and the 84 joint-success pairs the negative class, pooled without weights, with ties counted as one half. The AUROC is 0.49, close to chance. Its 95% interval, $[0.29, 0.70]$, is the percentile interval of 2,000 whole-task bootstrap draws that resample the 28 tasks with replacement and keep all pairs of a task together, so the two pairs of a setting keep their shared r_v. Every draw contained both classes. With families pooled, the raw distance gives 0.73 ($\pi_{0.5}$) and 0.64 (π_0), and the norm of the change in the first five commands at the first query gives 0.74 and 0.64; comparing pairs only within their family gives 0.62 $[0.48, 0.74]$ and 0.55 $[0.35, 0.70]$ for the ratio. Section S6 gives the values per family.

Outcome	$d < r$	$d = r$	$d > r$	Total
SS: both succeed	65	0	19	84
SF: lost success	12	0	6	18
FS: gained success	2	0	0	2
FF: both fail	4	0	2	6
All pairs	83	0	27	110

Table S10. **Pairs that lose a success and pairs that keep it both mostly move less than a clean rerun.** Counts for all 110 π_0 Sensor Noise pairs (55 settings on 28 tasks). Each pair's end-effector position distance d_{vn} after control 5 is compared with its own setting's clean-rerun distance r_v; the two pairs of a setting share r_v. All distances and outcomes are available, and all 55 rerun distances are positive.

All comparisons. The comparisons above are part of 720 listed in advance (two policies $\times$ two scopes $\times$ four families $\times$ fifteen metrics $\times$ three references). All are kept: 36 have no joint support and 684 have measured joint support. Each reports its exclusions, tasks, number of paired whole-task draws (2,000) and the reasons for any undefined ratio or interval.

Outcome groups and undefined references. The 39 unknown outcomes enter the bounds on success change and on changed outcomes over the planned denominator; motion means use the measured joint support, and the SF/SS groups are post hoc. A zero command distance between R_0 and its repeat R_0' leaves that ratio undefined. Final poses rarely share the same control step, so they form a separate, sparse comparison from control 60.

Table S9. **Relative to a clean rerun, Camera moves the end effector further after 60 controls in both policies, while several families change the commands less than a rerun does.** Primary panel per policy; each cell uses only settings with both noise-matched perturbed–clean distances and the R_0–R_1 clean-rerun distance in the same record (Joint N/tasks). TR: mean perturbed–clean distance; RR: clean-rerun distance; both averaged within task and over tasks equally. Columns: $TR - RR$ and ratio of task means for the first five commands (translation, command units) and end-effector (EEF) position after control 60 (metres). Brackets: pointwise 95% intervals from paired whole-task draws on the support shown. Success estimates keep the fixed denominator of 202 settings. Each of the 720 comparisons listed in advance keeps its own measured support.

Policy	Family	Joint N/tasks	Commands $TR - RR$	Commands TR/RR	EEF at control 60 $TR - RR$ (m)	EEF at control 60 TR/RR
$\pi_{0.5}$	B	48/28	$-0.0038[-0.0165, 0.0083]$	$0.91[0.66, 1.24]$	$0.00524[-0.000006, 0.01041]$	$1.27[0.9997, 1.65]$
$\pi_{0.5}$	C	51/28	$0.0572[0.0359, 0.0806]$	$2.36[1.80, 3.04]$	$0.02281[0.01480, 0.03160]$	$2.29[1.74, 3.03]$
$\pi_{0.5}$	L	43/26	$-0.0208[-0.0314, -0.0109]$	$0.52[0.38, 0.71]$	$-0.00104[-0.00598, 0.00326]$	$0.94[0.72, 1.24]$
$\pi_{0.5}$	S	55/28	$0.0115[-0.0050, 0.0281]$	$1.27[0.90, 1.76]$	$0.01298[0.00519, 0.02091]$	$1.77[1.26, 2.48]$
π_0	B	48/28	$-0.0343[-0.0535, -0.0159]$	$0.58[0.43, 0.77]$	$-0.00594[-0.01769, 0.00569]$	$0.83[0.57, 1.20]$
π_0	C	51/28	$-0.0017[-0.0212, 0.0181]$	$0.98[0.77, 1.27]$	$0.01924[0.00128, 0.04139]$	$1.55[1.04, 2.27]$
π_0	L	44/27	$-0.0619[-0.0799, -0.0452]$	$0.24[0.17, 0.35]$	$-0.01855[-0.02842, -0.00915]$	$0.48[0.36, 0.67]$
π_0	S	55/28	$-0.0373[-0.0551, -0.0204]$	$0.55[0.39, 0.72]$	$-0.00120[-0.01258, 0.00978]$	$0.97[0.70, 1.34]$

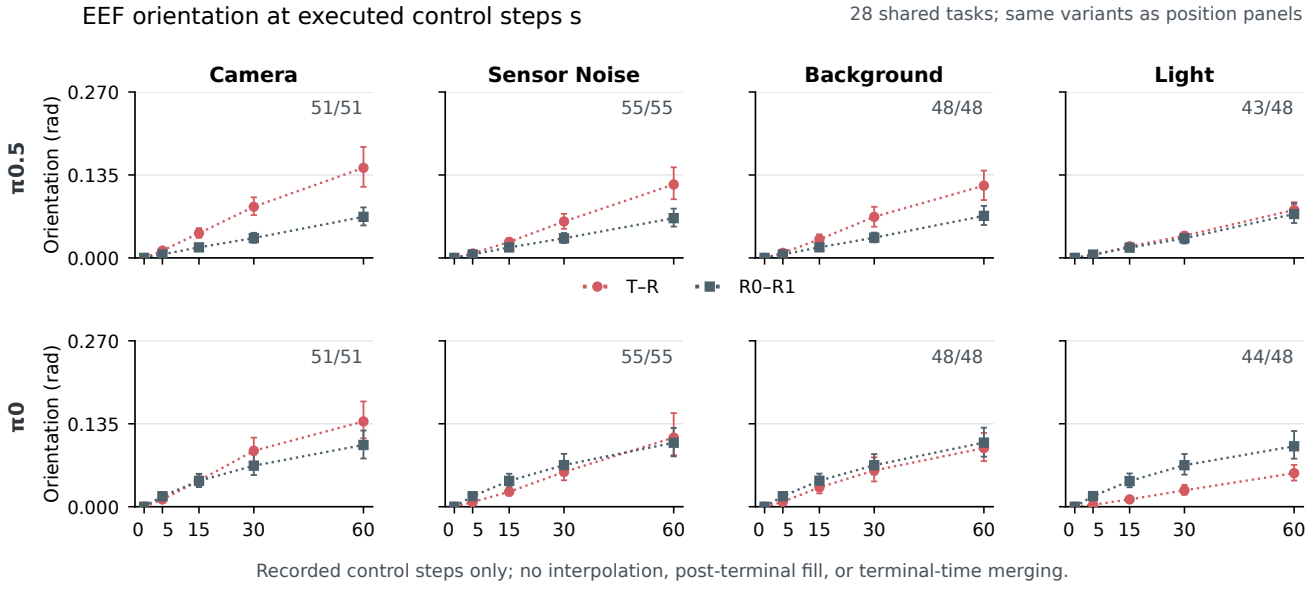


Figure S4. **End-effector orientation, a separate motion endpoint.** Perturbed–clean orientation distances in radians, with the same rule as the position panels (same setting, same control step); each control step has its own joint support. Lines connect the stored measurements; the two policies are shown in separate panels.

S5.1. Command changes reach the robot: strongly after 5 controls, still after 60

Do changes in the first five commands show in the robot's motion? The command change ranks with end-effector separation almost perfectly after five controls, as expected since those controls execute the commands, and more weakly but clearly after 60 (Table S11; Fig. S21). This supports Sec. 5.1 of the main paper. The comparison uses all 268 planned settings.

Same pairs, two control steps. We match the same policy, setting and noise draw after controls 5 and 60, requiring the same first-query command-change norm and a valid end-effector separation at both steps; every outcome group is included. The matched support is 516/536 pairs for $\pi_{0.5}$ and 526/536 for π_0, each on 40 tasks; unavailable measurements stay counted on the planned support.

Task-weighted rank correlation. Each eligible pair has weight $w_{tvn} = 1/(TV_tN_{tv})$, weighting tasks, their settings and their noise draws equally at each level. For a scalar z, define the weighted midpoint empirical CDF

$$F_w(z_i) = \sum_{j:z_j<z_i} w_j + \tfrac{1}{2} \sum_{j:z_j=z_i} w_j. \qquad \text{(S4)}$$

The rank correlation is the weighted Pearson correlation of these transforms of command-change norm and end-effector distance, pooled over the plotted families with equal task weight. Section S4 instead centers ranks within tasks. Constant tasks stay included; zero overall variance would leave the statistic undefined.

Uncertainty and units. The analysis rules were fixed before these calculations, on data already observed. Intervals come from 10,000 whole-task bootstrap draws, shared by both control steps, with the weighted CDFs recomputed in each draw; percentile intervals use linear interpolation, and all draws are defined. Command norms use the fifteen translation coordinates (command units); end-effector distances are in millimetres.

Table S11. **The command change ranks with end-effector separation more strongly after control 5 than after control 60, in both policies.** Weighted rank correlation on the same pairs at both steps. Brackets are pointwise 95% intervals for each policy and contrast.

Policy	Control 5	Control 60	Early minus late
$\pi_{0.5}$	0.953 [0.933,0.968]	0.474 [0.362,0.580]	0.479 [0.368,0.593]
π_0	0.974 [0.961,0.982]	0.409 [0.303,0.512]	0.566 [0.463,0.671]

S6. Paired Task Outcomes against a Repeat and a Clean Rerun

Which runs lose success under a perturbation, which gain it, and how does that compare with a clean rerun? Camera costs net success in both policies, and Sensor Noise in π_0; under Background, $\pi_{0.5}$ loses and gains successes equally, so its net success is unchanged while outcomes still change (Table S13). Lost successes exceed the clean-rerun level under Camera and Sensor Noise in both policies; gained successes exceed it in no family. This supports Sec. 5.2 and Fig. 5 of the main paper: net success hides lost and gained successes. Figure S22 gives the full comparison with the repeat and the clean rerun. These executions are separate from the internal recordings.

S6.1. Paired success, lost and gained successes

Design. Each policy–setting record holds five runs, R_0, T_0, R_1, T_1, R'_0, with policy-noise draws $0, 0, 1, 1, 0$, all from the setting's matched initial state: two clean runs, two perturbed runs that each share one clean run's noise, and a repeat of R_0. R_0 versus R_1 is the clean rerun; R_0 versus R'_0 is the repeat, which checks reproducibility. Each policy keeps its own definition of the policy noise, and every comparison is formed within one record. Estimates average the two perturbed–clean comparisons equally, then settings within tasks, then tasks equally. Whole-task resampling keeps both noise draws and all five runs together, so intervals describe variation over tasks given the recorded noise draws. The episode budget is fixed within each comparison; in the spatial suite it is 220 controls for π_0 and 280 for $\pi_{0.5}$.

Coverage. The study plans 268 settings per policy (536 noise pairs per policy), 536 policy–setting records across

Table S12. **The paired executions cover all planned settings; 39 of 2,680 run outcomes are unknown.** 268 planned settings per policy, five runs each; counts on the planned support, no interval. Recorded settings and validated outcomes are counted separately; unknown outcomes enter the bounds over missing outcomes.

Coverage	$\pi_{0.5}$	π_0
Recorded policy–setting records	268/268	268/268
All five outcomes validated	257	263
Validated run outcomes	1,314/1,340	1,327/1,340
Unknown outcomes	26	13
Technically missing	17	8
Failed validation	9	5

both policies and 2,680 run outcomes. Every planned record, including the six specified $\pi_{0.5}$ Light settings, exists. Table S12 keeps the full denominator: 2,641 outcomes are validated and 39 unknown, 25 through technically missing runs and 14 through failed validation.

The primary comparison uses the 28 common tasks and the 202 settings per policy that LIBERO-Plus assigns a difficulty level: 48 Background, 51 Camera, 48 Light and 55 Sensor Noise. Each noise draw has weight one half. The families share tasks, and each keeps its own selected settings and difficulty distribution; results describe this panel.

With χ the probability of a changed outcome and ΔSR the success change, lost (harm) and gained (recovery) successes, defined below, give the paired identities

$$\begin{aligned} \chi &= \Pr(\text{harm}) + \Pr(\text{recovery}), \\ \Delta\text{SR} &= \Pr(\text{recovery}) - \Pr(\text{harm}). \end{aligned} \tag{S5}$$

Definitions. For binary outcomes Y_R, Y_T of a clean and a perturbed run, a lost success (SF, harm) has probability $\Pr(Y_R = 1, Y_T = 0)$ and a gained success (FS, recovery) $\Pr(Y_R = 0, Y_T = 1)$, both unconditional paired probabilities under the stated weights. The success change (RD) is

$$\text{RD} = \text{SR}(T) - \text{SR}(R) = \text{recovery} - \text{harm}, \tag{S6}$$

and the rate of changed outcomes is their sum. Unknown outcomes take every possible value on the planned support, giving bounds over missing outcomes; sampling intervals are separate.

Net success hides lost and gained successes, measured against the clean rerun. Under Background, $\pi_{0.5}$ loses and gains 3.57% of outcomes each: its success change is zero while 7.14% of outcomes change. The adjusted intervals use 2,000 whole-task draws with the fixed setting and noise-draw weights; the two Camera point estimates coincide, and each policy has its own interval. The table below compares the rate of changed outcomes between perturbed and clean runs (T_n vs. R_n) with that of the clean rerun (R_0 vs. R_1), in task-equal percentages on complete records with intervals from 2,000 whole-task draws. Under Camera and Sensor Noise, perturbations change more outcomes than a clean rerun in both policies; under Background they do not (Light is incomplete). Split by direction, with the rerun's changed outcomes divided equally between the two directions, lost successes exceed the rerun level under Camera and Sensor Noise in both policies (excess $+10.71$ to $+23.21$ points, lower interval limits $+0.89$ to $+11.61$), and gained successes exceed it in no family (-0.96 to $+2.68$, every interval including zero). Table S14 adds the repeat.

Policy / family	Perturbed	Clean rerun	Excess [95% interval]
$\pi_{0.5}$ / B	7.14	3.57	$+3.57$ $[-2.68, +8.93]$
$\pi_{0.5}$ / C	29.46	3.57	$+25.89$ $[+13.39, +39.29]$
$\pi_{0.5}$ / S	20.54	7.14	$+13.39$ $[+2.68, +25.00]$
π_0 / B	10.71	7.14	$+3.57$ $[-5.36, +12.50]$
π_0 / C	24.11	5.36	$+18.75$ $[+4.46, +32.14]$
π_0 / S	17.86	0.00	$+17.86$ $[+9.82, +26.79]$

S6.2. Commands and motion on the same executions

Camera changes the commands and moves the end effector most in both policies; Light has the smallest means and incomplete support (Table S15). Plans hold ten actions ($\pi_{0.5}$) and fifty (π_0); the comparison uses the first five commands, end-effector positions at fixed control steps, and each run's final observation. Each row uses the settings where the metric and control step are available for T_0 vs. R_0, T_1 vs. R_1, and R_0 vs. R_1; outcome estimates keep the full planned denominator.

End-effector positions at controls 0, 5, 15, 30 and 60 are read when the corresponding environment call returns; controller commands and contact histories are separate measurements. Final observations follow each run's own last step rather than a shared step; missing poses are neither interpolated nor replaced by final poses.

Internal states in the paired executions. The $\pi_{0.5}$ internal states stored here are recomputed (prefix H before each block, action-expert H after each block), unlike the states recorded during the policy's own generation in the internal recordings; their availability is checked per run. The archived π_0 runs stored no internal states; Sec. S15 regenerates first-query states for 369 pairs. The other comparisons between the policies concern commands, motion and paired outcomes.

A family can change commands more yet cost less success. A ranking comparison listed in advance asks how

Table S13. **Camera costs net success in both policies, and Sensor Noise in π_0; under Background, $\pi_{0.5}$ loses and gains equally and its net success is unchanged.** Primary support of 28 tasks and 202 settings per policy. Success, lost and gained successes in percent; success change and its interval in percentage points. Bracketed estimates, including all incomplete Light estimates, are bounds over missing outcomes; the last column is the 98.75% whole-task bootstrap interval, Bonferroni-adjusted over the four families within each policy.

Policy/family	SR(R)	SR(T)	Lost	Gained	Success change	Adjusted interval for success change
$\pi_{0.5}$ / B	94.64	94.64	3.57	3.57	0.00	[-7.14,8.93]
$\pi_{0.5}$ / C	94.64	74.11	25.00	4.46	-20.54	[-39.29,-1.79]
$\pi_{0.5}$ / L	91.07	[91.07,99.11]	[0.00,5.36]	[5.36,8.04]	[0.00,8.04]	Not identified
$\pi_{0.5}$ / S	92.86	84.82	14.29	6.25	-8.04	[-23.21,5.36]
π_0 / B	89.29	85.71	7.14	3.57	-3.57	[-13.39,5.36]
π_0 / C	86.61	66.07	22.32	1.79	-20.54	[-35.71,-6.25]
π_0 / L	87.50	[84.82,92.86]	[1.79,6.25]	[3.57,7.14]	[-2.68,5.36]	Not identified
π_0 / S	92.86	78.57	16.07	1.79	-14.29	[-28.57,-2.68]

Table S14. **Perturbed runs change more outcomes than the repeat of the first clean run, which changes no validated outcome, under Background, Camera and Sensor Noise.** Population and weights as above; rates of changed outcomes (lost plus gained) in percent, brackets are bounds over missing outcomes, no intervals. Columns: perturbed run, repeat of the first clean run (same noise), clean rerun (different noise). Light is reported with its bounds.

Policy/family	Perturbed T vs. R	Repeat R_0 vs. R_0'	Clean rerun R_0 vs. R_1
$\pi_{0.5}$ / B	7.14	0.00	3.57
$\pi_{0.5}$ / C	29.46	0.00	3.57
$\pi_{0.5}$ / L	[5.36,13.39]	[0.00,10.71]	10.71
$\pi_{0.5}$ / S	20.54	0.00	7.14
π_0 / B	10.71	0.00	7.14
π_0 / C	24.11	0.00	5.36
π_0 / L	[5.36,13.39]	[0.00,8.93]	3.57
π_0 / S	17.86	0.00	0.00

often one family has the larger command or motion change while another has the larger net loss of success. All six ordered comparisons among Background, Camera and Sensor Noise are stored for the primary command measure; each fraction is the share of 2,000 resampled task sets in which the reversal holds strictly (ties count as none). For π_0, Background against Sensor Noise, the fractions are 69.9% for commands, 67.75% for EEF5 and 2.8% for EEF60, so the reversal depends on the measure and the control step.

Estimators that could not be computed. A planned estimator of the difference between correlations is undefined on these executions: for both policies and all five fixed primary command and end-effector metrics, none of the 2,000 draws defines it on the fixed support. The aggregation and lost-success comparisons that can be estimated use the studies in Secs. S10 and S11.5. The timing of exposure is examined in the separate 552-episode study (Sec. S9), which compares

Table S15. **Camera changes commands and moves the end effector most in both policies; Light has the smallest means and incomplete support.** Primary-panel settings whose perturbed–clean and clean-rerun records share metric and control step (N/tasks); no row replaces the planned outcome denominator. Task-equal perturbed–clean means, no intervals. Translation RMS over the first five commands in command units; EEF5/EEF60 are end-effector position distances after controls 5 and 60, stored in metres, shown in centimetres. Supports for each metric, control step and reference, including unequal final steps, are recorded separately.

Policy/ family	Joint N/tasks	Translation RMS	EEF5 (cm)	EEF60 (cm)
$\pi_{0.5}$ / B	48/28	0.03813	0.291	2.441
$\pi_{0.5}$ / C	51/28	0.09916	0.836	4.046
$\pi_{0.5}$ / L	43/26	0.02269	0.181	1.588
$\pi_{0.5}$ / S	55/28	0.05344	0.411	2.975
π_0 / B	48/28	0.04750	0.374	2.849
π_0 / C	51/28	0.08016	0.625	5.424
π_0 / L	44/27	0.01933	0.144	1.713
π_0 / S	55/28	0.04465	0.351	3.349

$C/F_{30}/R_{30}/N_{30}$ schedules within $\pi_{0.5}$ on its own cases.

Motion size and lost success, by family (post hoc). The table below gives the AUROC of Sec. S5 (motion ratio after control 5; lost against joint successes) per policy and family, with lost/joint-success pair counts in parentheses; the last row uses motion after control 60. Light has too few lost-success pairs. New draws make the π_0 Sensor Noise interval differ slightly from Sec. S5.

Family	$\pi_{0.5}$	π_0
Background	0.39 [0.13, 0.68] (4/89)	0.67 [0.25, 0.90] (8/80)
Camera	0.57 [0.40, 0.73] (27/71)	0.57 [0.38, 0.74] (24/65)
Sensor Noise	0.74 [0.57, 0.89] (16/86)	0.49 [0.29, 0.69] (18/84)
Pooled	0.72 [0.62, 0.82] (47/328)	0.64 [0.48, 0.77] (51/309)
Pooled, control 60	0.60 [0.47, 0.72]	0.68 [0.57, 0.77]

S7. Repeatability of Command Changes across Policy-Noise Draws

Does a perturbation change the commands in the same way under a different draw of policy noise? For the arm it does: across the four families, the agreement of the translation changes is 0.884–0.920 in $\pi_{0.5}$ and 0.799–0.889 in π_0, where 1 means identical changes (Table S16). This supports Sec. 5.1 of the main paper: the first five commands are a repeatable readout of a perturbation. Gripper changes agree much less.

Rotation repeats nearly as well; the gripper does not. Rotation agreement spans 0.818–0.858 in $\pi_{0.5}$ and 0.719–0.822 in π_0. Agreement of the continuous gripper command is lower, 0.143–0.371 and 0.077–0.342. These ranges span the family point estimates; Table S16 gives the intervals for each component.

The same perturbation under two noise draws. We use the commands of the paired executions (Sec. S6). Within each policy–setting record, R_0/T_0 and R_1/T_1 share a policy-noise draw. All four runs must have five commands at the first query ($q = 0$), with verified initial state, source, noise draw and command positions; a command stays missing if its environment call did not return. All outcome groups, including joint failure, are eligible.

A signed agreement score. The score divides the inner product of the two noise-matched changes by their mean squared size, so equal changes score one, unequal changes in the same direction score below one, and opposing changes score below zero. Flatten the first five commands and write $d_n = T_n - R_n$, $n \in \{0, 1\}$; translation and rotation have 15 coordinates each and the gripper five. Define

$$D = \tfrac{1}{2}(\|d_0\|^2 + \|d_1\|^2), \tag{S7}$$

$$M = \left\|\tfrac{1}{2}(d_0 + d_1)\right\|^2, \quad V = \tfrac{1}{4}\|d_0 - d_1\|^2, \tag{S8}$$

$$U = d_0^\top d_1 = M - V, \qquad D = M + V. \tag{S9}$$

In compact notation, with $\delta u_n = d_n$ the change in the first five commands under noise draw n, the numerator and denominator are

$$U = \delta u_0^\top \delta u_1, \quad D = \tfrac{1}{2}\big(\|\delta u_0\|^2 + \|\delta u_1\|^2\big). \tag{S10}$$

The agreement for one setting is

$$s = \frac{U}{D} = \frac{2 d_0^\top d_1}{\|d_0\|^2 + \|d_1\|^2} = \frac{2r\cos\theta}{1 + r^2}, \quad s \in [-1, 1], \tag{S11}$$

where the last form requires both norms to be positive and $r = \|d_1\|/\|d_0\|$. If both changes are zero, s is undefined; if only one is zero, $s = 0$ while the cosine is undefined. Ratios use the observed changes directly, without an epsilon or clipping. Reading U as a squared population mean would require independent, identically distributed noise; the ratio is a descriptive score for the two recorded draws, not an unbiased estimator of a population ratio.

Support and weighting. The primary plan has 202 settings on 28 common tasks per policy. All four runs have the first five commands in 197/202 settings for $\pi_{0.5}$ and 198/202 for π_0; Light has 43/48 settings on 26 tasks and 44/48 on 27, while the other families keep all primary settings and 28 tasks (Table S16). The full sensitivity scope keeps 258/268 and 263/268 settings (stored full-scope table); its families originally cover 34 Background, 32 Camera, 40 Light and 39 Sensor Noise tasks. Command eligibility and outcome availability are counted separately; the 39 unknown outcomes stay in the denominator of 2,680 runs.

Per-setting scores are averaged within task, and tasks equally, per policy, family and component; the estimate is a mean of ratios. The 2,000 whole-task bootstrap draws share their task resampling across series, keep settings and noise pairs nested, and are all valid; the 95% percentile intervals are pointwise and condition on the two noise draws and the selected settings.

Opposing changes and zero changes stay in the comparison. The continuous gripper changes in opposite directions ($U < 0$) in 36/197 and 62/198 primary settings, and the π_0 Light gripper interval crosses zero in both scopes. Translation never has $U < 0$; π_0 Sensor Noise rotation has it once. All scores are defined and no setting has only one zero change; the rules for zero changes remain in the protocol. In the full scope the gripper has $U < 0$ in 52/258 and 75/263 settings; this larger scope is a sensitivity analysis, not an independent replication.

Light has different eligible settings in the two policies, so a direct comparison of the policies requires their common settings. The gripper command can also be read as an event: Sec. S7.1 tests whether the timing of its first opening-to-closing transition adds information about lost success at a fixed cutoff.

S7.1. Gripper timing does not clearly improve prediction of lost success

Does the timing of the first opening-to-closing gripper command add information about lost success beyond the size of the command changes? At a cutoff of 60 controls, fixed in advance, adding timing lowers the held-out Brier loss slightly in both policies, with intervals that include zero (Table S17). This is an additional, exploratory study; the main paper does not rely on it. The feature is a command event, not physical closure or object contact.

Table S16. **Translation and rotation changes repeat across policy-noise draws; the gripper agrees less.** Primary 28-task plan per policy; Settings and Tasks give defined/planned support. Entries are task-equal means of the per-setting agreement U/D with 95% whole-task intervals. These are normalized inner products of the changes under the two draws, not energy percentages.

Policy	Family	Settings	Tasks	Translation	Rotation	Continuous gripper
$\pi_{0.5}$	Background	48/48	28/28	0.898 [0.843, 0.946]	0.858 [0.793, 0.913]	0.371 [0.270, 0.467]
$\pi_{0.5}$	Camera	51/51	28/28	0.920 [0.872, 0.957]	0.818 [0.733, 0.892]	0.362 [0.282, 0.442]
$\pi_{0.5}$	Light	43/48	26/28	0.884 [0.810, 0.943]	0.848 [0.778, 0.907]	0.143 [0.009, 0.279]
$\pi_{0.5}$	Sensor Noise	55/55	28/28	0.909 [0.870, 0.942]	0.849 [0.788, 0.908]	0.371 [0.250, 0.491]
π_0	Background	48/48	28/28	0.865 [0.810, 0.909]	0.807 [0.720, 0.877]	0.342 [0.219, 0.453]
π_0	Camera	51/51	28/28	0.889 [0.840, 0.928]	0.814 [0.756, 0.866]	0.325 [0.185, 0.460]
π_0	Light	44/48	27/28	0.799 [0.711, 0.880]	0.719 [0.633, 0.797]	0.077 [−0.090, 0.245]
π_0	Sensor Noise	55/55	28/28	0.844 [0.780, 0.896]	0.822 [0.762, 0.876]	0.228 [0.078, 0.388]

Table S17. **Adding gripper timing to the command-change sizes does not clearly lower the loss at a fixed cutoff of 60 controls.** Cutoff pairs (of 404 per policy) reach the cutoff with valid command records and outcomes; the subset whose clean run succeeds (N/tasks) is analysed. Improvement is baseline minus augmented out-of-fold Brier loss, positive when timing lowers the loss, with a 97.5% paired whole-task interval. The denominator 404 is the fixed plan per policy; 375/360 pairs are analysed.

Policy	Cutoff pairs	Clean-success N/tasks	Baseline Brier	Augmented Brier	Improvement [97.5% interval]
$\pi_{0.5}$	396/404	375/27	0.09664	0.09555	0.00109[−0.00224, 0.00448]
π_0	397/404	360/26	0.12323	0.12172	0.00151[−0.00471, 0.00724]

Event reconstruction. Across all 50 runs, the Panda gripper command is positive for closing and negative for opening; zero leaves the accumulated command unchanged. Events are reconstructed only from complete, finite command records up to the cutoff with valid identifiers; the 39 unknown run outcomes, early terminations and technical failures are kept.

Baseline and augmented predictors. Of the 404 primary pairs per policy, both runs must reach the cutoff with valid command records and outcomes, and the clean run must succeed. The baseline predictor uses $\log(1 + \mathrm{RMS})$ of the translation, rotation and continuous-gripper command differences over the same first 60 controls, plus family indicators. The augmented predictor adds whether each run has an opening-to-closing transition by the cutoff, the difference of their times divided by 60 when both do, and each run's initial command direction; occurrence flags separate an absent event from a zero time difference. Both use only information available at the cutoff.

Fitting on held-out tasks and intervals. The ridge model, fixed in advance, uses penalty $\alpha = 1$, an unpenalized intercept, centering and scaling on the training data only, and leave-one-task-out prediction clipped to $[0, 1]$, with tasks weighted equally and families, settings and noise draws weighted equally within their parents. Both predictors use identical rows in each fold; a training fold with a single class would fall back to the training mean, and none does. Both initial-closing indicator columns are zero in all eligible rows and add no variation. The primary quantity is baseline minus augmented out-of-fold Brier loss. Paired whole-task resampling (2,000 draws) gives 97.5% intervals per policy; the predictors are not refitted inside the bootstrap.

Outcomes of the analysed pairs. On the primary support the outcome counts are $328/47/17/4$ SS/SF/FS/FF with eight unknown pairs for $\pi_{0.5}$ and $309/51/10/27$ with seven unknown pairs for π_0; every validated pair is kept whether or not a transition occurs. Pairs whose clean run succeeds number $93/98/82/102$ (Background/Camera/Light/Sensor Noise) for $\pi_{0.5}$ and $88/89/81/102$ for π_0. The estimates in Table S17 apply to this conditional support; cutoff, features and penalty were fixed in advance.

S8. Response Structure across Observations and Generation Steps

This section follows the response beyond the first generation step of the first query. Sec. S8.1 shows that shared directions fitted at one observation of an episode still retain most of the response at a later observation, far more than clean-feature or random directions; this supports the statement in main Sec. 4.1 that the result holds at a later observation. Secs. S8.2 and S8.3 are additional studies; the main paper does not rely on them.

Figures. Later observation: Fig. S6 (episodes, step $j = 0$) and Table S18 (condition means). Generation: Figs. S7–S9 (layer 18) and S10–S11 (layer 1) on all recordings; Figs. S12, S13 and S19a–c on the subset matched to commands. Execution: Figs. S14–S15, motion aligned by executed control where measured; outcomes keep planned denominators.

S8.1. Shared directions fitted at one observation hold at a later observation

Directions fitted at an earlier observation retain 50.8–64.2% (L1), 35.7–45.7% (L9) and 32.7–38.1% (L18) of the centered response energy at a later observation on 16 held-out tasks, against at most 11.0% for clean-feature directions and at most 1.8% for random directions (Table S18). The advantage over both controls holds in all 30 condition–layer comparisons. Unlike the fit in main Sec. 4.1, this one holds out tasks but not perturbation conditions: the ten test conditions are also used for fitting.

Data: two observations per episode. The study runs $\pi_{0.5}$ on original LIBERO: 3,360 episodes with two observations each, 6,720 records. Features exist for 6,554 records; 83 missing episodes (166 records) stay missing. The two observations are taken after a suite-specific number of executed controls, Spatial 20/40, Object 25/55, Goal 30/60 and LIBERO-10 50/100, with no feature recording in between. At each observation, clean and perturbed images of the same visited state pass through the policy with the same initial noise; we record clean features and responses at L1, L9 and L18, first generation step ($j = 0, t = 1$), first five planned actions ($3 \times 5 \times 1024$ per observation). Between the observations, state, images and noise all change.

Fit: held-out tasks, seen conditions. Of the 40 tasks, 24 are used for fitting and 16 are held out. Only the first observations of the fitting tasks under the ten persistent conditions enter the fit, 1,404/1,440 available observations, so the test conditions are seen during fitting. Tasks, conditions, start states, noise repeats and the five plan positions are weighted equally within their parents. Each layer has its own mean μ_p for each position p and a channel scatter shared across positions,

$$C = \sum_{i,p} \frac{w_i}{5} (\Delta H_{ip} - \mu_p)^\top (\Delta H_{ip} - \mu_p). \qquad \text{(S12)}$$

Its top 16 eigenvectors form the shared directions $B \in \mathbb{R}^{1024\times 16}$; the largest absolute loading of each fixes its sign. The rank is fixed and nothing is selected by outcome. Clean-feature directions use the same observations and weights but are fitted to clean features; random directions are a fixed Gaussian basis orthonormalized by QR. All three reconstruct the same response about the same μ_p, with 16 coefficients per position computed from the test response, 80 per observation out of 5120 response entries. A mean-only baseline uses μ_p with zero coefficients.

Measures. For one observation, let $E = \|\Delta H\|_F^2$, $E_c = \|\Delta H - \mu\|_F^2$ and $S_a = \|\Delta H - \widehat{\Delta H}^{(a)}\|_F^2$ be the raw energy, the centered energy and the reconstruction error of directions a. The plotted centered retention and the tested advantage over a control are

$$R_a = 1 - \frac{S_a}{E_c}, \qquad G_{\text{response}-a} = \frac{S_a - S_{\text{response}}}{E}. \qquad \text{(S13)}$$

R_a pools the energy of all five positions before forming the ratio. Negative values of G are kept, and ratios with a zero denominator are undefined. Channel participation, the effective number of channels that carry the response energy, describes how widely the energy spreads; it is not a spectral rank or an intrinsic dimension.

The advantage over both controls holds in every comparison. All 30 condition–layer comparisons show a positive advantage over both controls, with positive 95% lower bounds (descriptive). The intervals come from 2,000 whole-task bootstrap draws, which keep the dependence within tasks and among comparisons on the same 16 tasks, and condition on the fitted directions and the observed support. Retention changes little between the two observations: for the paired first-to-second-observation difference in retention of the shared directions, 28/30 intervals cross zero; the remaining two are pointwise contrasts without multiplicity adjustment.

Retention holds even when the response changes direction. Fig. S6 keeps all 960 planned held-out episodes under persistent conditions, 940 available and 20 missing. The horizontal axis is the cosine between the full 5×1024 responses at the two observations; the vertical axes are the second-observation retentions of the three sets of directions. Even near zero cosine, the shared directions retain appreciably more energy than clean-feature and random directions: the

Table S18. **At the later observation, shared directions retain far more of the response than both controls; retention falls with depth while participation rises.** 16 held-out tasks, ten persistent conditions; each cell has 94/96 defined observations and 16/16 tasks. Ranges are over condition means, averaging available noises within start states, start states within tasks, and tasks equally. All three sets of 16 directions reconstruct the same position-centered response. Retention pools the five positions' energies before forming the ratio; participation pools position energies on 1024 channels.

Layer	Shared directions	Clean-feature directions	Random directions	Channel participation
L1	50.8–64.2%	3.6–6.3%	1.6–1.8%	31.0–39.9
L9	35.7–45.7%	0.8–1.8%	1.4–1.8%	63.9–117.9
L18	32.7–38.1%	8.2–11.0%	1.4–1.5%	186.2–229.0

response changes, but it stays within the shared span. The scatter is descriptive, with no fitted threshold or association model; colors mark conditions, one point per episode.

Channel and attention views. The channel view shows all five plan positions and 1024 channels in fixed pages of 128, not channels selected by amplitude. Its attention diagnostic recomputes the unconditional main-camera probability on $8 \times 2 \times 16 \times 16$ maps per layer, averaged over the first five action queries; camera mass, its signed change and the unconditional camera half-L_1 differ from the within-camera conditional TV, and no per-position attention, complete-key P, prefix V or W was recorded. Ten held-out examples, chosen by a fixed rule without outcomes, allow inspection. Three-dimensional plots use the first three directions with their energy fractions; positions are centered separately, so distances across positions are not hidden-state distances. Clean and three history controls, perturbed only at the first policy query, have zero raw response but centered coordinates $-\mu_p$; their histories retain the earlier exposure.

Implementation notes. The random directions and the bootstrap use a fixed seed (20261012). Raw flow is a 5×32 generated output; planned-command RMS describes the generated block, and command analyses use the values returned by the controller. An episode's clean–perturbed success pair uses the independent clean run from the same start state and noise, not the clean re-rendering of the same state behind its response.

S8.2. Generation profiles and command associations on LIBERO-Plus

This is an additional study; the main paper does not rely on it. Across the generation steps of the first query, the shift in attention probabilities shrinks while the raw difference in the L18 attention output grows; the raw response of the post-block state shrinks, but relative to its shrinking clean scale it grows (Table S20). Which of these measurements relates more strongly to the command change is not resolved: all four differences between the attention-output and attention-probability associations have intervals that include or touch zero.

Table S19. **All recorded LIBERO-Plus pairs have internal measurements; 404 also enter the command analysis.** Recorded $\pi_{0.5}$ LIBERO-Plus pairs by family; counts, no interval. The command column is the matched subset with a verified change in the first five commands; the other 132 lie outside that intersection and are not failed episodes.

Family	Recorded pairs	With commands
Background	118	96
Camera	112	102
Light	152	96
Sensor Noise	154	110
Total	536	404

Data. The analysis keeps all 536 recorded LIBERO-Plus pairs and all 4,824 rows indexed by case, layer and generation step at L1/L9/L18 and $j = 0, 5, 9$. The command target exists for the matched subset of 404 pairs (Table S19). Case, row, noise, task, clean reference, family, suite and outcome are checked before matching; planned actions never replace missing executed commands. All seven P/W/AE fields are present at every recorded site. Prefix V is fixed within a query and is not repeated across generation steps.

Measures. P-shift is the total variation of the attention probabilities over all visible keys, averaged over valid heads and action queries. W is the attention output, combining prefix and action-side values, after the output projection and before gating, over all ten plan positions; its raw difference RMS, clean RMS and relative L_2 have distinct denominators and keep the model's activation units. P and W are recomputed from saved states; H (AE) is the post-block state recorded during the policy's own generation (Table S1). Later generation steps use each branch's own x_j within the same first query.

Attention output grows while the post-block response shrinks in raw size. Figure S19 uses the 404 matched

pairs of Table S19, with the same cases and weights for every signal. Panel c shows the raw post-block H difference RMS and the clean-relative response on separate axes, without indexing to generation step 0; Table S20 gives the endpoint values. For every family, raw W increases across steps 0/5/9, while raw H decreases and relative H increases, because the clean RMS of H falls from approximately 46.8 to 16.6 over these steps. On all LIBERO-Plus recordings, the full-key TV decreases from the first to the last generation step in all 48 layer×perturbation-category–task-suite groups; at L18 the relative attention-output response increases in all 16 groups and its raw difference RMS in 15.

Associations with the command change. The target is the RMS difference over the first five commands and three translation coordinates, in command units. It repeats across each case's nine layer/step rows, so the 4,824 rows are repeated internal measurements of shared targets, and the scatterplots are descriptive. Association estimates and intervals come from a task-centered analysis at the first generation step that keeps all four sites and families (Sec. S4). For Camera, the prefix-H/P/W/AE associations with the command change are $-0.072/0.348/0.467/0.400$; for Sensor Noise, $0.620/0.453/0.554/0.568$. Camera W has association -0.284 $[-0.484, -0.044]$ with the signed outcome and an AUROC for lost success of 0.842 $[0.683, 0.967]$. The four W-minus-P intervals include or touch zero. The outcome groups are post hoc, and the estimates describe the observed tasks.

Wrist-camera values respond at L9 and L18 but not at L1. The Camera and Sensor Noise subsets contain 102 and 110 recorded pairs, each on the same 28 tasks. At the first query, we measure the norm of the wrist-value difference relative to the clean wrist-value norm. These prefix values are recomputed once per query, so they have no generation-step axis. Means use task-equal weighting over two policy-noise draws per setting. Brackets are pointwise 95% intervals from 2,000 whole-task bootstrap draws (descriptive, not simultaneous across rows).

Family	Layer	Relative response	Pointwise 95% interval
Camera	L1	0.000	$[0.000, 0.000]$
	L9	0.514	$[0.480, 0.550]$
	L18	0.275	$[0.252, 0.299]$
Sensor Noise	L1	0.000	$[0.000, 0.000]$
	L9	0.415	$[0.380, 0.451]$
	L18	0.232	$[0.207, 0.259]$

Both families have zero mean wrist-value response at L1 and nonzero responses at L9 and L18. This extends the layer comparison to the matched LIBERO-Plus panel; the separate fixed-state experiment in Sec. S10.4 shows the response with the wrist input explicitly unchanged.

Value responses concentrate on fewer tokens and channels at depth. Raw size, relative size and participation describe the amplitude of the value response, its scale relative to the clean values, and how widely its energy spreads. Channel participation is $(\sum_d e_d)^2 / \sum_d e_d^2$ for $e_d = \sum_k (\Delta V_{kd})^2$; token participation is $(\sum_k e_k)^2 / \sum_k e_k^2$ for $e_k = \sum_d (\Delta V_{kd})^2$. Both are undefined when the response energy is zero. For the Camera main-camera V, the difference RMS rises from 0.895 at L9 to 8.053 at L18 while the relative magnitude falls from 0.942 to 0.451; token participation falls from 190.17 to 62.46 and channel participation from 212.97 to 86.31 of 256 axes. Both cameras follow these directions in all four families. Prefix H and normalized Z also show different relative trends despite increasing raw differences, and in the action expert participation increases as retention by the shared directions decreases.

An idealized flow shows how generation alone separates two conditions. Take a fixed endpoint map $a(c)$ for condition c and velocity $v(x, t, c) = (x - a(c))/t$. The state is $x(t) = tz + (1 - t)a(c)$, so two conditions that share the noise z separate as $\Delta x(t) = (1 - t)[a(c') - a(c)]$, from zero at $t = 1$ to the full endpoint difference at $t = 0$. The states separate progressively although the endpoint map is fixed and the velocity difference is constant. The example describes state evolution only; the measured W responses also involve the policy's attention.

Related results elsewhere (each with its own directions, support and target).

Normalized responses at different sites. For Camera main-camera tokens, the relative response of prefix H rises $0.246 \to 0.615$ and that of Z falls $0.756 \to 0.484$ from L9 to L18, although both raw differences increase. In the fixed-state camera study, deep relative wrist-value norms are $0.400/0.364$ with an unchanged wrist image (Secs. S11.2 and S10.4).

Retention on the full LIBERO-Plus panel. Centered shared directions retain 36.7–84.2% of centered energy, against 0.5–10.6% for clean-feature and 1.3–1.9% for random directions, with 160 coefficients from the observed response over ten plan positions; uncentered shared directions reconstruct 32.4–75.3% of raw energy (Sec. S3). Denominators and the ten-position budget differ from Sec. S8.1.

A fixed coordinate subset. A separate digital-RGB analysis with a fixed subset of 64 coordinates captures about 3.7% of standardized energy on 2048 axes; a fixed coordinate subset and a shared span are different objects.

Angle prediction. Predicting the 30° response from the 15° one keeps its 41.2%/40.4% gains over matched simple rules,

Table S20. **Over generation, the attention-output response grows; the post-block response shrinks in raw size but grows relative to its clean scale.** L18 of the first query on the same 404 matched LIBERO-Plus pairs: 96/102/96/110 pairs for Background/Camera/Light/Sensor Noise, each on 28 tasks. Entries are task-equal means, averaging noises within settings and settings within tasks. Raw: per-case Frobenius norm over the square root of tensor size; relative: divided by the clean tensor norm before averaging. W is recomputed from saved states, before gating; H is recorded after the block.

Family	Attention output W: raw RMS $j=0\to 9$	Post-block H: raw RMS $j=0\to 9$	Post-block H: relative L_2 $j=0\to 9$
Background	$1.387\to 1.528$	$3.436\to 2.434$	$0.073\to 0.147$
Camera	$2.225\to 2.513$	$5.555\to 3.791$	$0.119\to 0.228$
Light	$0.895\to 1.035$	$2.218\to 1.609$	$0.047\to 0.097$
Sensor Noise	$1.434\to 1.587$	$3.993\to 2.662$	$0.085\to 0.160$

but it receives the observed smaller-angle response and has seen the target angle in training (Sec. S10.1).

Gains over one feature set versus absolute accuracy. In the separate matched-lighting probe of signed vectors, adding H18 improves on the visual predictor, yet the combined predictor's absolute error is 8.39% above the training-target mean for flow and 0.328% below it for candidate actions (Sec. S10.6). On LIBERO-Plus, the earlier uncentered readout compares response coordinates with the paired first flow as an existing predictor (Sec. S11.3); none of its original gain intervals is positive. The current comparison with the same directions is in Sec. S3.4.

S8.3. From the first five commands to motion in the paired executions

This is an additional study; the main paper's link from commands to motion (main Sec. 5.1) rests on Secs. S5.1 and S7. Here the same paired executions are followed to a later control: after 60 controls, Camera separates the perturbed run from its clean run by 2.29 $[1.74, 3.03]$ times the clean-rerun distance in $\pi_{0.5}$ and 1.55 $[1.04, 2.27]$ times in π_0, so Camera's effect on motion exceeds the clean-rerun scale in both policies at this later point.

Data. The cases are those of the command-consistency analysis at the first query: two policies, 268 planned settings each, 2,680 planned runs. Commands and motion are matched by policy, setting and execution; all 39 unknown outcomes stay unknown.

Measures. For each command component, the plot shows the size of the command change at the first query and the cosine between the two noise-matched changes; U/D remains the main paper's direction-and-magnitude statistic. End-effector position separation is the Euclidean distance after 5 or 60 executed controls; orientation is a separate angular endpoint. Missing values at the fixed steps are not filled from terminal poses, another noise or a neighboring observation. The command–position plots show 516/536 planned noise pairs for $\pi_{0.5}$ and 526/536 for π_0; a setting's two noises remain dependent, and outcome colors play no role in selection. The reference panels use 257/268 and 263/268 settings at each fixed step and keep T–R, R_0–R_1 (clean rerun) and R_0–R_0' (repeat of the first clean run) distinct; the plots show all planned rows and their missingness. The 268-setting views describe the full panel; main estimates and intervals use the primary 28-task panel.

Table S21. **Three repeats of the first clean run differ slightly in position after 60 controls but share their outcomes.** Qualified R_0–R_0' pairs in both policies; rows are the three recorded settings with nonzero repeat distance after executed control step 60, all with zero distance at step 5. Exact recorded distances after control 60, shown in millimetres; outcomes list R_0 then R_0'; no interval. The source of these small differences is unknown.

Policy	Setting	Control 60 (mm)	Outcomes
$\pi_{0.5}$	Example A	0.0319815382	SS
$\pi_{0.5}$	Example B	1.1713919779	FF
π_0	Example C	0.1556652612	SS

Camera exceeds the clean-rerun scale after 60 controls in both policies. The Camera ratios above are primary-panel estimates. Background's $\pi_{0.5}$ ratio is 1.273 $[0.9997, 1.6505]$, with a paired-distance difference interval that crosses zero; Light keeps its incomplete observed support. Sec. S5 gives the motion comparison.

Repeats are exact except in three pairs. The repeat R_0' checks that runs are reproducible. Every other qualified repeat pair has zero distance after 60 controls; the three in Table S21 have no recorded qualification issue, differ by small distances after 60 controls and keep their outcomes. Zero denominators and unavailable comparisons remain in the source tables.

Figures for Sec. S8. Figures S5 and S6 show response size, concentration and retention at a later observation. For atten-

tion, read the layer-18 relative response, raw difference and clean scale together (Figs. S7–S9); all three show the same 536 recorded LIBERO-Plus pairs. The layer-1 views give a shallow-layer comparison (Figs. S10–S12). Figure S13 shows the 404 command-matched cases used in Fig. S19. In the attention panels, 0/5/9 index generation steps within a query, not physical frames. Figures S14 and S15 show the direction of the command change and the clean-rerun reference. The full spatial attention example is in Fig. S16.

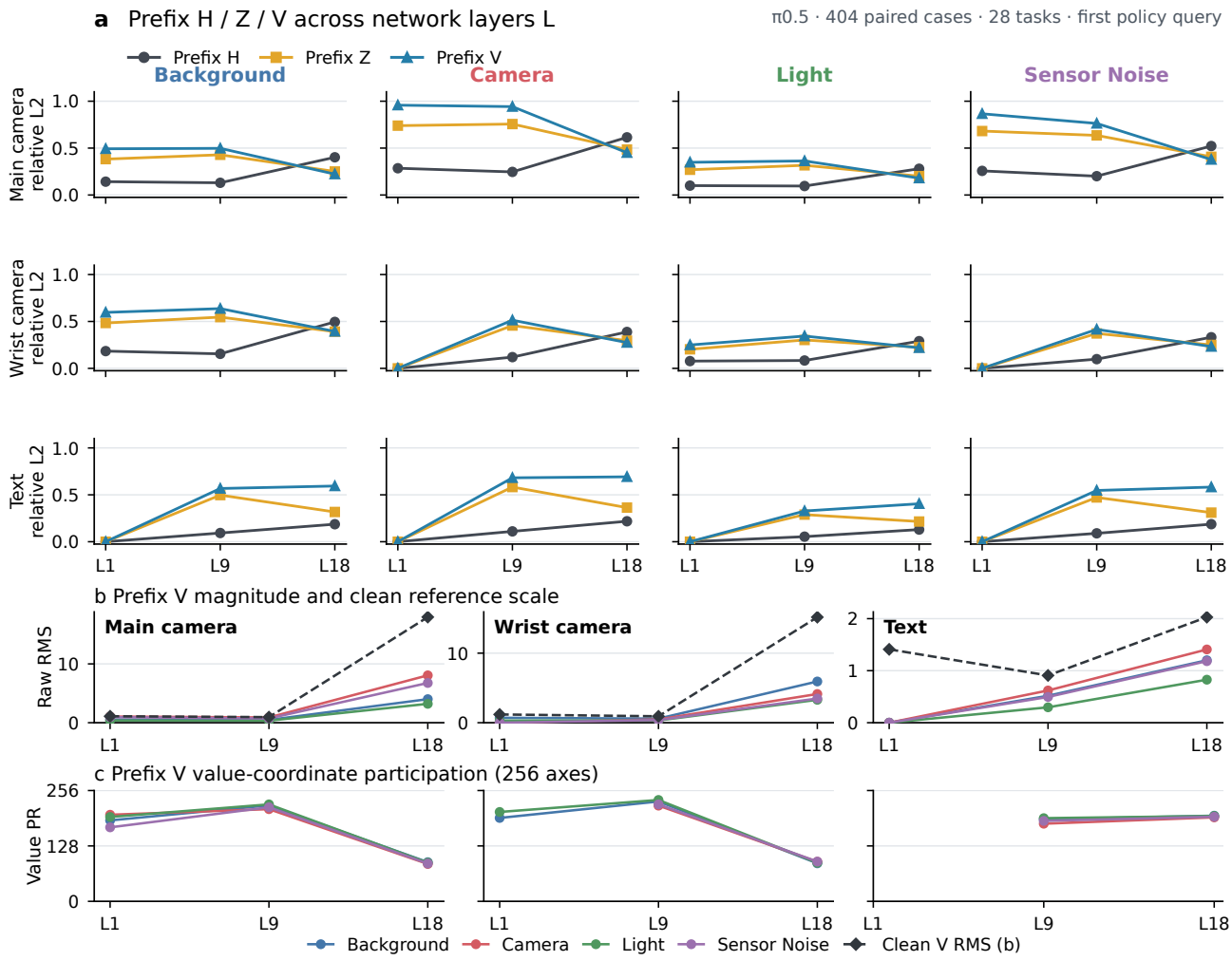


Figure S5. **Response size and concentration across channels follow different layer patterns.** (a) Clean-relative H/Z/V responses by region, (b) raw value-response RMS beside its clean scale, and (c) the effective number of participating V channels (256 axes). 404 matched LIBERO-Plus pairs on 28 tasks; descriptive family aggregates without intervals.

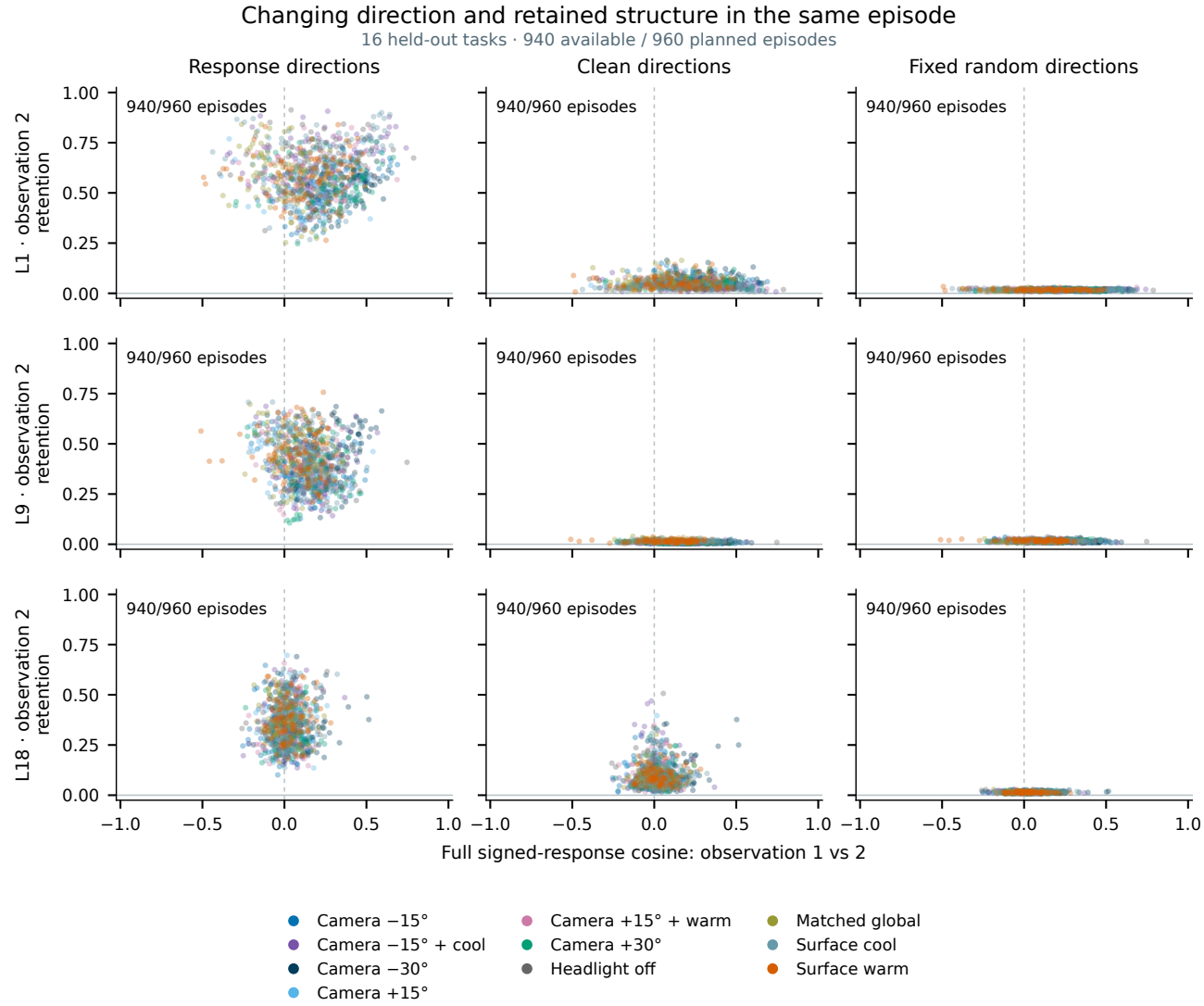


Figure S6. **Shared directions fitted at an earlier observation retain more of the later response than both controls, even when the response changes direction.** For each episode, the cosine between its full responses at the two observations (horizontal) against centered retention at the second observation (vertical). Shared, clean-feature and random directions use equal coefficient budgets. Cross-observation study (Sec. S8.1): 16 held-out tasks, ten conditions, 940/960 planned episodes observed; one point per episode, no interval.

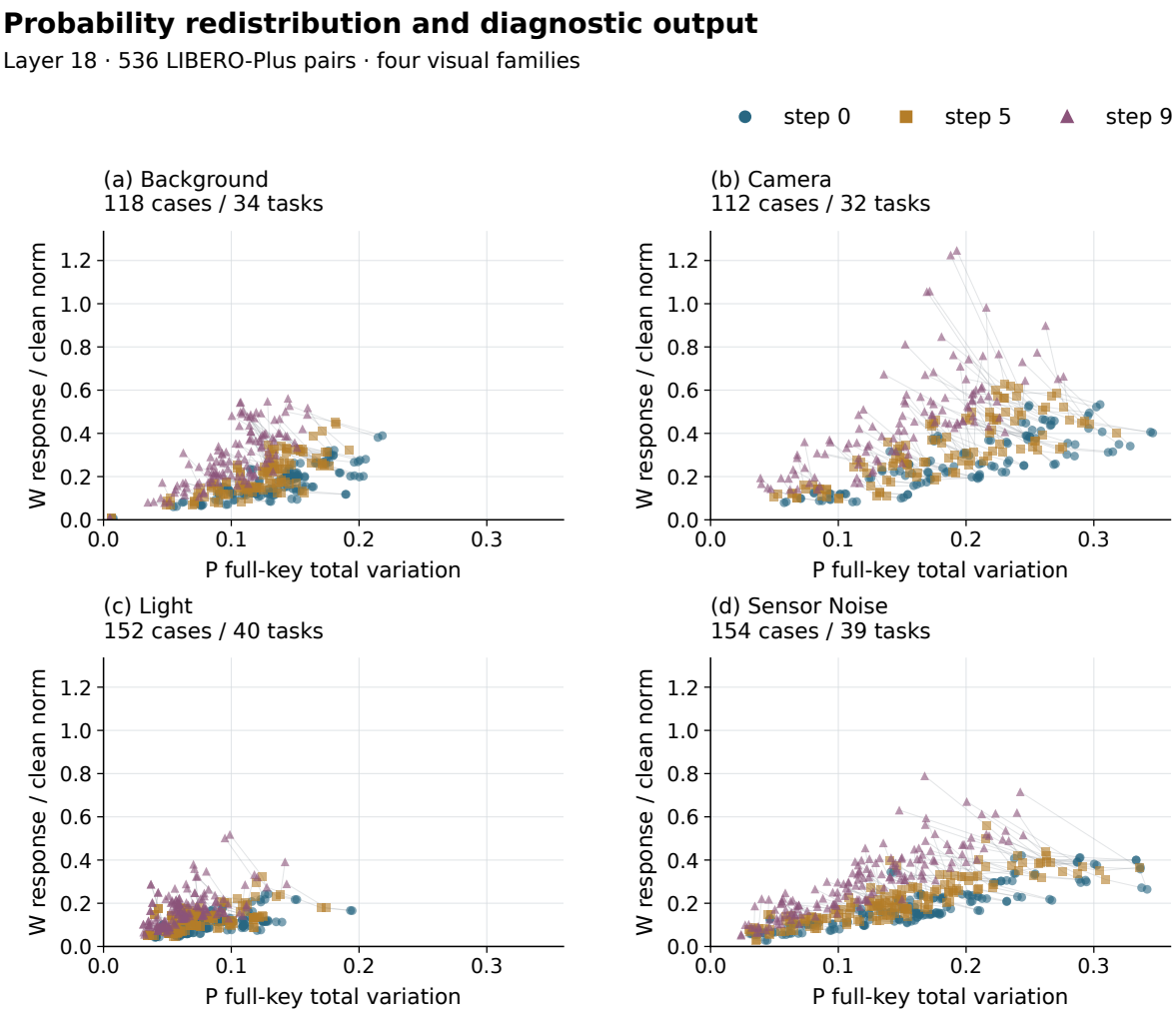


Figure S7. **At layer 18, attention-probability shifts shrink and relative attention-output responses grow over generation.** Horizontal: probability TV over all visible keys; vertical: the attention-output difference norm divided by the clean W norm. Both are recomputed at each branch's saved generation state, after output projection and before gating for W. Lines join generation steps 0/5/9 of the same case within the first query. All 536 recorded LIBERO-Plus pairs; Fig. S19 uses the 404 command-matched cases. One point per case and step, no interval. Figures S8 and S9 retain these cases and trajectories while showing the raw difference and clean scale.

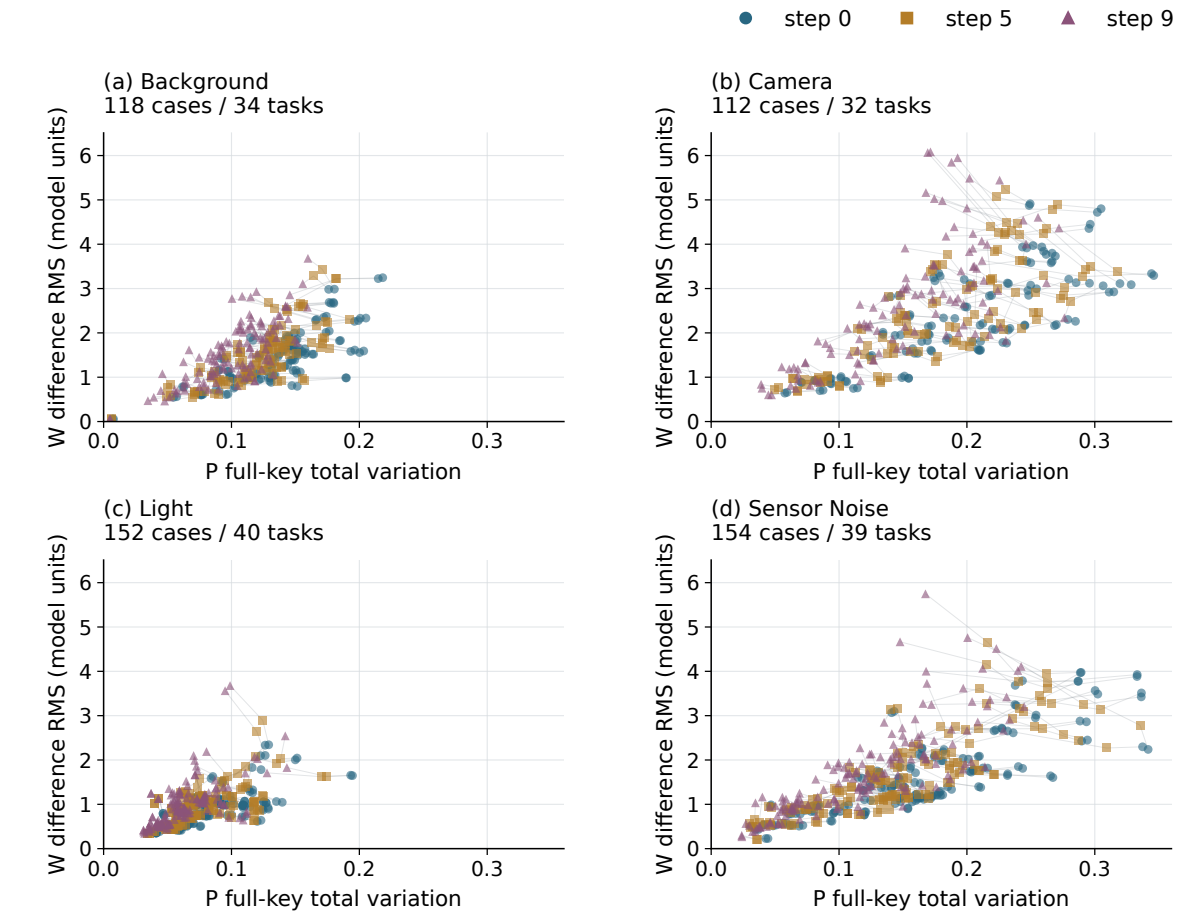


Figure S8. **The layer-18 attention-output difference on its raw scale.** Vertical: perturbed–clean W difference RMS in model units. The 536 cases, horizontal axis and lines joining generation steps 0/5/9 as in Fig. S7. Recomputed at each branch's saved generation state; one point per case and step, no interval. The corresponding clean W scale is shown in Fig. S9.

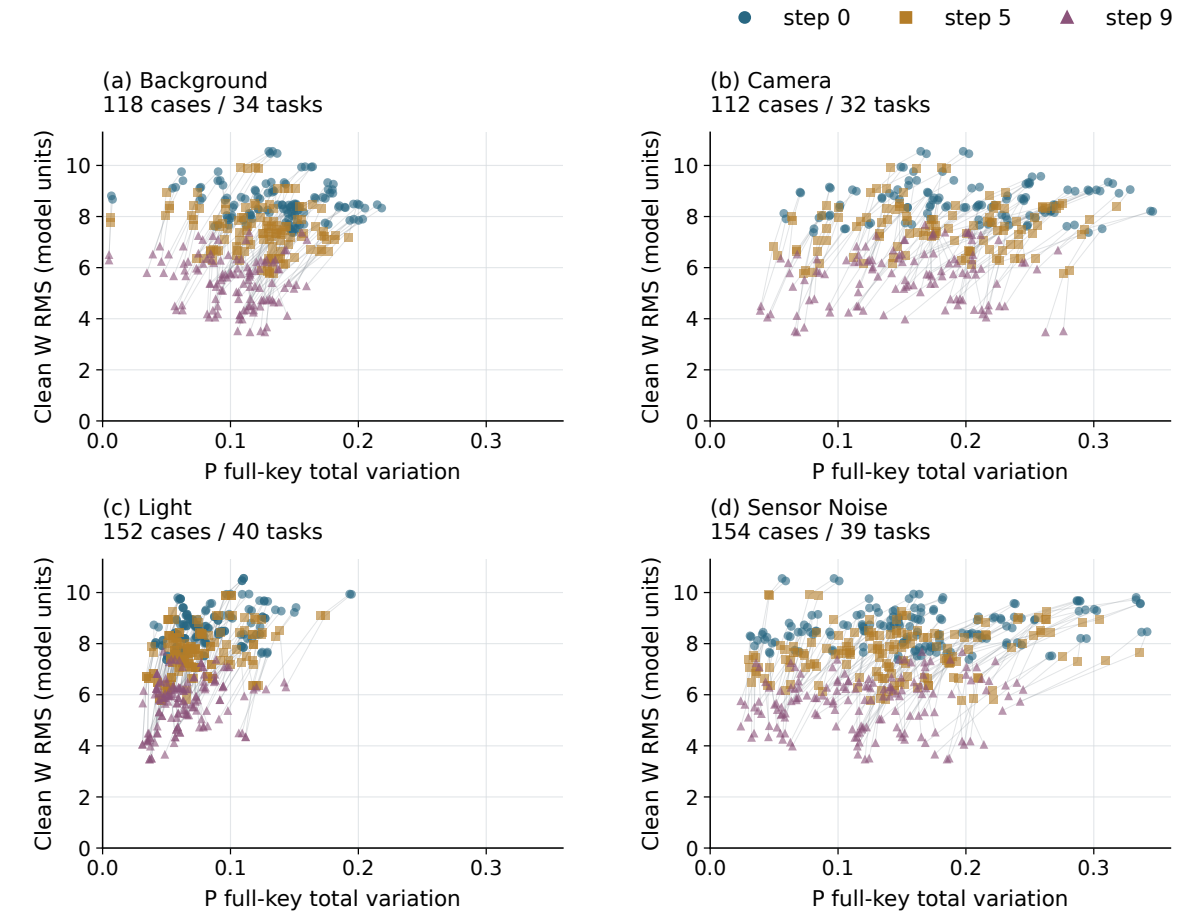


Figure S9. **The clean output scale also changes during layer-18 generation.** The vertical axis shows clean W RMS in model units, the reference scale for the relative response in Fig. S7. The 536 cases, horizontal axis and lines joining generation steps 0/5/9 are as in Figs. S7 and S8. Recomputed at the saved generation states; one point per case and step, no interval. The relative response must be read against this changing reference.

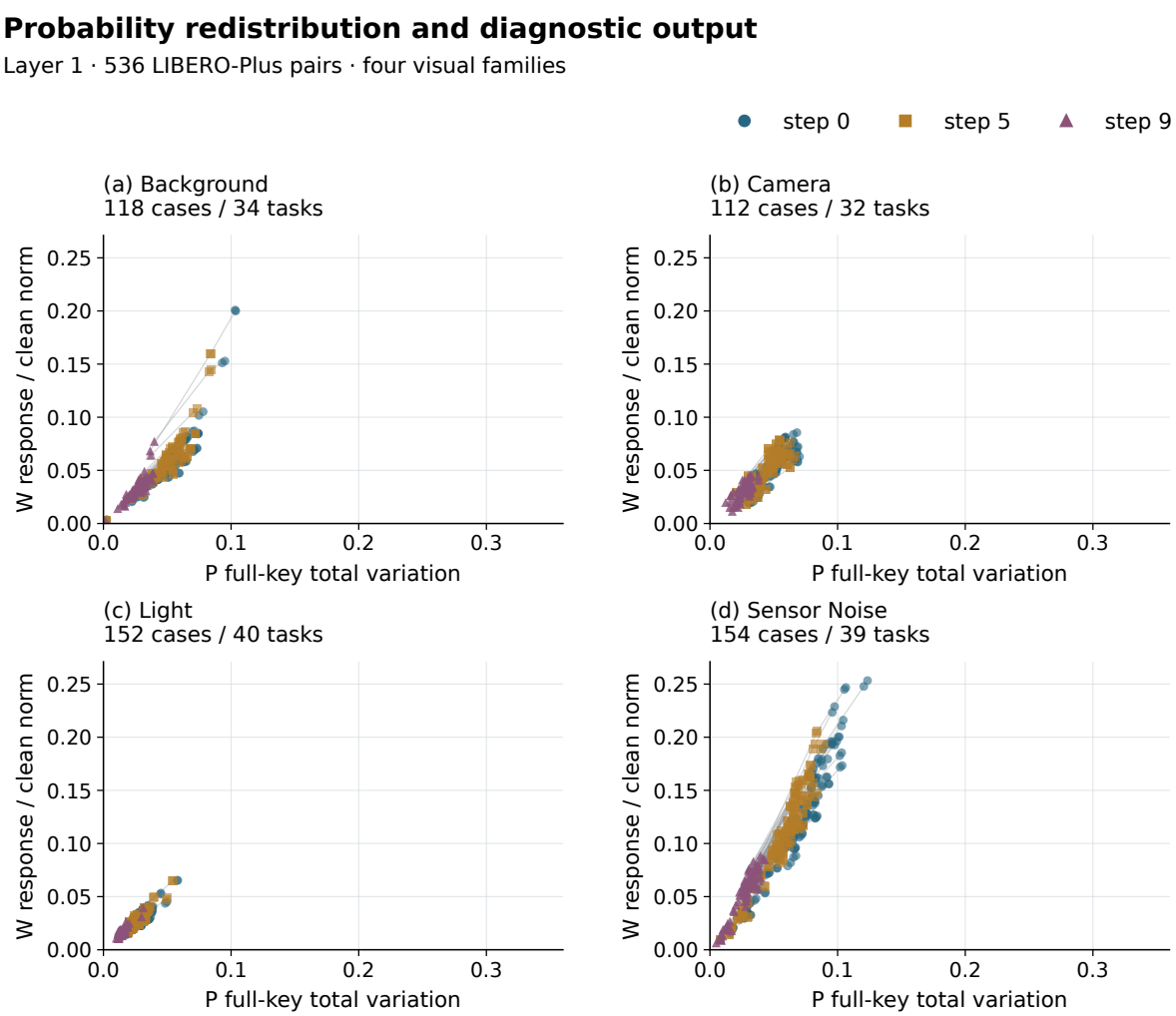


Figure S10. **Probability shifts and relative attention-output responses over generation at layer 1.** Lines join generation steps 0/5/9 of the same case within a query. Horizontal: probability TV over all visible keys; vertical: attention-output response W divided by its clean norm. Both recomputed, all 536 LIBERO-Plus pairs; one point per case and step, no interval.

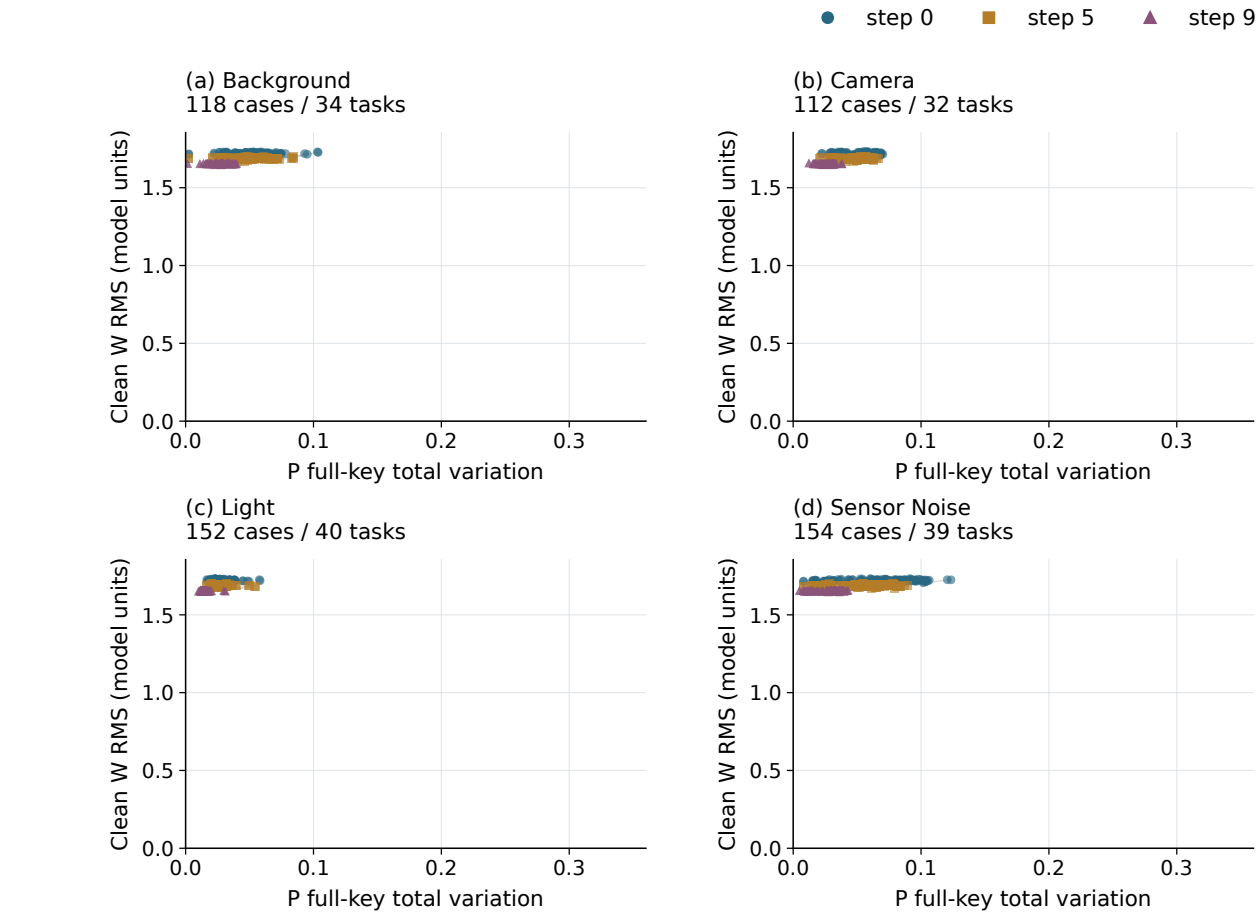


Figure S11. **A nearly constant clean output scale leaves the response difference to account for the relative spread.** The cases and horizontal axis match Fig. S10; the vertical axis now shows clean W RMS (model units), the denominator of the relative response.

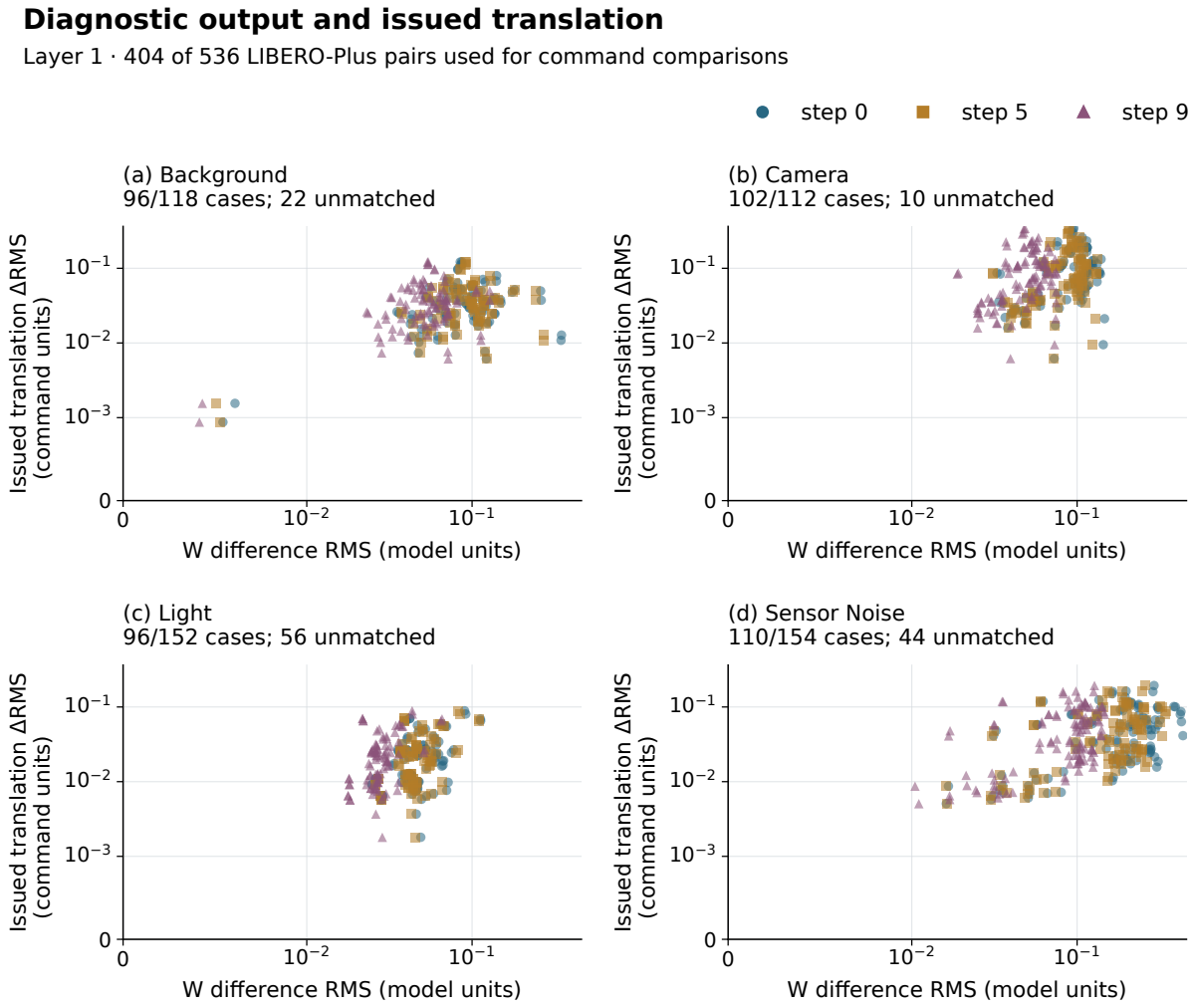
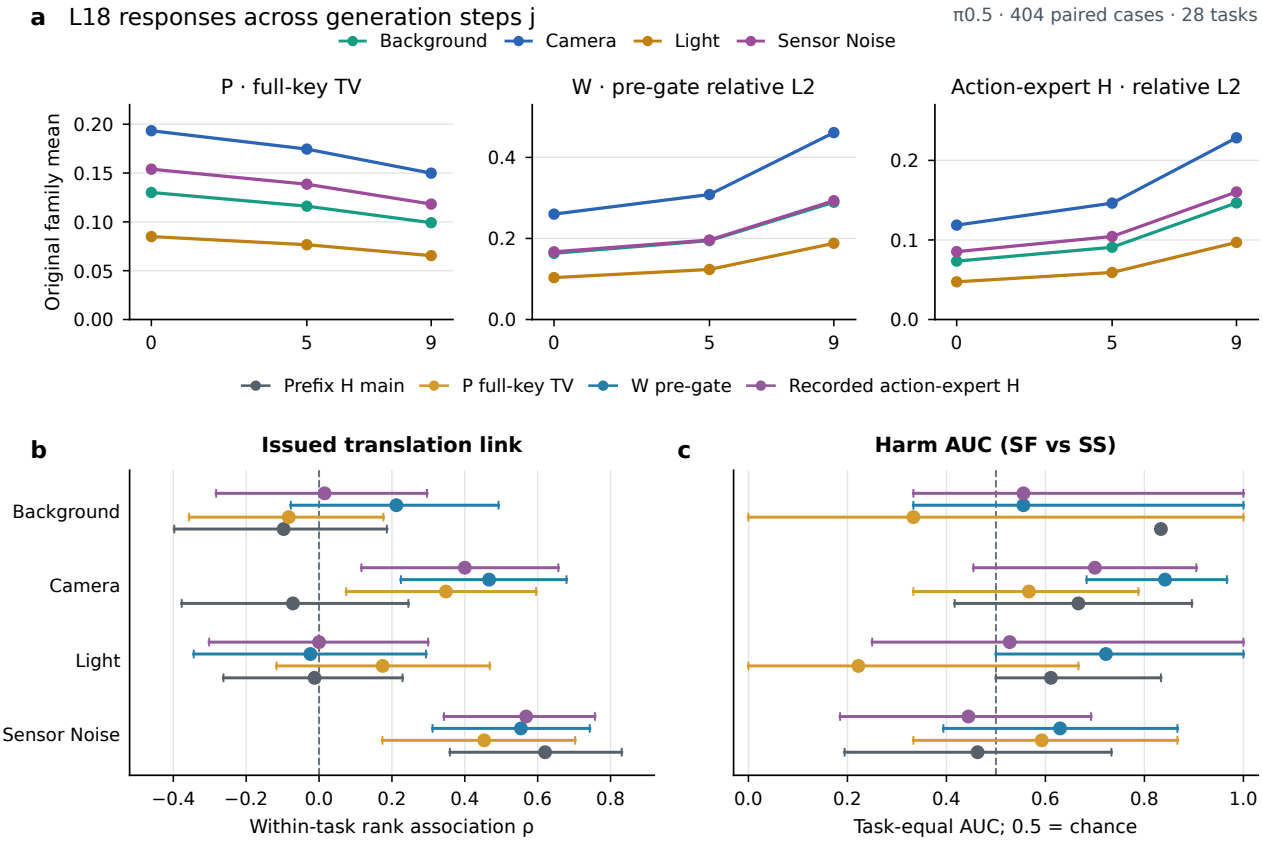
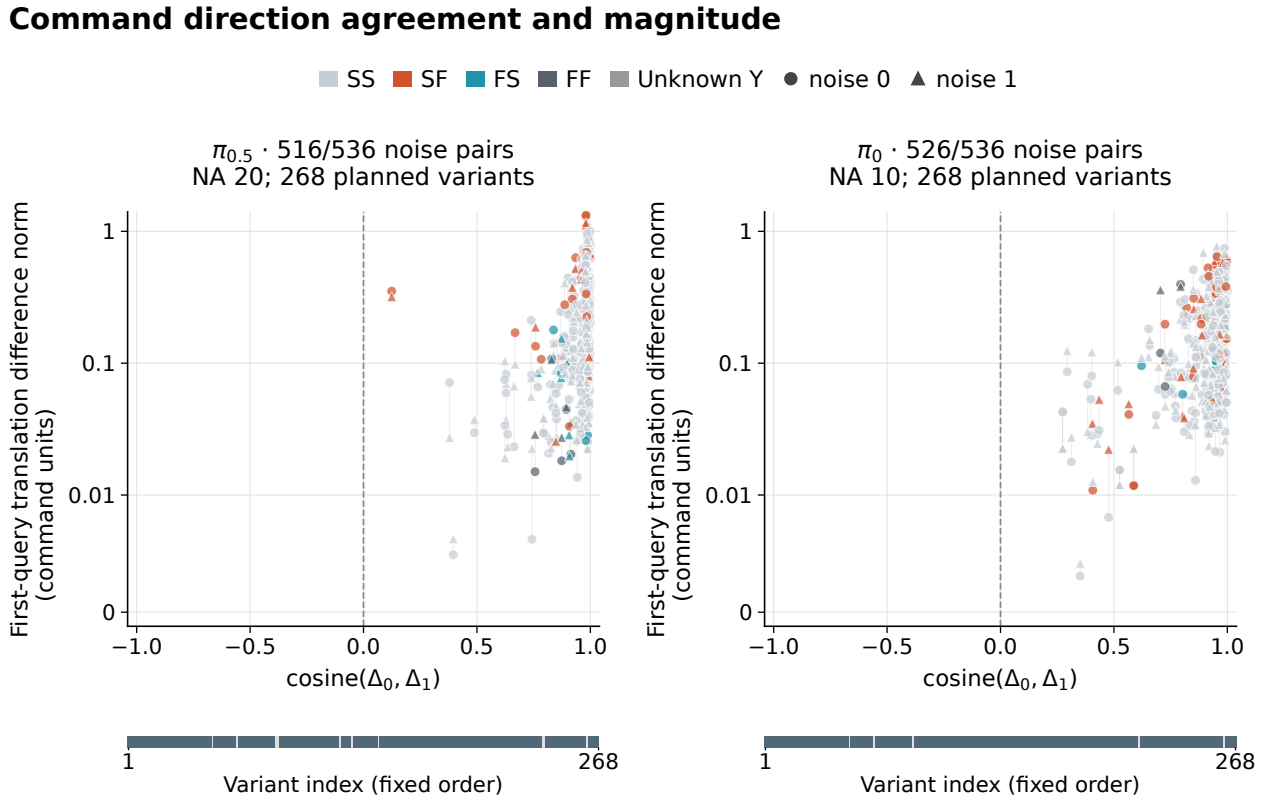


Figure S12. **Layer-1 attention-output responses against the change in the first five commands.** Horizontal: raw W difference RMS (model units) at generation steps 0/5/9; vertical: RMS change in the first five commands (translation, command units), one value per case. The matched subset contains 404 of 536 pairs with recorded commands; 132 unmatched pairs remain in the full recording count. Symlog axes; no interval.



Harm AUC: Camera W 0.842 [0.683, 0.967], 10 tasks; B/L: 3 each. W−P issued-association CIs all include 0.

Figure S13. **Association with the command change and separation of lost successes ask different questions of the same internal measurements.** On 404 matched LIBERO-Plus pairs, (a) L18 family means over generation steps 0/5/9; (b) within-task rank associations of first-step L18 measurements with the change in the first five commands (translation); (c) task-equal AUROC for lost success (SF versus SS), where 0.5 is chance. The AUROC uses the eligible tasks with both outcomes; pointwise descriptive 95% whole-task intervals and definitions are given in Table S7.



Lines pair two noises of one variant; they are not physical trajectories.

Magnitude axis: symlog linear below 0.01 command units; cosine is linear.

Coverage: dark = 2 available noises; tan = 1; grey = 0. Missing is not zero.

Figure S14. **Direction of the translation command change across the two policy-noise draws, by size of the change.** At the first query, horizontal: cosine between the two noise-matched changes; vertical: size of the change in the first five commands (command units, symlog). Lines join a setting's two noises; color gives the paired outcome. Of 268 planned settings per policy, 516/536 and 526/536 noise pairs are plotted for $\pi_{0.5}$ and π_0, respectively; the remainder appears in the coverage strip. No interval.

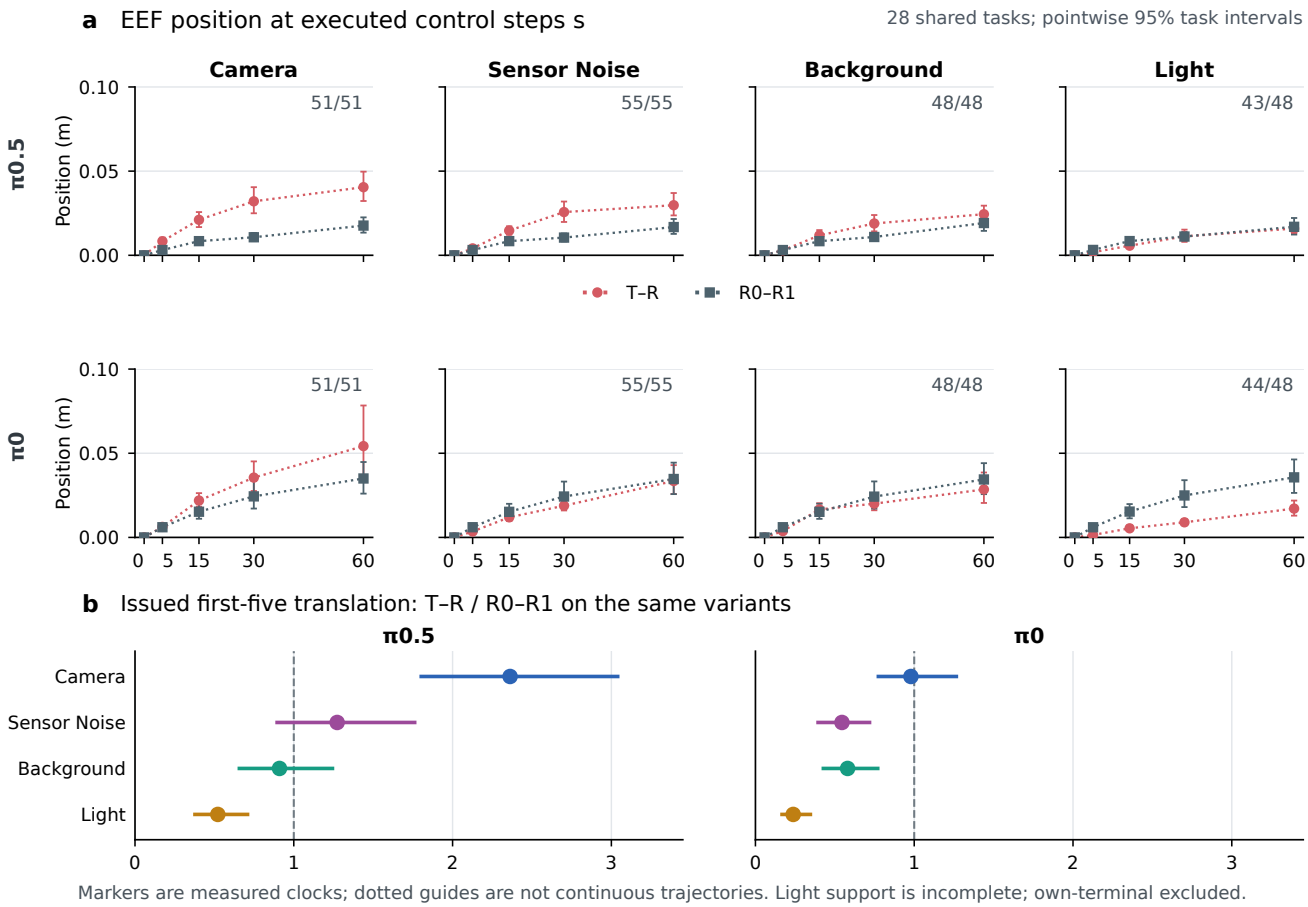


Figure S15. **The clean rerun puts command and motion changes on a common reference scale.** Primary panel of the paired executions, supports printed. (a) Task-equal mean end-effector position distance (metres), T–R against the clean rerun R0–R1, after executed control steps 0, 5, 15, 30, 60; 95% whole-task intervals. (b) T–R/R0–R1 for the first five commands (translation), same intervals. Light has incomplete fixed-step support; terminal observations are analyzed separately. Intervals describe mean distances in (a) and command-distance ratios in (b), on their respective matched supports.

S9. Timing of Camera Exposure and Terminal Position

This additional study, on which the main paper does not rely, asks whether it matters when a camera perturbation is seen. In $\pi_{0.5}$, moving an equal-length exposure window from an earlier stage to just before clean completion changes neither final position nor success detectably: the primary difference in terminal distance is -2.925 mm $[-9.833, +1.776]$ and in success 0.000 points (Table S22). Exposure for the whole episode does lower success: the difference from the clean run is -7.971 points $[-18.841, -0.725]$, in a secondary comparison in which the clean run always ran first.

Design: four runs per group, compared within groups. The study uses 23 tasks (11 single-support, 12 single-containment), two fixed start states (24 and 25) and three shared policy-noise samples per task and state: 138 matched groups of four runs, 552 episodes. Each group starts with a clean run C, which sets the exposure windows. The other three runs apply a $+30°$ camera rotation for the whole episode (F_{30}), near clean completion (R_{30}), or during the earlier window (N_{30}); the attached headlight follows the camera. The subscript is the rotation in degrees. The runs follow a preallocated order. The primary comparison is the task-equal mean of the three paired $R_{30} - N_{30}$ differences at start state 24 on all 69 assigned groups; start state 25 repeats the rule on another 69 groups, and the pooled 138-group analysis is secondary. Pointwise 95% intervals come from 10,000 shared whole-task bootstrap draws that keep the paired groups within each task.

Timing does not detectably matter; persistent exposure costs success. Observed successes for C/F_{30}/R_{30}/N_{30} are 69/61/68/68 of 69 at state 24 and 68/65/65/66 of 69 at state 25. The terminal-distance intervals span zero in the primary, replication and pooled analyses (Table S22). $F_{30} - C$ is a different contrast: C always runs first and sets the timing, so this ordered comparison of persistent exposure differs from the randomized comparison of R_{30} and N_{30}.

The clean run sets both windows. Both windows span four complete policy queries, 20 executed controls. Let T count the clean run's executed controls and $e = 5\lfloor T/5 \rfloor$. The completion window covers $[e - 20, e)$. For the earlier window, let g be the zero-based index of the first transition of the gripper command from nonpositive to positive. Among windows of 20 executed controls aligned to five-step boundaries between $g + 5$ and $e - 25$, we choose the one whose midpoint is closest to the midpoint of that range, breaking ties toward the earlier window. If none exists and $g \geq 45$, the fallback is $[5, 25)$, provided it does not overlap the completion window. Both windows require a successful clean run and four complete reference queries; otherwise both assigned runs keep the unchanged camera. Query indices are fixed before the perturbed runs and are not retimed as those runs evolve.

Endpoints: terminal position and task success. The position endpoint is measured after each run's last executed control, at its own first success, done signal or fixed horizon: the world-XY distance from the named target's root-body origin to the *current* origin of the named receiver body, or to the declared center of the receiver site, evaluated whether or not the task succeeded. A positive $R_{30} - N_{30}$ means R ends farther from that center; for task success, a positive $\mathrm{SR}(R_{30}) - \mathrm{SR}(N_{30})$ means *more* R successes. A run that stops early stays in its assigned group even if this shortens its exposure.

All groups stay in the analysis, including no-op and shortened exposures. Failures and time limits are observed outcomes; there are no technical failures. The 552 unique episodes cover every planned group, with terminal position and success usable in all 552 episodes, so both intention-to-treat (ITT) contrasts are identified. Each start state has 58 groups in which both windows are defined and 11 in which neither is, none of unknown status. In those 11 groups both R and N apply the declared zero-angle no-op and stay in the denominator; their observed R–N contrasts are zero, and the analysis keeps their actual outcomes rather than imposing equality. Of the 58 eligible groups, N receives all four scheduled queries and 20 executed controls in 58/58 at each state, R in 52/58 at state 24 and 51/58 at state 25; the remaining six and seven R runs have shorter exposure and stay in the ITT analysis. The estimates and exposure records cover all 138 assigned groups; the exploratory subset of 38 groups and the paired executions of the main paper are separate populations; neither replaces this analysis.

Implementation note. The bootstrap uses a fixed seed (20261008).

Table S22. **Moving the exposure window does not detectably change terminal distance or success; persistent exposure lowers success.** All planned groups (69 per state, 138 pooled), no missing run. Task-equal means of paired within-group differences, left minus right; distances in millimetres; success in percentage points; 95% intervals from 10,000 whole-task draws. State 24 $R_{30} - N_{30}$ is primary.

Population and contrast	Terminal XY distance difference, mm [95% CI]	Task success difference, pp [95% CI]
State 24 $R_{30} - N_{30}$, primary; 23 tasks, 69/69 pairs	-2.925 $[-9.833, +1.776]$	0.000 $[-4.348, +4.348]$
State 25 $R_{30} - N_{30}$, fixed-state replication; 69/69 pairs	-1.601 $[-6.788, +2.317]$	-1.449 $[-8.696, +4.348]$
Both states $R_{30} - N_{30}$, secondary; 138/138 pairs	-2.263 $[-8.068, +1.640]$	-0.725 $[-6.522, +4.348]$
Both states $F_{30} - C$, secondary; 138/138 pairs	$+1.488$ $[-2.914, +5.533]$	-7.971 $[-18.841, -0.725]$

S10. Further Studies on Original LIBERO

This section reports seven studies of $\pi_{0.5}$ on original LIBERO, each with its own tasks, start states and unit of analysis. Two back Sec. 5.2 of the main paper (the size of a change is not its cost): the ten-condition study (Sec. S10.2) and the headlight-versus-matched-dimming comparison (Sec. S10.6). The others are additional studies; the main paper does not rely on them.

S10.1. Predicting a larger-angle camera response from a smaller-angle one

Can the response to a smaller camera rotation predict the response to a larger one? Partly: the observed response beats copying it and an unconditional template, and keeping all ten plan positions beats compressing them to their mean (Table S23). This is an additional study; the main paper does not rely on it.

Data. The first cohort has 234 complete curves among 240 planned, from 80 start states (indices 12/13) on 40 tasks; the six unavailable curves remain missing; inclusion depends on source availability, not success. A second cohort has 240 cases from start states 22/23 on the same tasks, recorded separately, and reuses the fitted predictors. Each curve has a clean and four camera conditions ($\pm 15°, \pm 30°$); the camera's attached headlight moves with it. Five task-held-out folds use 32 training and eight test tasks; training includes the target angle. Responses are taken at the first query and generation step, over all ten plan positions, after L1, L9 and L18.

Loss and averaging. Let $\mathcal{H}$ average noise within state, states within task, tasks within suite, and suites equally. For case prediction error e_i and copying error c_i, $R = \mathcal{H}(e_i/c_i)$ differs from $Q = \mathcal{H}(e_i)/\mathcal{H}(c_i)$. The loss L averages mean squared error normalized by the target response. The full representation alone has $R = 0.794/0.814$ and beats direct copying in 36/40 tasks in each cohort. Keeping full positions lowers aggregate L by 41.15%/40.44% relative to the compact mean (mean-only correction); taskwise percentages would define a different quantity.

Representations compared. Write $X = \mathcal{Q}_B(D_{15} - \mu_{15})$, with means fitted on training tasks. The mean-only representation repeats the position mean of X across ten rows; the mean–contrast representation adds the fixed contrast between the first five and last five positions; the full representation keeps all rows. In real-valued notation the full base is $B = D_{15} + \mu_{30} - \mu_{15}$; compact bases replace case-specific position detail with their reconstruction. Every method observes the full smaller-angle response, so the comparison measures retained information, not a saving in observation.

Inputs. A context block holds 350 coordinates: the 15 smaller-angle translation commands, their 15-coordinate difference from clean, and 320 policy-noise coordinates. A separate block holds three norms of the smaller-angle response. The held-out D_{30} and larger-angle commands are used only for scoring. For the second cohort, inputs come from the five-condition LIBERO study and targets from the larger-angle run, with its own clean reference.

One weighted ridge fit serves all representations. Tasks are weighted equally, then states and noises within their parents. Context and norm coordinates are standardized on training data, each block scaled by the square root of its number of varying coordinates. A representation block is centered and scaled by one training RMS Frobenius norm, without channel whitening. Weighted linear ridge has a free intercept; at the primary penalty $\lambda = 10$, the dual system is

$$(\sqrt{W}K\sqrt{W} + 10I)\alpha = \sqrt{W}(Y - \bar{Y}), \qquad \text{(S14)}$$

where K sums the normalized block kernels and W contains weights scaled to the training case count. All preprocessing stays within the training fold.

Implementation notes. L1/L9/L18 are post-block residuals of zero-based blocks 0/8/17. The constructed mean-plus-contrast base and directly computed half means use different operation orders and can differ at the bit level.

S10.2. Command change ranks conditions by success change, but not task–condition cells

Does a larger command change mean a larger loss of success? Across conditions, yes: a condition's mean command change ranks its mean success change with rank correlation -0.903 $[-0.952, -0.309]$ over the ten condition means. Across the 400 task–condition cells, the correlation is -0.075 $[-0.195, +0.043]$, an interval that includes zero (Table S24). This supports Sec. 5.2 of the main paper: the command change orders conditions but hardly tells which tasks lose success.

Table S23. **Compressing the observed response to fewer positions loses predictive information.** Response-normalized loss L (lower is better) at $\lambda = 10$, on the same 40 tasks, first 234-curve and second 240-case cohort, task-held-out folds. The mean-only representation keeps the position mean; the joint mean–contrast representation uses the mean-plus-half block; compact bases use their reconstruction. Point losses without intervals; the six unavailable curves are outside the measured data.

Base and correction	First 234	Second 240
Full, repeated context	0.255363	0.266813
Full, mean only	0.241199	0.250142
Compact mean, mean only	0.409871	0.420015
Full, joint mean–contrast	0.238654	0.248661
Compact mean–contrast, joint	0.326003	0.340889
Full, separate mean/contrast	0.23725	0.24779
Full, repeated mean	0.24184	0.25031
Full, all positions	0.237290	0.248179

Data: 2,344 of 2,400 planned pairs. The study runs ten persistent visual conditions on 40 tasks, two start states and three policy-noise draws. Paired internal measurements and outcomes are available for 2,344/2,400 planned perturbed/clean pairs: 2,110 joint successes, 104 lost successes, 83 gained successes and 47 joint failures. Of the 400 task–condition cells, 364 have six pairs, 20 have four and 16 have five; missing pairs remain missing. The 236 observed clean episodes are reused across conditions with their original outcome labels.

What is measured, and when. Unlike the main paper's first-query measure, features and commands are taken at two scheduled mid-episode controls: 20/40 for Spatial, 25/55 for Object, 30/60 for Goal and 50/100 for LIBERO-10, at the first generation step and relative to a clean evaluation of the same state. The command change uses the first five planned commands (translation part, 15 numbers). The L18 response magnitude is the relative RMS over five positions and 1,024 channels. Main-camera conditional total variation (TV) of attention requires all eight heads. The two steps are averaged within each episode, and each outcome is counted once. Condition means weight episodes equally within condition; cell means are ranked equally. The lost-success AUROCs compare lost successes (SF) with joint successes (SS) within conditions and weight the 21,405 SF–SS comparisons, not tasks.

Within conditions, response size barely separates lost from kept successes. All three lost-success AUROCs are near chance (0.5); the command-change AUROC is below it, so within a condition, pairs that lose a success have, if anything, smaller command changes than pairs that keep it.

Table S24. **The command change ranks condition means by success change, but not task–condition cells.** Ten-condition study, 2,344 perturbed/clean pairs, 40 tasks, ten conditions; each scalar is averaged over two observations. First two rows: rank correlation between the mean command change (translation part) and the mean success change, across the ten condition means and across the 400 task–condition cells. Last three: lost-success AUROC, comparing lost with kept successes within conditions and weighting SF–SS pairs, unlike the task-equal AUROC of Sec. S4. Estimates use case-level scalars; the correlation intervals in the text come from the same 1,000 whole-task draws (40 tasks resampled with replacement, each carrying its start states, noises, conditions and shared clean episodes; ranks recomputed; percentile intervals with linear interpolation).

Comparison	Estimate
Command change–success change, condition means	-0.903
Command change–success change, task–condition cells	-0.075
L18-response lost-success AUROC	0.535
Main-camera conditional-TV lost-success AUROC	0.581
Command-change lost-success AUROC	0.421

After removing task and condition effects, the internal measures co-vary but hardly predict success. We rank each variable globally with average ties and remove additive task and condition effects. Pearson correlation of the residual ranks is 0.722 between L18 response and command change, 0.401 between L18 response and main-camera TV, and 0.286 between main-camera TV and command change. Their correlations with the success change ΔY are -0.0009 $[-0.0422, 0.0395]$, -0.0280 $[-0.0694, 0.0194]$ and $+0.0542$ $[0.0093, 0.0998]$ for L18 response, main-camera TV and command change. Intervals use 1,000 draws of 40 task clusters that keep nested states, noises, conditions, missing pairs and reused clean episodes, without suite stratification or multiplicity adjustment. These coefficients describe association in the observed data after removing additive task and condition effects.

Attention mass and attention shape carry different associations. A separate first-query comparison on the same 2,344 pairs centers percentile ranks within condition. The signed change of main-camera attention mass correlates -0.00351 with the command change; a Jensen–Shannon (JS) summary of the attention distribution correlates $+0.493$. Both average the first five positions over ten generation steps, three layers and eight heads; unlike conditional TV, JS pools layers. Both are point estimates without intervals.

Implementation notes. Commands are in adapter coordinates; acknowledgements of execution are recorded sepa-

Table S25. **Translation and rotation changes agree more across three noise draws than continuous-gripper changes.** 764 complete three-draw groups of the ten-condition study (40 tasks, ten conditions); group ratio U/D for the first five planned commands of the first query, averaged within task and over suites equally per condition. Columns: range of the condition means; number of groups with negative U, which are kept. Point ranges; intervals are in the component tables.

Planned component	U/D range	$U < 0$ groups
Translation	0.909–0.967	1
Rotation	0.816–0.860	0
Continuous gripper	0.096–0.437	187

rately. Both bootstrap analyses use seed 20261008.

S10.3. Command changes are consistent across three policy-noise draws

Is a perturbation's command change systematic, or does it depend on the policy noise? For the arm, it is systematic: across three noise draws, translation and rotation changes share a large common component, while continuous-gripper changes agree less and more often oppose one another (Table S25). This is an additional study of the first query's planned commands; the two-draw agreement of executed commands cited in the main paper is in Sec. S7.

Data. The analysis uses 764 complete groups (one start state under one condition) over 40 tasks and the ten conditions of the ten-condition study. Each group has three noise-matched perturbed–clean differences of the first query's first five planned commands, with its outcome records.

Common component and noise-dependent variation. For one command component, write the flattened differences as d_1, d_2, d_3, with $\bar{d} = (d_1 + d_2 + d_3)/3$. Define

$$D = \tfrac{1}{3}\sum_{s=1}^{3} \|d_s\|^2, \qquad M = \|\bar{d}\|^2, \tag{S15}$$

$$V = \tfrac{1}{3}\sum_{s=1}^{3} \|d_s - \bar{d}\|^2, \qquad D = M + V, \tag{S16}$$

$$U = M - \tfrac{1}{2}V = \tfrac{1}{3}\sum_{s<t} d_s^\top d_t. \tag{S17}$$

D is the energy of the command change. Under independent, identically distributed noise, U estimates the squared mean change without the finite-sample bias in M; U/D does not inherit that unbiasedness. Negative group values are kept.

Gripper changes alter amplitude, not sign. Across all 11,460 planned gripper positions in these groups, no perturbed–clean sign change was observed. This concerns planned signs at the first query, not executed closing or contact. The component intervals use 10,000 suite-stratified whole-task draws that keep all nested groups.

Large camera command changes do not imply lost success. Camera command changes are several times the spread between clean runs, also when that spread uses $n - 1$ instead of $n = 3$: medians over seven coordinates are 2.996, 3.841, 5.238 and 5.291 for $-15°, +15°, -30°, +30°$, without a common physical unit. Groups above this reference that lose no success across their three repeats include joint successes, gained successes and joint failures. No lost success can also mean the clean run already failed, so outcome codes must be read together.

Implementation notes. The component intervals use seed 20261001.

S10.4. Wrist attention: how much it receives and how it is spread

When only the main camera turns, attention within the unchanged wrist image is redistributed more at L18 than at L9, but weighting by the wrist's share of all attention reverses the ranking (Table S26). Deep wrist features also change although the wrist pixels do not. This is an additional study for the attention analysis (Sec. S20); the main paper does not rely on it. Fig. S16 shows one example.

Four measures of one regional change. The measures separate allocation (how much attention the region receives) from redistribution (how it is spread within the region). At each head and plan position, let $m = \sum_{k\in\mathcal{C}} P_k$ be the probability mass on camera keys among all keys (the allocation), and $\widetilde{P}_k = P_k/m$ for $m > 0$. Camera-conditional total variation (TV), the redistribution, is $\frac{1}{2}\sum_{k\in\mathcal{C}} |\widetilde{P}^1_k - \widetilde{P}^0_k|$; its mass-weighted version is $w = \frac{1}{2}(m_0 + m_1)\,\mathrm{TV}$, computed before averaging. The camera-restricted difference is $n_{\mathcal{C}} = \sum_{k\in\mathcal{C}} |P^1_k - P^0_k|$, without the factor of one half; it differs from both the signed mass change and the sum of separate allocation and redistribution norms.

Mass weighting reverses the layer ordering in every task. After averaging over phases and noise, L18 has larger conditional TV and smaller weighted TV than L9 on every task, across four lighting and camera changes and three generation steps; the wrist-restricted L_1 change is also smaller in all 432 task/generation-step/change measurements. Finer strata contain 11 conditional-TV reversals among 1,728 comparisons; the subsets of the first five and last five positions keep five and six weighted-TV reversals. All are repeated measurements of 36 tasks.

Table S26. **Different attention measures rank the wrist layers differently.** Fixed-state camera study, 36 tasks, main camera $0^\circ \rightarrow 15^\circ$ at the baseline headlight, generation step 0; equal-weight means over tasks of per-task averages across all eight heads, ten positions, demonstrations, phases and the two recorded noise definitions. Conditional TV is a percentage of wrist mass; the other rows are percentage points of probability over all keys. Point values, no intervals.

Wrist quantity	L1	L9	L18
Conditional TV (%)	0.084	15.753	25.180
Mass-weighted TV (pp)	0.0139	5.453	2.333
Absolute mass change (pp)	0.407	4.767	2.300
Wrist-restricted L_1 change (pp)	0.4071	12.9226	5.2600

Deep wrist values respond to an identical wrist image. The L1 wrist-value response is zero under the camera change, while the mean per-patch relative norms at L9/L18 are 0.400/0.364 at the baseline headlight: a deep token at a fixed image position carries context from the main view.

Text-token states respond under a fixed instruction. A companion readout also records camera responses in the text-token states H and Z. On its common scoring positions, the mid-observation responses at L9/L18 are nonzero in all 72 demonstration pairs from 36 tasks, and the L1 responses are all zero. Instruction, tokenizer and pipeline settings are fixed for the 144-state set; token bytes were not compared directly. The shared positions cover 6–9 text slots, or 42.86–100% of the target slots.

Splitting the change into probability and value terms depends on the reference. In the L18 main-camera slice at generation step 9, the value term exceeds the probability term on 7/36 tasks with one reference and on 28/36 with the reverse. The split isolates terms of the saved attention read; an intervention would also involve the gate and the rest of the policy.

S10.5. Attention moves within an identical wrist image

Does attention to the wrist image change when only the main view changes? In one fixed example it does, at L9 and L18 while L1 stays near zero, although the wrist pixels are identical (Fig. S16). This is an additional illustration for the attention analysis; the main paper does not rely on it. The signed row shows where conditional attention rises or falls on average; the magnitude row keeps changes that cancel in the signed average.

One fixed example, all heads and positions. The example is the first, in lexicographic order, of an existing eight-state gallery from the fixed-state camera study: LIBERO-10 task 01, demonstration 7, early frame 1, with the instruction "put both the cream cheese box and the butter in the basket." All of L1/L9/L18 and every head–position pair are shown, without selection by response size or outcome. Raw wrist pixels and the wrist inputs the policy consumes are verified identical, and the main-camera inputs differ. The example uses the common-noise record at $j = 5$; Table S26 reports the separate $j = 0$ comparison over 36 tasks.

Reading direction, magnitude and mass together. The magnitude map averages $|\Delta q|$ at each token before display, and the mean TV below each layer sums absolute changes before averaging, whereas the signed map allows cancellation across heads and positions. The regional mass m places each conditional distribution within attention over all keys. L1 stays near zero on the shared scale, while L9 and L18 show spatial redistribution. All three layers use one scale, symmetric about zero for the signed row and starting at zero for the magnitude row; no values are clipped or rescaled by layer. A rounding-interval check at L1 is compatible for all 80 head–position entries; it is a compatibility check, not a measured bound on numerical error.

Spatial addresses. Each map is a 16×16 grid in the policy's image orientation: local index $k = 16y + x$ maps to wrist prefix key $256 + k$, with x increasing rightward and y downward. The original 256×256 RGB image and the consumed 224×224 image cover the same square field; each model patch is 14×14 consumed pixels. The maps keep this orientation without flip, rotation or crop. They show conditional probabilities from a replay of the saved attention.

Implementation notes. The color scale has asinh linear width 0.0009429195779375732 and absolute limit 0.060346852988004684, taken from the original gallery.

S10.6. Headlight removal and a dimming matched in image statistics

Removing the headlight changes the first five commands 5.12 times as much as a global dimming matched in initial image statistics, yet both succeed on 77/80 cases, each on two cases where the other fails. This supports Sec. 5.2 of the main paper: a fivefold larger command change comes with the same success count, and equal counts hide changed outcomes.

Fivefold command change, equal success counts, exchanged failures. The command change is the RMS of the 15 translation coordinates of the first five commands. Its mean is 0.0570 for headlight removal and 0.0111 for

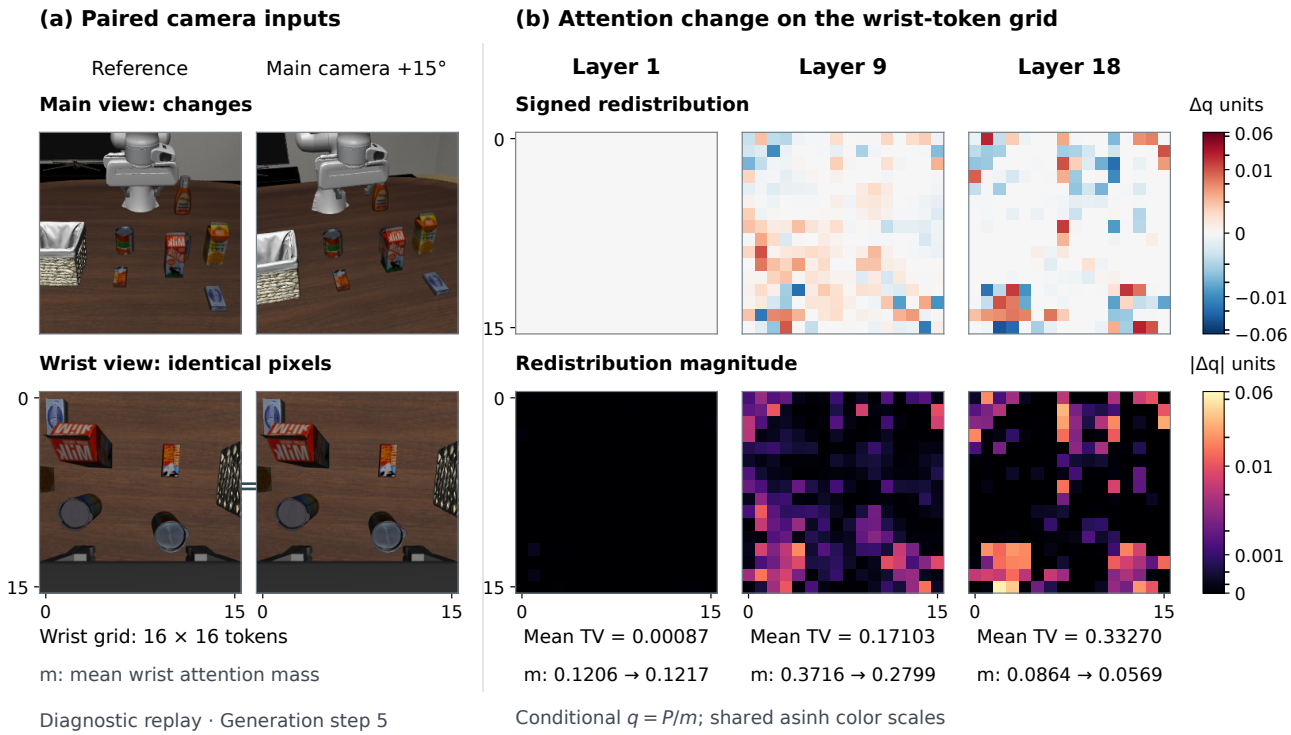


Figure S16. **Conditional attention redistributes over identical wrist pixels.** (a) The physical state and instruction are fixed. Main-camera yaw changes from 0° to 15° at the baseline headlight setting; the wrist images are pixel-identical. (b) Conditional attention at generation step $j = 5$, with matched state, time and noise. For each of eight heads and ten plan positions, $q_k = P_k/m$, where $m = \sum_{k \in \text{wrist}} P_k$, and $\Delta q = q_{\text{changed}} - q_{\text{reference}}$. Rows show $\langle \Delta q \rangle$ and $\langle |\Delta q| \rangle$ over all 80 head–position entries (eight heads $\times$ ten plan positions) in this one fixed example, taking absolute values before averaging in the second row. Each row shares one asinh color scale across layers, with ticks in probability units. Mean TV is $\langle \frac{1}{2} \sum_k |\Delta q_k| \rangle$; mass values read reference $\rightarrow$ changed. The top-left image origin and token coordinates are shared by all panels.

matched dimming (ratio of means 5.12); the paired difference is 0.0459 $[0.0363, 0.0562]$ (95% interval over task clusters), and the headlight change is larger on 76 cases and smaller on four. The paired success difference is 0 $[-5, 5]$ pp; this interval describes the observed comparison, and an equivalence test was not prespecified.

Data and runs. The study has 80 cases from 40 tasks and start states 10/11, each run clean, with the headlight off and with matched dimming. Each run succeeds on 77 of 80 cases, and each pair of runs shares one failure (75 joint successes, 2 and 2 single failures, 1 joint failure). The runs share the start state and ten settling steps, checked before the first query.

The dimming is matched on initial image statistics only. Initial RGB images have shape $256 \times 256 \times 3$. For each camera, matching uses the image mean and the mean absolute difference from clean. A gain $g \in \{0.25, 0.275, \ldots, 0.75\}$ is chosen to minimize the largest of the four statistic mismatches, ties going to the larger gain. Dimming scales the ambient, diffuse and specular RGB of the active lights, including the headlight, and uses no policy output or outcome. All 80 cases meet the criterion of five code values and stay in the analysis: the largest mismatch is 3.94, gains lie in [0.35,0.55], and each gain stays fixed for the episode. Spatial lighting patterns and later images can still differ.

A separate prediction of policy outputs from response features. A probe on these cases tests response features as predictors of the policy's outputs (results in Sec. S8). It uses fixed random projections with 32 coordinates per condition, concatenated into a 64-dimensional pair and divided by the saved norms. Flow and final-plan targets cover all ten positions and seven action coordinates; the plan target is taken before the robot adapter. Averaging the two start states within each task gives 40 samples. Four leave-one-suite-out folds use 30 training and ten test tasks and a fixed ridge penalty 10, with no hyperparameter search.

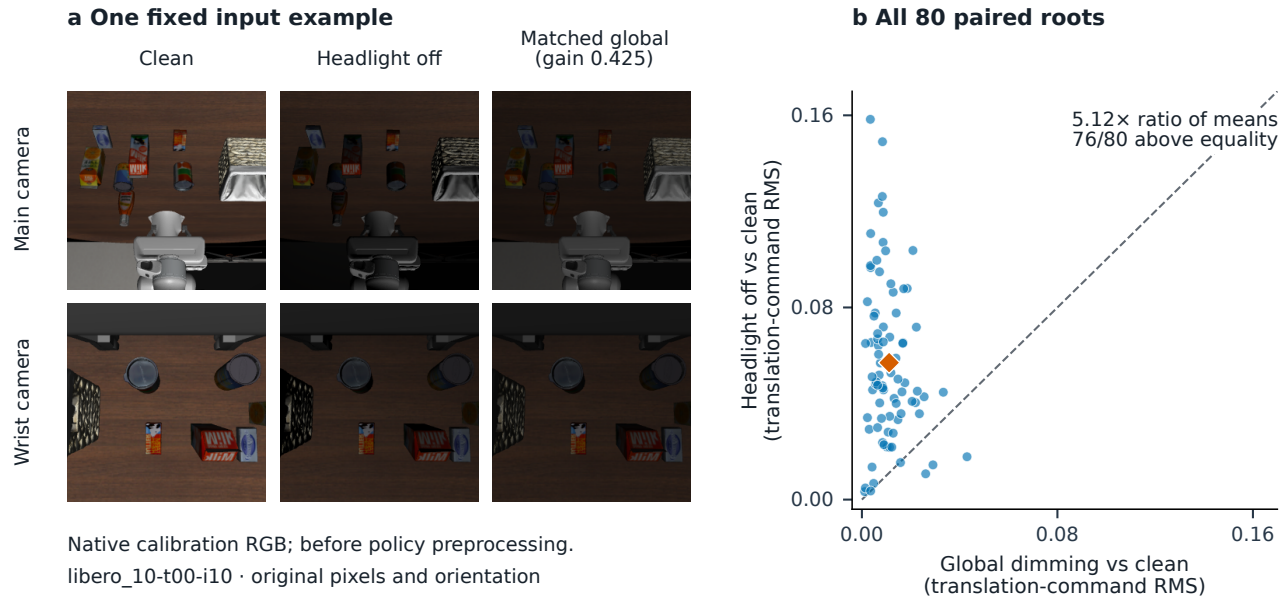


Figure S17. **Images matched in summary statistics can still change the commands differently.** Main and wrist images of the first complete case in fixed order (LIBERO-10 task 0, start state 10, global gain 0.425), chosen by order, not by response or outcome; pixels and orientation as recorded. The command comparison in the text uses all 80 cases and the RMS over 15 translation coordinates, in command units rather than displacement.

S10.7. From commands to later motion and task outcomes

How do first-query command differences carry into motion and outcome? In the 480-pair camera study, arm-command differences come with different end-effector poses in every pair, while gripper signs stay matched; a brief visual change can leave a changed physical history after it ends, with either outcome. This is an additional study; the main paper does not rely on it.

Shared clean runs; most pairs succeed in both conditions. The larger-angle run contains 720 unique episodes and 480 perturbed/clean pairs: 418 joint successes, 40 lost successes, 14 gained successes and 8 joint failures, a net change of $-26/480 = -5.4$ points. The two angles share their clean runs, so the 480 pairs are not 480 independent tasks. Each first-query plan is 10×32 before and 10×7 after the robot adapter. The first twenty controls come from four successive plans, not from the first plan; every pair reaches control twenty.

Matched gripper signs coexist with different arm poses. At each physics substep, the Panda gripper controller updates its persistent two-finger command by a bounded term proportional to $[-1, +1]\, 0.01\, \mathrm{sign}(g)$. Equal initial controller state, equal command signs and equal substep counts therefore give equal normalized gripper output, and the first twenty command signs agree in all 480 pairs. Controller state was not logged at every substep, so this combines observed sign agreement with a property of the controller code. Saved finger positions still differ after five controls in every pair. The first five commands keep all six arm coordinates within the controller's input bounds, so clipping preserves their differences, and the next saved observations show different end-effector positions and orientations in every pair. Sensor timestamps precede query timestamps by about 2 ms. These observations link the arm commands to early state differences.

The same first query can lead to different outcomes. A pulse analysis applies a visual change only for the first five controls and compares it with the persistent change and with clean. It keeps 697/720 planned three-run comparisons: 233 headlight, 232 camera $+15°$ and 232 warm-surface cases. The 697 pulse/persistent pairs match on 17 recorded checks of initial input and history, including the first commands and boundary physics; despite the shared first query, outcomes differ in 24 cases. Paired internal traces are unavailable at query 1. At the two later scheduled steps, the pulse run has zero visual response in raw and adapter coordinates, yet its physical state differs from clean: all 657 pulse/clean

joint-success cases have different physical snapshots, and three other cases succeed clean and fail with the pulse. A restored image can have zero response while the robot keeps a changed physical history.

The pulse-minus-clean success change is $+3.75$ pp with observed 95% interval [0.97,7.08], computed post hoc. It averages noise within start state and mechanism, mechanisms within start state, start states within task, and tasks and suites equally; it is not $(18-3)/697$. The interval uses 20,000 suite-stratified task draws. The separate range [-0.83,4.03] pp gives bounds over missing outcomes in the 720-case population, not sampling uncertainty; both are reported.

The last query can precede successful completion. A geometry reconstruction uses the five-condition LIBERO and larger-angle runs: 2,160 episodes across 40 tasks, with 4,320 query slots and 5,400 goal-clause slots, of which 4,104 clause slots have measurements. Site measures are unavailable for three LIBERO-10 tasks. Containment and contact follow the benchmark's definitions; object–support XY distance measures position.

On the 12 single-containment tasks, the last recorded query fails containment in 69/70 successful clean episodes of the five-condition study and 69/70 of the larger-angle run, 138/140 in total; the episode ends one to five controls later. Across all conditions, including clean, the counts are 413/417 and 209/210 successful episodes. The gap reflects timing between the last stored query and the successful end; task success is evaluated at the later terminal state.

On the same 12 tasks, each visual condition has 72 paired episodes. At the last query, target–receiver XY distance changes by $+2.04/-1.37$ mm for camera $-15°/+15°$ and by $+5.09/-0.182$ mm for $-30°/+30°$. Query times can differ between runs. These distances describe query-time geometry, with asymmetric changes and opposite outcome shifts across conditions.

Implementation notes. The pulse interval uses seed 2026092817.

S11. Relating LIBERO-Plus Responses to Commands and Task Outcomes

This section describes the paired clean and perturbed $\pi_{0.5}$ recordings on LIBERO-Plus to which the shared directions are fitted, and what these pairs show about commands, motion and outcomes. On the common tasks, Camera and Sensor Noise cost success; perturbed runs end farther from their clean runs than a clean rerun does, even when both succeed; and a prespecified test of whether response size identifies lost successes reaches neither prespecified conclusion. This describes the data of main Sec. 4.1 and is consistent with main Sec. 5.2; the main paper's outcome figures come from the separate paired executions (Sec. S6). Related analyses: Secs. S3.4 and S4.

S11.1. Task outcomes of the recorded pairs

The recordings: 536 pairs on 40 tasks. The recordings contain 268 settings, each run under two policy-noise draws: 536 clean–perturbed pairs on 40 tasks, covering the LIBERO-Plus categories Background Textures, Camera Viewpoints, Light Conditions and Sensor Noise. Two inspected Light settings change texture assets as well as lighting, so the Light family denotes the benchmark's prescribed settings, not an isolated lighting factor. Each pair has its own executed clean run; the 80 clean-run recipes fix initialization and noise settings and are not 80 trajectories reused across pairs. Outcomes (clean→perturbed) are 447 joint successes (SS), 56 lost successes (SF), 20 gained successes (FS) and 13 joint failures (FF). Pairs are nested within tasks.

The common-task comparison: 404 pairs on 28 tasks. The common-task comparison uses the 28 tasks shared by all four families and excludes 18 Light pairs without a LIBERO-Plus difficulty level, leaving 202 settings and 404 pairs (Table S27). All feature, command, motion and outcome quantities below use these same pairs. The estimates describe the selected settings; full-panel summaries and full-benchmark rankings refer to different populations.

Table S27. **Pairs per family in the full panel and on the common tasks.** $\pi_{0.5}$ LIBERO-Plus recordings: pairs/tasks in the full panel, pairs/settings on the common tasks, and common tasks per family. The full Light panel includes 18 pairs without a difficulty level. The common tasks are shared, but families still differ in settings, difficulty levels and number of settings per task.

Family	Full: pairs/tasks	Common: pairs/settings	Tasks
Background	118/34	96/48	28
Camera	112/32	102/51	28
Light	152/40	96/48	28
Sensor Noise	154/39	110/55	28

Camera and Sensor Noise cost success. The success change, perturbed minus clean, is $+2.679$, -17.857, $+2.679$ and -12.500 percentage points for Background, Camera, Light and Sensor Noise on the common tasks, with pointwise 95% intervals $[-5.357, 11.607]$, $[-30.379, -6.250]$, $[-7.143, 13.393]$ and $[-22.321, -4.464]$, and simultaneous 95% intervals over the four families (Bonferroni-adjusted percentiles) of $[-7.143, 13.393]$, $[-34.821, -3.571]$, $[-8.929, 16.071]$ and $[-25.893, -2.679]$. The Camera and Sensor Noise intervals exclude zero; Background and Light have positive point estimates with intervals that include both gains and losses. Estimates average the two noise draws within a setting, settings within a task–family, and then tasks equally; intervals use 10,000 bootstrap draws of whole tasks. Each change is measured against the family's own clean runs and mix of difficulty levels.

S11.2. Definitions of the response and attention measures

This subsection defines the measures: absolute size measures a difference, relative size compares it with the clean tensor, and attention total variation (TV) measures a change in how attention is allocated.

Recording sites. The first policy query is $q = 0$; j indexes generation steps and p plan positions. We record L1, L9 and L18 (zero-based block indices 0, 8, and 17). Prefix measurements replay the prefix computation with masks for valid tokens and defined queries. H is the block input and Z the pre-attention RMSNorm output supplied to attention, so a larger H response at depth need not mean a larger change in what attention receives. Action-expert (AE) hidden states follow the policy's own generation and keep the ten ordered plan positions at each generation step. Channel indices belong to their own representation and layer; equal indices in different spaces do not imply a shared direction.

Absolute and relative responses can move in opposite directions. For a selected set of N scalar coordinates, let X_R and X_T denote the clean and perturbed tensors and $\Delta X = X_T - X_R$. We report

$$d_{\mathrm{RMS}}(X) = \|\Delta X\|_F / \sqrt{N}, \tag{S18}$$

$$r(X) = \|\Delta X\|_F / \|X_R\|_F. \tag{S19}$$

In these recordings, the absolute AE L18 response falls from the first to the last generation step in every family–suite cell, while the relative response rises because the clean norm falls, also in every cell. A relative trend therefore mixes response size with the clean scale. Ratios are formed per pair before averaging and are undefined for a zero clean norm; a ratio of family-average norms is a different estimator.

Attention change is measured over all visible keys. The attention probabilities P come from a separate observational probe at the recorded generation states. For each valid action query and attention head, the total variation over all visible keys is

$$\mathrm{TV}(P_T, P_R) = \tfrac{1}{2} \sum_{k \in \mathcal{V}} |P_T(k) - P_R(k)|, \qquad \text{(S20)}$$

where $\mathcal{V}$ is the common visible-key set. We average over valid queries and heads and keep the probability mass on visual, text and other visible keys. The regional net mass change sums the signed probability change in a region; the regional conditional-distribution change first normalizes each run by its regional mass. Neither substitutes for full-key TV. The text region contains text tokens; the recorded configuration does not append the numerical joint state as discrete text.

Attention output combines allocation and transformed values. The content readout recomputes the attention output from prefix and action-side values, after projection and before the residual gate. Values and their weighted combination can change even when probability shifts decrease. This diagnostic replay measures each branch at its saved generation state, separately from the recording of the policy's own generation (Table S1).

Commands and motion keep their own units. The policy output record separates the policy's 32-coordinate actions from the seven-coordinate LIBERO commands after the adapter. We score translation, rotation and gripper components over the first five and over all ten plan positions. Command RMS uses saved command units; end-effector (EEF) distance uses simulator observations in the simulator's length units, without conversion to metres or millimetres. Orientation distance is the quaternion geodesic angle in radians. Gripper-state distance is the Euclidean difference of two recorded joint coordinates, not an opening width.

S11.3. An earlier readout: direction, size and the first generation update

This earlier, exploratory analysis asks whether response coordinates carry command information beyond response size, clean-feature coordinates and the policy's first generation update (the first flow). Shared-direction coordinates predict translation and rotation better than response size, but not clearly better than clean-feature coordinates, and add no measurable gain to the first flow. It uses uncentered directions on the full panel; the main paper does not rely on it (its readout is Sec. S3.4).

Held-out tasks and perturbation families. The analysis uses all 536 pairs and 72 fixed, split-specific sets of 16 directions from the uncentered LIBERO-Plus fits. Its primary evaluation holds out a perturbation family and a whole-task component; five folds give one held-out prediction per task, for 60 family–layer–fold splits. Twelve known-task, held-out-family splits form the secondary analysis. The primary split keeps tasks that share a clean-run recipe together. These directions differ from the separate five-position LIBERO directions of Sec. S8.1; the centered refit is described in Sec. S3.

Compact representations and prediction controls. Inputs come from the first policy query and first generation step; each layer is fitted separately with the ten plan positions in order. Four compact representations use 16 coordinates per position, or 160 coefficients each: shared directions, clean-feature directions, fixed random directions and the largest response coordinates. Clean-feature coordinates project the response onto directions fitted to clean features. The other baselines are zero prediction, the training target mean, ten positionwise magnitudes and the full 10,240-coordinate response. The 320-coordinate paired difference in the first flow is used alone and together with the shared-direction coefficients. Only the four compact representations match in coefficient count; the full response and the first flow have different capacities.

Direction beats size for translation and rotation; the choice of directions is unresolved. At L18, shared-direction coefficients give lower error than magnitude in all eight family-by-scope translation comparisons and all eight rotation comparisons; magnitude gives lower error in all eight gripper comparisons. Comparing shared and clean-feature coordinates of equal size tests the choice of directions: all eight translation intervals include zero, one rotation interval favors the shared directions and seven include zero. For the gripper, shared-direction coefficients improve on the full response in all 24 layer–family–scope comparisons, while magnitude gives lower error in 21 and the paired-difference intervals cross zero in three. The best representation depends on the command component.

Response coordinates add nothing measurable to the first flow. The first flow already contains a downstream prediction of the generation update. Adding the compact response to it gives 29 higher-error intervals and 43 that include zero among the 72 component–scope–layer–family comparisons. The secondary known-task comparison gives 26 higher-error intervals and 46 that include zero. None of the gain intervals is positive. These 72 overlapping comparisons describe the additional prediction of this linear estimator on shared data; their counts are descriptive, with no multiplicity-adjusted global test.

Estimator. We fit weighted affine ridge regression with an unpenalized intercept, fixed penalty 0.01, training-set centering and one training RMS scale per feature block. Nested task-component, family, setting and noise weights sum to one. Flow-only and augmented models share the flow-block scale. Targets are paired differences in the final generated plans, scored separately in model coordinates and in command units after the adapter. Pointwise intervals come from 2,000 bootstrap draws of task components.

S11.4. Motion compared with the clean rerun

After 60 controls, perturbed runs are farther from their clean runs than a clean rerun is in every family, and under Camera and Sensor Noise also when both runs succeed. Two references give the separation a scale: the clean rerun, which changes only the policy noise, and a repeat of the clean run with the same inputs and noise.

Command differences are followed by motion. All 404 common-task pairs start at zero EEF-position, orientation and gripper-state distance, and all issue their first five commands. After 5 controls, the task-equal EEF distances are 0.002911, 0.008362, 0.002027 and 0.004108 for Background, Camera, Light and Sensor Noise, with pointwise intervals above zero. After 60 controls the means are 0.024468, 0.041819, 0.018961 and 0.030180. Each family keeps all 28 tasks over this window. Later differences also reflect the different states the two runs reach. The repeat matches the recorded initial inputs, configuration, random-number state and policy noise; for each case it averages all eligible, independently executed repeats, whose remaining variation need not be pure floating-point error. The clean rerun compares the two policy-noise draws within each setting. All 202 settings support the clean-rerun comparison; the repeat supports 396 of the 404 cases. Both references keep the case's clean–perturbed pair.

After 60 controls, separation exceeds the clean rerun. Let $\overline{d}_{TR,f}(60)$ be the task-equal clean–perturbed EEF distance for family f and $\overline{d}_{RR,f}(60)$ the corresponding distance between the two clean runs. After 60 controls their ratios are 1.51, 2.39, 1.17 and 1.68 for Background, Camera, Light and Sensor Noise. Each ratio divides two family means; it is not an average of per-task or per-setting ratios, and no interval for it has been estimated. The paired distance-difference intervals (2,000 bootstrap draws of the fixed 28 tasks, pointwise 95%) are in the released analysis package.

Extra motion and net success measure different effects. Background has a positive interval for motion beyond the clean rerun while its success interval includes zero. With the repeat as reference, the Background/Camera/Light/Sensor Noise reference distances are 0.002518/0.001873/0.002512/0.002565 on 28/26/26/28 tasks; the two references keep their own populations and means.

Runs that both succeed still move apart. Outcome groups refer to the case's own clean–perturbed pair: ALL includes SS, SF, FS and FF, and SS and SF use that pair's final outcomes, not those of every repeat. After 60 controls, with the clean rerun as reference, Camera has $d_{TR} = 0.04360$ in SS (75 cases, 25 tasks) and 0.04291 in SF (23 cases, 12 tasks); Sensor Noise has 0.02872 in SS (91 cases, 27 tasks) and 0.03456 in SF (16 cases, 10 tasks). The SS intervals for motion beyond the clean rerun are positive for Camera, 0.02619 $[0.01678, 0.03596]$, and Sensor Noise, 0.01047 $[0.00181, 0.01859]$, while those of Background, 0.00730 $[-0.00019, 0.01414]$, and Light, 0.00228 $[-0.00348, 0.00756]$, include zero. These outcome-defined groups have different support. Both runs of a pair can thus succeed along different paths. Beyond 60 controls, finished episodes shrink the paired set, so later values need their task and case counts.

S11.5. Does response size identify lost successes?

This prespecified test asks whether response size ranks lost successes (SF) above joint successes (SS), within task. Its adjusted intervals cross the thresholds of both prespecified criteria, so it supports neither conclusion; the secondary pooled AUROCs are modest. This is consistent with main Sec. 5.2 but does not by itself establish it.

Four directed scores fixed before the test. The comparison starts from all 518 labelled pairs, separately from the display on the 28 shared tasks, and uses four fixed scalar scores: the relative AE L18 response over the ten plan positions at the first generation step, the relative prefix main-camera L18 response, the L18 full-key attention TV at the first generation step, and the RMS change of the first five commands (translation). Larger values always rank as a more likely lost success; this direction was fixed in advance and kept after the results were seen.

Within-task discrimination with prespecified criteria. The primary AUROC compares SF with SS within each task, with half credit for ties, weighting eligible tasks equally; a task lacking either outcome has no primary AUROC. Camera contributes ten eligible tasks with 15 SF and 23 SS cases; Sensor Noise contributes 13 with 18 SF and 32 SS cases. The 98.75% intervals for this restricted support (20,000 whole-task bootstrap draws shared across quantities and families, Bonferroni-adjusted over the four scores within each family) are in the released analysis package. The positive criterion requires the same score's adjusted lower bound to exceed 0.60 under both Camera and Sensor Noise; the alternative

criterion of no practical discrimination requires all eight adjusted upper bounds to lie below 0.70. The adjusted intervals cross the thresholds of both criteria.

A secondary pooled comparison uses different support. The pooled secondary analysis keeps all SF and SS cases, takes the logarithm of each score with a fixed 10^{-12} floor, subtracts its task–family mean and computes a pooled AUROC. Camera has 25 SF and 83 SS cases, with AUROCs 0.623/0.636/0.587/0.648 for the four scores; Sensor Noise has 22 SF and 125 SS, with 0.548/0.527/0.581/0.608. The adjusted intervals of the command score are [0.501,0.817] and [0.408,0.802]. Comparing higher with lower LIBERO-Plus difficulty levels under Camera and Sensor Noise, several responses increase while the within-task success-change intervals cross zero.

S12. Removing the Shared L9 Directions without a Clean Observation

Restoration (Sec. S13) needs a clean observation of the same state. This study needs none: in $\pi_{0.5}$ on original LIBERO, it removes part of the L9 state's component along the shared directions, measured from a clean training mean, under a camera change and a headlight removal, in 6,048 planned rollouts. The edit lowers the response at L18 by 8.8% (Camera) and 7.6% (headlight); clean-feature directions with the same mean edit size raise it ($-2.5/-3.6\%$ reduction), and random directions lower it by 0.1–0.3%. It moves the first five commands by only a small fraction of the change the perturbation causes, and no success change against the unedited policy excludes zero. This supports the second test in main Sec. 4.2.

Design. The directions are the L9 shared directions of the cross-observation study (Sec. S8.1), not the LIBERO-Plus fits. The design has 16 held-out tasks $\times$ 6 start states $\times$ 3 policy-noise draws $=$ 288 cases; each case runs 3 visual conditions (clean, Camera $+15°$, headlight off) $\times$ 7 configurations (Unedited, Shared-small, Shared-large, Clean-feature, Random-1/2/3) $=$ 6,048 planned main episodes, plus 16 technical repeats. The edit is $H' = H - \lambda[(H-\mu_H)B]B^\top$ at the action-expert L9 post-block state for all ten plan positions, applied only at the first generation step of each policy query; later steps and all other layers run unchanged. The larger shared-direction edit is matched to the clean-feature and random edits by their training-set mean norm over the three conditions; the smaller one uses half that dose. Matching within each condition did not meet the 2% tolerance, so the actual edit L_2 per condition is reported with the comparison. Table S29 gives μ_H, each configuration's λ, the directions, the schedule and when the L18 response is measured.

The L18 response falls along the shared directions. The L18 response reduction is $1 - \|\Delta_{\text{edited}}\|^2/\|\Delta_{\text{unedited}}\|^2$ per case, averaged with equal task weights. Beyond the values above for Shared-large and the controls, the half-dose edit (Shared-small) lowers it by $4.2/3.6\%$.

The command change is small relative to the perturbation's. The edit footprint, the edited-minus-unedited RMS of the first five commands (translation part) on the same input, is 0.0006 (Camera) and 0.0006 (headlight), about 0.5% and 1.2% of the change the perturbation itself causes (unedited perturbed-minus-clean RMS 0.128 and 0.051). Rotation and the continuous gripper value are separate endpoints. In the 120-case interim analysis the formatted gripper output was identical in all 3,840 non-self pairs although the continuous gripper value differed in 3,837 (an interim finding, not recomputed on the full population). This edit acts only at the first generation step; restoration, which acts at every step, moves the commands far more (Sec. S13.5).

No success change against the unedited policy excludes zero. Success is reported as bounds over missing outcomes, because the prespecified rule withholds point estimates and ordinary intervals whenever a planned outcome is missing (Table S28). Within each condition, the known lost and gained successes from unedited to Shared-large are $2.4/1.4$, $2.4/3.1$ and $1.4/3.1\%$ of planned pairs (clean/Camera/headlight; unknown pairs 1.7–2.1%). Table S30 gives the nine prespecified contrasts: Shared-large minus Unedited (D), minus Clean-feature (Q), or minus the mean of the three random sets (R), under each visual condition.

Over the 111 missing planned outcomes, only Camera Q keeps its sign under every assignment: it is positive, favoring the shared directions over the clean-feature directions. The other eight bounds include zero. The Clean-feature edit's own bounds exclude zero in each condition (Table S28): it lowers success under clean input and Camera and raises it under headlight removal. For the unedited policy, Camera's success bounds are 91.0–93.1%, against 92.4–93.4% under clean input, whereas headlight removal costs 2.8–5.9 points.

Post hoc sampling intervals on the known outcomes. A post hoc bootstrap of the 5,937 known outcomes over the 16 held-out tasks (2,000 draws, task-equal means inside each draw) assesses sampling variation. Table S30 reports its point estimates, pointwise 95% intervals and Bonferroni-adjusted intervals at $1 - 0.05/9$ beside the bounds. Camera Q and R have positive pointwise and adjusted intervals; the other seven intervals include zero. For Camera Q, the adjusted interval $[+0.3, +13.2]$ and the bounds $[+2.8, +6.6]$ both favor the shared directions over the clean-feature directions. For Camera R, the adjusted interval is positive

Table S28. **Edits along the shared directions lower the L18 response; matched controls do not; all command footprints are small.** Prespecified 288-case plan, task-equal means. Edit L_2: actual edit norm per condition; L18 red.: per-case reduction of the L18 response energy (%); footprint: edited-minus-unedited RMS of the first five commands (translation) as a percentage of the unedited perturbed-minus-clean RMS; SR: closed-loop success over all planned episodes; ΔSR: edited minus unedited. SR and ΔSR are bounds over the 111 missing planned outcomes (lower–upper, %): they enumerate every possible assignment of the missing outcomes, so a range that excludes zero keeps its sign under every assignment; sampling uncertainty needs a separate analysis (Table S30). Random rows give the range over the three random direction sets.

Condition	Configuration	Edit L_2	L18 red. (%)	Footprint (%)	SR	ΔSR [bounds]
clean	Unedited	–	–	–	92.4–93.4	–
	Shared-large	148.4	–	–	90.6–92.4	$[-2.8, 0.0]$
	Shared-small	74.2	–	–	92.4–93.4	$[-1.0, +1.0]$
	Clean-feature	141.0	–	–	89.9–91.7	$[-3.5, -0.7]$
	Random-1/2/3 (range)	139.9–142.2	–	–	89.9–93.8	see text
Camera	Unedited	–	–	–	91.0–93.1	–
	Shared-large	132.8	8.8	0.5	91.7–93.8	$[-1.4, +2.8]$
	Shared-small	66.4	4.2	0.4	89.2–91.3	$[-3.8, +0.3]$
	Clean-feature	141.0	−2.5	0.6	87.2–88.9	$[-5.9, -2.1]$
	Random-1/2/3 (range)	139.7–141.9	0.2–0.3	0.5	89.2–93.1	see text
headlight	Unedited	–	–	–	87.5–89.6	–
	Shared-large	143.0	7.6	1.2	89.2–91.3	$[-0.3, +3.8]$
	Shared-small	71.5	3.6	1.1	89.2–91.3	$[-0.3, +3.8]$
	Clean-feature	141.0	−3.6	1.4	90.3–92.7	$[+0.7, +5.2]$
	Random-1/2/3 (range)	139.8–142.1	0.1–0.2	1.1	89.2–92.0	see text

Table S29. **Definition of the edit: training mean, directions, dose, schedule and response measurement.**

Quantity	Definition	Role
μ_H	Clean action-expert L9 post-block H at generation step 0, averaged over the 1,404 training-task records of the cross-observation study with that study's record weights (summing to one); the 16 held-out tasks are excluded. One 1024-vector per action position (10×1024), subtracted position by position. Its first five positions agree with the cross-observation clean fit center to 4.9×10^{-4} (declared tolerance 2.6×10^{-3}).	Training mean
B	16 directions for every edit: the cross-observation L9 shared directions (Shared-small/large), its L9 clean-feature directions (Clean-feature) and three random orthonormal sets from PCG64 seeds 20261014–20261016 (Random-1/2/3); all use the same μ_H.	Directions
λ	Shared-large 0.7521; Shared-small 0.3760 (half); Clean-feature 0.1721; Random-1/2/3 0.9427, 0.9826, 1.0000. Rule: on training first queries, each set's mean edit L_2 (equal weight per task, then per condition) equals a common budget of 141.1 that every set attains with $\lambda \leq 1$ (Random-3 at $\lambda = 1$); fixed before the evaluation outcomes were observed. Matching within each condition exceeded the 2% tolerance and remains unmatched.	Matched mean edit
Schedule	The edit is applied at the first generation step ($j = 0$) of every policy query, at all ten action positions; later steps and all other layers run unchanged.	Intervention time
L18 response	Action-expert L18 post-block H at the first generation step of the episode's first policy query (before the first environment step), all ten positions and 1024 channels. Per case and condition, ΔH is the condition minus clean under the same configuration, and the reduction is $1 - \sum(\Delta H_{\text{edited}})^2 / \sum(\Delta H_{\text{unedited}})^2$; means average noise draws within start state, start states within task and tasks equally.	Response endpoint

while the bounds include zero, so the comparison with random directions depends on the analysis of known outcomes.

Table S30. **Success contrasts: what the edit costs against the unedited policy, and how it compares with the control edits.** Every row is Shared-large minus the named comparator, in percentage points. Bounds enumerate the 111 missing outcomes of the 6,048-episode plan. Points and intervals use the 5,937 known outcomes (post hoc whole-task bootstrap, 16 tasks, 2,000 draws). The 99.44% intervals are Bonferroni-adjusted across the nine contrasts. Bounds and sampling intervals describe different sources of uncertainty. Random mean averages the three random direction sets.

Condition	Comparator	Bounds over missing outcomes	Known-outcome point	95% interval	99.44% interval
Clean	Unedited (D)	$[-2.8, 0.0]$	-1.0	$[-3.0, +1.0]$	$[-3.8, +1.7]$
Clean	Clean-feature (Q)	$[-1.0, +2.4]$	$+0.7$	$[-1.9, +3.3]$	$[-3.0, +4.8]$
Clean	Random mean (R)	$[-2.2, +1.2]$	-0.8	$[-2.5, +0.8]$	$[-3.5, +1.3]$
Camera	Unedited (D)	$[-1.4, +2.8]$	$+0.7$	$[-0.7, +2.1]$	$[-1.2, +2.8]$
Camera	Clean-feature (Q)	$[+2.8, +6.6]$	$+5.0$	$[+0.7, +10.2]$	$[+0.3, +13.2]$
Camera	Random mean (R)	$[-0.6, +3.2]$	$+1.6$	$[+0.6, +2.7]$	$[+0.3, +3.2]$
Headlight	Unedited (D)	$[-0.3, +3.8]$	$+1.4$	$[-1.6, +4.0]$	$[-2.8, +5.3]$
Headlight	Clean-feature (Q)	$[-3.5, +1.0]$	-1.4	$[-5.9, +1.4]$	$[-8.0, +2.4]$
Headlight	Random mean (R)	$[-2.5, +1.6]$	-0.7	$[-2.5, +0.9]$	$[-3.2, +1.4]$

Shared-large against Unedited includes zero in all three conditions.

Missing outcomes. Acquisition resumed in the prespecified order after a pause; interim results had been observed, but case selection did not depend on outcomes. 281 cases completed all 21 runs, 6 completed partially (90 missing runs), and one was interrupted after creation; none remained unexecuted. In total, 5,937 of 6,048 main outcomes are known. All planned outcomes remain in the estimand.

Development check of the schedule. With edits only at the first generation step, the L18 difference at the last step retains 3.7–5.5% of the first-step difference (12 development inputs); editing every generation step changes the commands about ten times more, with a reduction of about 2% in the visual difference (Spatial task 7, Camera).

Edits and readouts answer different questions. A readout asks what the coordinates predict about the commands; this study measures what changes when they are edited inside the policy.

S13. Restoration of the Shared L9 Directions

Restoration copies clean values back along the shared directions at L9: at every generation step of every query, it computes the L9 state under a clean observation of the same physical state and writes the clean values along the shared directions into the perturbed run. It needs that clean observation, so it is a controlled test, not a deployable method. Fixed in advance were the site, task success as the primary endpoint, the error of the first five commands as the second, and six simultaneous comparisons per endpoint: restoration against the unedited run and two controls (clean-feature and random directions), under Camera and Sensor Noise, on held-out tasks. Command error falls against the unedited run and both controls; the shared directions reach most of what replacing the whole L9 state does; the controls write less, and with edit size matched in development restoration is better under Camera but not under Sensor Noise; observed success is higher with intervals that include zero, and replacing the whole L9 state raises success (post hoc, pointwise). This supports main Sec. 4.2, Table 2 and Figs. 1b and 4.

S13.1. Directions, tasks and cases

Directions and site. The policy is the LIBERO-finetuned $\pi_{0.5}$ of the representation study. The edit acts on the action expert's L9 post-block state (ten ordered action positions, 1,024 channels). L9, the middle recorded layer, was fixed before the confirmation runs as a compromise between the response coverage of L1 and the command prediction of L18 (main Sec. 4.1); one site, and two of the four families (Camera and Sensor Noise), keep six runs per case tractable. The shared and clean-feature directions are the first sixteen directions of the position-centered LIBERO-Plus fits of Sec. S3, learned at the first query and first generation step; the confirmation uses the Camera and Sensor Noise fits from folds 1, 2 and 4 of that analysis. A set has sixteen channel directions, shared by all positions.

Held-out tasks. Development uses folds 0 and 3, confirmation folds 1, 2 and 4: nine and 31 distinct tasks, with no overlap. Each task is assigned the fit from its fold of the original joint task-component/family split, so its own task component and the tested family are excluded from that fit's training data. Camera confirmation has 27 tasks and 48 settings; Sensor Noise has 30 tasks and 60 settings; the two share 26 tasks. Each setting has one start state, two policy-noise draws and six runs, giving $108 \times 2 \times 6 = 1{,}296$ episodes. All planned command measurements and outcomes are recorded.

S13.2. Restoration and the six runs

A clean observation of each run's own current state. Let s_q^a be the physical and controller state of run a at query q, and x_{qj}^a, t_j its current plan and time at generation step j. Both observations are rendered from this s_q^a. For Camera, the clean observation restores the setting's complete clean visual configuration; for Sensor Noise, it removes the image transform from the same underlying image. Rendering it leaves the simulator, controller and random-number states unchanged.

Two forward passes with these observations and the same x_{qj}^a, t_j give $H_T, H_C \in \mathbb{R}^{10\times1024}$ at L9, so H_C is the clean value for the run's current state. Restoration sets

$$D = H_T - H_C, \qquad H' = H_T - \alpha D B B^\top, \qquad \alpha = 1, \tag{S21}$$

where $B \in \mathbb{R}^{1024\times16}$ has orthonormal columns. Up to the precision of the write, $H'B = H_C B$, while the orthogonal complement keeps its perturbed value. Position centering is used only to fit B; the edit projects the full current difference D onto the fixed directions. After the write, later blocks continue with the perturbed image prefix.

The six runs. The clean run N_C and the perturbed run N_T are unedited. Restoration R_{resp} and the controls R_{clean} and R_{rand} apply Eq. (S21) with the shared, clean-feature and random directions, at all ten generation steps of every policy query, with $\alpha = 1$. R_{full} replaces the whole L9 state, $H' = H_C$, at the same site and schedule; its later blocks also read the perturbed prefix.

The random control uses a fixed mixture of three random sets of 16 orthonormal directions, assigned within task and family, with both noise draws of a setting receiving the same set. One set is the first sixteen columns of the 32-direction random set of Sec. S3; the other two come from the same generator with separately fixed seeds. All six runs start from the same simulator and controller state and use the same policy noise at each query. Once an edit changes generation, each run follows its own plan and trajectory, and each later clean observation follows that run's own state.

S13.3. Endpoints and simultaneous intervals

Command error. Let $A_{tvn}^a \in \mathbb{R}^{5\times3}$ hold the translation parts of the first five commands of the first query, for task t, setting v, noise draw n and run a. The command error is

$$e_{tvn}^a = \left\| A_{tvn}^a - A_{tvn}^{N_C} \right\|_F, \tag{S22}$$

the Euclidean norm over fifteen numbers, in command units after the adapter. All 1,296 records contain the five commands, each logged after the environment step returned, and they match the first five positions of the corresponding ten-position plans exactly. The clean target is paired by setting, start state and noise draw.

Task-equal contrasts and fixed scales. Write $\langle z \rangle_f$ for the average over the two noise draws, then over settings within a

Table S31. **Restoration against the unedited run and both controls: command error falls in all six comparisons; success intervals all include zero.** Restoration R_{resp} against each comparator. Positive values favor restoration: first five commands closer to those of the clean run N_C (in units of a fixed family scale S_f), or more successes. Each endpoint has its own family of six simultaneous 95% whole-task bootstrap intervals. Camera: 27 tasks, 48 settings and 96 cases; Sensor Noise: 30 tasks, 60 settings and 120 cases. Every success interval includes zero; against R_{clean} under Sensor Noise its lower end is zero.

Family	Comparator	Command-error reduction $G_{f,b}$ [simultaneous interval]	Success change (pp) [simultaneous interval]
Camera	N_{T}	0.3068 [0.1573, 0.4821]	+1.85 [−8.33, 11.11]
Camera	R_{clean}	0.2908 [0.1423, 0.4641]	−1.85 [−12.04, 7.41]
Camera	R_{rand}	0.2999 [0.1530, 0.4724]	+1.85 [−9.26, 12.96]
Sensor Noise	N_{T}	0.0523 [0.0179, 0.0985]	+3.33 [−1.67, 9.17]
Sensor Noise	R_{clean}	0.0337 [0.0035, 0.0740]	+3.33 [0.00, 8.33]
Sensor Noise	R_{rand}	0.0516 [0.0165, 0.0996]	+1.67 [−2.50, 6.67]

task, then equally over the tasks of family f. For comparator $b \in \{N_T, R_{\text{clean}}, R_{\text{rand}}\}$, the command-error reduction and the success change are

$$G_{f,b} = \frac{\langle e^b - e^{R_{\text{resp}}} \rangle_f}{S_f}, \tag{S23}$$

$$\Delta^Y_{f,b} = 100 \left\langle Y^{R_{\text{resp}}} - Y^b \right\rangle_f, \tag{S24}$$

where Y is the final binary outcome. The family scales, fixed in advance, are $S_{\text{Camera}} = 0.316943$ and $S_{\text{Sensor}} = 0.200099$ in the same command units. They are constants, so G is in units of S_f, not a percentage of each case's error. Main Figs. 1b and 4a show these contrasts in command units, multiplying each gain and its interval endpoints by the same S_f: Fig. 1b the contrast with N_T, and Fig. 4a those with R_{clean} and R_{rand}.

Six simultaneous comparisons for each endpoint. Each endpoint has its own family of six comparisons: three comparators under two perturbation families. The prespecified procedure uses 10,000 bootstrap draws of whole tasks, keeping all paired observations of each drawn task. Within a perturbation family, the same task draws serve all three comparators and both endpoints. The simultaneous 95% intervals use percentile tails $0.05/(2 \times 6)$ and $1 - 0.05/(2 \times 6)$, with linear interpolation between ordered draws. Table S31 reports all twelve intervals.

S13.4. Results: command error and task success

Command error falls against the unedited run and both controls. All six command intervals are above zero (Table S31). Against N_T, restoration lowers the task-equal command error from 0.4093 to 0.3120 under Camera and from 0.2091 to 0.1987 under Sensor Noise. In units of S_f, it reduces the error by 0.2908 and 0.0337 more than the clean-feature control, and by 0.2999 and 0.0516 more than the random control.

The reduction is broad across tasks. The table below summarizes the task-level command-error reductions behind Table S31, as comparator minus R_{resp} in command units (the units of main Figs. 1b and 4a), with tasks weighted equally. Against N_T, the error falls in 25 of 27 Camera tasks and 24 of 30 Sensor Noise tasks. Medians lie below the means: the three largest task reductions account for 37% (Camera) and 48% (Sensor Noise) of the summed reduction. These are descriptive task statistics; the simultaneous intervals are those of Table S31.

	Camera (27 tasks)			Sensor Noise (30 tasks)		
Comparator	Mean	Median	> 0	Mean	Median	> 0
N_T	0.0972	0.0557	25	0.0105	0.0060	24
R_{clean}	0.0922	0.0538	23	0.0067	0.0041	23
R_{rand}	0.0951	0.0552	25	0.0103	0.0057	24

Under Camera, restoration moves the commands mainly toward the clean commands. Write the translation parts of the first five commands as $A^a \in \mathbb{R}^{15}$. For each setting, start state and noise draw, let $c = A^{N_C} - A^{N_T}$ be the direction toward the clean commands and $d^a = A^a - A^{N_T}$ the change the edit causes. Its signed component along c and the size of the rest are

$$d^a_{\parallel} = \frac{\langle d^a, c \rangle}{\|c\|_2}, \qquad d^a_{\perp} = \left\| d^a - d^a_{\parallel} \frac{c}{\|c\|_2} \right\|_2. \tag{S25}$$

For each run, the command error satisfies $(e^a)^2 = (\|c\|_2 - d^a_{\parallel})^2 + (d^a_{\perp})^2$: $d_{\parallel}$ is progress toward the clean commands and $d_{\perp}$ the change in other directions. Both are computed per record and then averaged like the command error.

All 1,296 records contain the five commands, and all 216 clean-target distances are positive. Table S32 reports all six runs as a descriptive decomposition. Restoration has mean along/orthogonal components $0.10365/0.04660$ under Camera, so most of its change points toward the clean commands, and $0.01139/0.01229$ under Sensor Noise, where the change in other directions is as large as the progress toward

the clean commands. Only the command-error contrasts carry intervals (Table S31).

The 16 directions reach most of what replacing the whole L9 state does; the controls write less. Replacing the whole L9 state lowers the mean command error to 0.2973 (Camera) and 0.1943 (Sensor Noise), only somewhat below restoration (Table S33); the remaining error enters through the later blocks, which still read the perturbed image. These values are a reference, not one of the six comparisons. From Tables S33 and S32, replacing the whole state removes 27% (Camera) and 7% (Sensor Noise) of the command error of N_T, restoration removes 24% and 5%, and restoration thus reaches 86.9% and 70.7% of the whole-state reduction (computed from unrounded values); these are the percentages quoted in the main paper. Restoration's edit size is 62% and 71% of the whole-state edit and five to seven times that of the controls, so per unit of edit it removes 1.4 times as much command error as replacing the whole state under Camera and as much under Sensor Noise. Restoration and the two controls share the number of directions, the coefficient and the schedule, but the controls write far less.

Observed success is higher, with intervals that include zero. Restoration raises observed success from 73.15% to 75.00% under Camera and from 83.33% to 86.67% under Sensor Noise; the clean runs succeed in 98.15% and 97.50% (Table S33). The paired gains over N_T are +1.85 and +3.33 percentage points. All six simultaneous success intervals include zero, the one against the clean-feature control under Sensor Noise with its lower end at zero (Table S31). Under Camera, the clean-feature control has higher observed success than restoration.

Lost and gained successes in every run; replacing the whole L9 state raises success. The table counts, unweighted, cases whose outcome differs from the unedited run N_T with the same noise draw: failures that become successes (gained) and the reverse (lost). Gained successes grow with how much of the clean state is copied, while every edited run also loses some. The counts are descriptive; the simultaneous intervals are in Table S31. In a post hoc analysis with pointwise intervals, outside the six simultaneous comparisons, R_{full} minus N_T is +7.41 [+0.93, +14.81] (Camera) and +5.83 [0.00, +12.50] (Sensor Noise): this layer matters for the outcome, and restoration delivers part of that gain.

	Camera (96 cases)			Sensor Noise (120 cases)		
Run	Successes	Gained	Lost	Successes	Gained	Lost
N_T	69	–	–	100	–	–
R_{rand}	68	3	4	102	5	3
R_{clean}	72	6	3	100	3	3
R_{resp}	71	7	5	104	7	3
R_{full}	76	10	3	107	10	3
N_C	94	26	1	117	19	2

S13.5. Development checks: generation schedule and matched edit size

On the development tasks, restoring at every generation step matters, and with edit size matched, restoration beats clean-feature directions under Camera but not under Sensor Noise. These are descriptive means without intervals.

Saved states and plan endpoint. Camera covers eight settings on five tasks; Sensor Noise covers seventeen settings on nine tasks. Each setting has two noise draws and four saved query states, $q = 0, 2, 5, 10$, from the unedited perturbed run, giving 64 Camera and 136 Sensor Noise cases, all complete. Clean and perturbed observations share the saved physical state, every candidate generation starts from the same recorded policy noise, and each case keeps its assigned directions.

The endpoint is the first five positions of the generated translation plan, and its improvement is

$$g^a = \frac{\|A^{N_T} - A^{N_C}\|_F - \|A^a - A^{N_C}\|_F}{S_f}, \qquad \text{(S26)}$$

with the fixed family scales defined above. Means average the four saved queries and both noise draws within settings, then settings within tasks, with equal weight per task (Table S34). This endpoint is a plan at a saved state; the confirmation endpoint above is the first five commands of a complete rollout.

Restoring at every generation step matters. Both schedules use Eq. (S21) with the shared directions. Moving from restoring only at $j = 0$ to restoring at $j = 0, \ldots, 9$ raises the mean gain from 0.00475 to 0.15468 for Camera and from 0.00188 to 0.06687 for Sensor Noise. The schedule changes both the sequence of writes and the later generation states. The edit on original LIBERO, which acts only at the first step, is reported in Sec. S12.

A common edit size at each position. For the shared, clean-feature and assigned random directions B_k, let $p_k = (H_T - H_C)B_k B_k^\top$ at one action position. The matched-size rule sets

$$s = \min_{k \in \{\text{resp,clean,rand}\}} \|p_k\|_2, \qquad H'_k = H_T - s\frac{p_k}{\|p_k\|_2}. \qquad \text{(S27)}$$

Table S32. **How each run changes the first five commands: progress toward the clean commands and change in other directions.** Descriptive task-equal means on all confirmation cases: 96 Camera cases (48 settings, 27 tasks) and 120 Sensor Noise cases (60 settings, 30 tasks), two noise draws per setting. Each quantity is computed per record from the translation parts of the first five commands, then averaged. $d_{\parallel}, d_{\perp}$ are defined in Eq. S25; e is the command error. Command units; means without intervals. The geometric identity holds per record, not for the means.

	Camera			Sensor Noise		
Run	Along $d_{\parallel}$	Orthogonal $d_{\perp}$	Error e	Along $d_{\parallel}$	Orthogonal $d_{\perp}$	Error e
Clean (N_C)	0.40928	0.00000	0.00000	0.20913	0.00000	0.00000
Perturbed, unedited (N_T)	0.00000	0.00000	0.40928	0.00000	0.00000	0.20913
Restoration (R_{resp})	0.10365	0.04660	0.31205	0.01139	0.01229	0.19867
Clean-feature directions (R_{clean})	0.00529	0.01001	0.40422	0.00388	0.00633	0.20541
Random directions (R_{rand})	0.00221	0.00391	0.40710	0.00016	0.00246	0.20900
Whole L9 state (R_{full})	0.11974	0.05176	0.29733	0.01671	0.01918	0.19433

Table S33. **The six runs: command error, task success and edit size.** Descriptive task-equal means on the confirmation cases. e: command error (Eq. S22), zero for N_C by definition. Edit size: actual change written into the L9 state (tokenwise ℓ_2), summed over every query of the run's own episode, so it depends on both the writes and the episode length. Restoration and the controls share the number of directions, coefficient and schedule, yet write different amounts. R_{full} (whole L9 state) is a reference, not one of the six prespecified comparisons.

	Camera			Sensor Noise		
Run	Error e	Success (%)	Edit size (10^3)	Error e	Success (%)	Edit size (10^3)
N_{C}	0.0000	98.15	0.00	0.0000	97.50	0.00
N_{T}	0.4093	73.15	0.00	0.2091	83.33	0.00
R_{resp}	0.3120	75.00	46.20	0.1987	86.67	51.47
R_{clean}	0.4042	76.85	7.79	0.2054	83.33	7.74
R_{rand}	0.4071	73.15	9.65	0.2090	85.00	9.64
R_{full}	0.2973	80.56	74.20	0.1943	89.17	72.77

Table S34. **Development: plan gains depend on the schedule and on the edit-size rule.** Descriptive task-equal means of Eq. S26 on all 64 Camera and 136 Sensor Noise cases: 8/17 settings on 5/9 tasks, two noise draws and saved queries $q = 0, 2, 5, 10$. First/all: edit at $j = 0$ or at $j = 0, \ldots, 9$. Matched: Eq. S27 applied at each run's current state. N_T has zero gain by definition.

Rule	Camera / S_f	Sensor Noise / S_f
Clean (N_C)	1.20070	1.19909
Unedited (N_T)	0.00000	0.00000
Shared, first	0.00475	0.00188
Shared, all	0.15468	0.06687
Clean-feature, all	0.02056	0.02137
Random, all	0.00364	0.00118
Whole L9 state, all	0.22249	0.12432
Matched: shared	0.02539	0.00975
Matched: clean-feature	0.01763	0.01881
Matched: random	0.00308	0.00044

A zero target leaves the state unchanged. The actual written changes, $H_{\text{after}} - H_{\text{before}}$, are recorded; their shared-minus-clean-feature norm differences take both signs. Under complete generation, each run reapplies this rule at its own current state.

Clean-feature edits change the local output more. Applying each candidate write at the same state of the unedited perturbed generation, we measure the change in the 32-dimensional normalized flow velocity, $\delta v_k = v_k - v_{N_T}$, separately for plan positions 0–4 and 5–9. Clean-feature edits give a larger $\|\delta v\|_2$ than shared-direction edits in every recorded task–query–generation-step–position-group mean: $400/400$ for Camera ($5 \times 4 \times 10 \times 2$) and $720/720$ for Sensor Noise ($9 \times 4 \times 10 \times 2$). These grids repeat measurements of the same five and nine tasks; all norm comparisons are defined. The clean local velocity was not recorded.

With edit size matched, the ranking differs between families. Under complete generation with matched edit size, the gains for shared/clean-feature directions are $0.02539/0.01763$ for Camera and $0.00975/0.01881$ for Sensor Noise. Restoration is thus better than clean-feature directions under Camera and worse under Sensor Noise, although clean-feature directions change the local output more in both: local output size is not plan correction.

Implementation notes. Projections and edit sizes are computed at higher precision than the policy's own execution, which writes the edited state at its native precision; writes that round to zero at that precision remain in the data.

S14. How the Readout Estimator Changes Command Comparisons

Does the finding that commands are read best at L18 depend on the linear readout? We refit the readout with two nonlinear estimators on the same coordinates, pairs and splits. The layer ordering holds: the random-feature readout of the shared coordinates has higher translation error than predicting no change in 4 of 4 families at L1, 1 of 4 at L9 and 0 of 4 at L18 (Table S35). Which directions predict better, however, depends on the estimator: of 216 comparisons, the random-feature readout changes the interval category of 102 and the MLP of 76. This supports Sec. 4.1 of the main paper (Fig. 3b): a linear readout predicts commands best at L18 and poorly at L1 and L9, and rankings depend on the readout.

What was compared. The main paper uses a fixed affine ridge readout (Sec. S3.4). Here a random Fourier feature (RFF) ridge readout and a one-hidden-layer MLP read the same 16 coordinates along the shared and the clean-feature directions, on the same 404 pairs and held-out splits. Predicting no change does not depend on the readout. Table S36 adds rotation and gripper. The 216 comparisons per readout are the 36 layer–family–command cells, each comparing the shared directions with the full hidden state, the magnitude baseline, the centered clean-feature directions, random directions and the uncentered shared directions, plus first flow with the shared coordinates against first flow alone.

The ranking of shared over clean-feature directions depends on the estimator. Under the random-feature readout the L9 advantage of the shared directions becomes a zero-spanning difference in Background, Camera and Sensor Noise ($+49.6\% \to -0.1\%$, $+27.4\% \to -3.7\%$, $+56.3\% \to -5.0\%$) and stays positive in Light ($+71.0\% \to +49.8\%$; Table S36). At L1 the affine advantage ($+20.3$ to $+74.7\%$, all four intervals above zero) stays positive under random features in Light ($+21.9\%$) and Sensor Noise ($+17.1\%$), and under the MLP in all four families. For Camera translation at L18, the random-feature readout favors the shared directions by 13.4% $[7.1, 20.5]$, where the affine interval spans zero; its error is 10.909×10^{-3}, against 3.737×10^{-3} for the affine readout and 15.414×10^{-3} for no change (Table S35). At L18 under the MLP, the shared directions have higher translation error than the clean-feature directions in Background and Camera, and higher rotation error in every family.

Absolute accuracy limits what the nonlinear readouts show. The MLP has higher error than predicting no change in every layer–family cell for every feature, first flow included (point estimates). Its comparison therefore describes this estimator and cannot determine how much command information the coordinates hold. For this reason the main paper reports the fixed affine readout, and relies on the removal test (Sec. S12) and restoration (Sec. S13) to measure directly how edits along the directions change the computation and the commands.

Implementation notes: matched fitting and resampling. The nonlinear-readout analysis was specified before its results were inspected. It uses the same 404 matched pairs, the same fixed position-centered shared, clean-feature and random directions, the same 60 joint task/family holdout splits and 2,000 whole-task bootstrap draws, with the same pair weights and scoring. Each input block was centered by its outer-training weighted mean and divided by its scalar RMS, using all outer-training rows before any inner split; flow and response blocks were standardized separately before concatenation. The random map, the inner-fold assignment and the MLP were each seeded deterministically from the readout, family, layer, fold and feature block.

Random-feature ridge. The map was $\phi(z) = \sqrt{2/D}\cos(z\Omega/\sigma + b)$, with standard-normal entries in Ω, phases uniform on $[0, 2\pi)$, and $D = 2{,}048$ except for the ten-dimensional magnitude baseline ($D = 40$). The bandwidth σ was the median off-diagonal pairwise Euclidean distance among standardized outer-training rows, using one if the median was zero. Five-fold cross-validation over outer-training task groups, keeping tasks that share a clean scene together, selected $\lambda \in \{0.001, 0.01, 0.1, 1, 10\}$ by mean validation error; exact ties kept the smaller value. The map was fixed before inner fitting; ridge then re-centered and scalar-scaled the mapped features within each inner training fold, with an unpenalized intercept.

MLP fitting and stopping. The MLP had one hidden layer of 256 ReLU units and a linear 390-dimensional output in raw/adapter units, trained by full-batch Adam with learning rate 10^{-3}, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$ for at most 500 updates. A seeded permutation reserved approximately 20% of outer-training task groups for validation (rounded to the nearest integer, at least one); the targets were centered using the remaining fitting rows and their normalized pair weights. Initial weights were Gaussian with standard deviations $\sqrt{2/d}$ and $\sqrt{1/256}$ for the input and output layers (d is the input width), with zero biases. Validation MSE was checked after each update; training stopped after 25 consecutive updates without an improvement greater than 10^{-12}, and predictions used the best validation weights. The

Table S35. **Along the shared directions, the affine and random-feature readouts beat predicting no change at L18 in every family; the MLP never does.** Translation error (MSE $\times 10^3$) of the first five commands, task-equal on the same 404 matched pairs across 28 tasks; lower is better. Every estimator uses the same position-centered 16 shared and clean-feature directions; No-change is common to all. RFF ridge: random Fourier feature ridge readout. Error reductions with pointwise 95% intervals, including rotation and gripper: Table S36.

		Affine		RFF ridge		MLP		
Layer	Family	Shared	Clean	Shared	Clean	Shared	Clean	No-change
L1	Background	4.297	10.213	2.450	2.723	16.543	129.003	2.084
L1	Camera	16.280	20.428	16.404	17.002	21.546	100.024	15.414
L1	Light	1.875	7.400	1.303	1.668	4.418	76.428	1.063
L1	Sensor Noise	18.564	27.219	4.914	5.930	139.285	366.620	4.609
L9	Background	3.825	7.597	1.929	1.927	16.318	58.358	2.084
L9	Camera	14.407	19.831	18.427	17.776	89.104	104.278	15.414
L9	Light	1.365	4.713	0.968	1.930	4.285	25.489	1.063
L9	Sensor Noise	6.331	14.484	4.028	3.836	26.858	132.264	4.609
L18	Background	1.035	0.964	0.786	0.787	33.596	22.576	2.084
L18	Camera	3.737	3.457	10.909	12.590	155.875	90.506	15.414
L18	Light	0.433	0.496	0.321	0.307	11.097	9.487	1.063
L18	Sensor Noise	1.538	1.480	1.953	2.056	67.980	52.938	4.609

archive keeps task-level scores and aggregate results, but not the bandwidth, selected penalty or stopping update of each individual fit.

S15. Response Geometry and Command Information in π_0

Do different perturbations also change the features of π_0 along shared directions? Yes. On held-out tasks and families, 16 shared directions retain 70–77% of the response energy at L9, 39–50% at L1 and 43–52% at L18, against 1–28% for clean-feature and about 1.5% for random directions, in every family (Table S37). This supports claim C1 of the main paper (Sec. 4.1), which states that the result holds in π_0. The command readout qualifies it: in π_0 the shared coordinates carry command information at L18 but predict commands no better than clean-feature coordinates.

What was compared. We recorded π_0 at the first query on its paired LIBERO-Plus inputs and compared shared, clean-feature and random directions, 16 each. In Table S37 each row fixes the layer and family; higher retention means a better reconstruction of the same centered response (task-equal means over 28 task components, 2,000 whole-task bootstrap draws).

The shared directions carry command information at L18 but do not beat clean-feature directions. At L18, a linear readout of the shared coordinates predicts the change in the first five commands (translation) better than the magnitude baseline and better than random coordinates, with intervals above zero in 4 of 4 families for each comparison; the corresponding counts are 0 of 4 at L1 and 0 of 4 at L9. Against the full hidden state or the centered clean-feature coordinates, no interval is above zero at any of the three layers (0 of 4 each), and adding the shared coordinates to first flow gives no interval above zero. The readout thus finds command information in the shared directions at L18 but does not favor them over clean-feature coordinates.

Recording π_0 at the same computation stage. The π_0 plan has 50 actions, and its action expert prepends one state token; the recorded state is $[10, 3, 1, 51, 1024]$, and the analysis uses the 50 plan positions. The original π_0 paired executions stored no internal states, so the policy was rerun on their recorded first-query inputs and policy noise, on different hardware. Regenerated raw actions differ from the recorded actions by at most 0.006–0.008 in absolute value. Input reconstruction is deterministic, and two independent reruns of the same run on the same hardware agree exactly.

Fitting the directions and the command readouts. We use the joint task/family split and affine readout of Sec. S3.4. For each held-out family, five task-component folds exclude both that family and the test fold from fitting; testing uses their intersection. Tasks that share a clean scene stay in one component, and common tasks keep the $\pi_{0.5}$ fold assignments. Each split fits position means and shared and clean-feature directions anew on its admitted π_0 fitting recordings. Fitting weights tasks equally, then families, settings, noise draws and positions within their parents.

At the first generation step, removing the state token leaves 50 ordered positions with 1,024 channels. Each set of 16 directions shares its channel basis across positions, giving

Table S36. **Which directions predict commands better depends on the readout.** Error reduction (%) of the shared over the centered clean-feature directions for the first five commands, task-equal over 28 tasks, with pointwise 95% whole-task bootstrap intervals (2,000 draws shared across readouts); 404 pairs with matched commands across the four families (96, 102, 96 and 110 in Background, Camera, Light and Sensor Noise). RFF ridge: random Fourier feature ridge readout. Affine values agree with the main-paper readout at the displayed precision.

Layer	Family	Command	Affine	RFF ridge	MLP
L1	Background	translation	+57.9 [+45.4, +70.3]	+10.0 [−7.2, +29.9]	+87.2 [+83.6, +90.6]
L1	Background	rotation	+31.1 [+17.9, +45.2]	−0.3 [−12.8, +10.1]	+90.6 [+88.8, +92.4]
L1	Background	gripper	+57.7 [+40.4, +69.1]	+4.4 [−8.5, +13.8]	+87.4 [+83.1, +91.6]
L1	Camera	translation	+20.3 [+1.0, +33.7]	+3.5 [−3.6, +9.0]	+78.5 [+72.3, +83.7]
L1	Camera	rotation	+21.5 [+9.2, +32.6]	+5.9 [−2.3, +11.5]	+91.9 [+90.5, +93.2]
L1	Camera	gripper	+24.7 [+12.2, +34.3]	−7.5 [−17.5, −0.2]	+92.3 [+90.4, +93.9]
L1	Light	translation	+74.7 [+64.4, +82.5]	+21.9 [+3.1, +37.1]	+94.2 [+92.4, +95.7]
L1	Light	rotation	+68.4 [+58.5, +76.9]	+13.3 [+0.7, +25.4]	+96.1 [+95.6, +96.5]
L1	Light	gripper	+53.2 [+41.1, +62.3]	+6.4 [−5.5, +16.1]	+96.3 [+95.1, +97.2]
L1	Sensor Noise	translation	+31.8 [+8.6, +48.3]	+17.1 [+2.3, +29.6]	+62.0 [+54.0, +68.8]
L1	Sensor Noise	rotation	+39.5 [+14.8, +56.6]	+6.1 [−3.2, +19.7]	+68.3 [+58.9, +74.6]
L1	Sensor Noise	gripper	+31.7 [−0.9, +55.0]	−23.9 [−34.2, −12.4]	+66.2 [+49.6, +77.0]
L9	Background	translation	+49.6 [+25.6, +67.9]	−0.1 [−31.0, +19.1]	+72.0 [+60.7, +80.8]
L9	Background	rotation	+34.2 [+11.3, +53.8]	−1.9 [−13.2, +8.6]	+79.7 [+73.2, +85.1]
L9	Background	gripper	+30.8 [+17.9, +42.1]	+0.9 [−13.7, +14.6]	+81.6 [+71.7, +88.6]
L9	Camera	translation	+27.4 [+5.7, +40.3]	−3.7 [−9.8, +5.5]	+14.6 [−8.1, +30.9]
L9	Camera	rotation	+5.0 [−12.9, +17.9]	−8.7 [−16.9, +2.2]	+15.5 [−9.9, +35.0]
L9	Camera	gripper	+25.9 [+5.6, +39.9]	−4.7 [−20.7, +9.2]	+20.7 [−6.2, +43.9]
L9	Light	translation	+71.0 [+63.3, +77.5]	+49.8 [+30.1, +59.5]	+83.2 [+78.4, +86.9]
L9	Light	rotation	+54.0 [+38.2, +66.3]	+19.7 [+5.0, +34.0]	+82.5 [+77.8, +86.2]
L9	Light	gripper	+42.0 [+29.4, +52.1]	+6.7 [+0.2, +12.8]	+82.6 [+77.0, +86.8]
L9	Sensor Noise	translation	+56.3 [+33.1, +72.1]	−5.0 [−26.8, +9.4]	+79.7 [+75.9, +83.2]
L9	Sensor Noise	rotation	+59.2 [+50.6, +69.5]	−12.6 [−46.1, +2.6]	+80.0 [+77.0, +82.6]
L9	Sensor Noise	gripper	+51.5 [+40.7, +61.5]	+0.4 [−7.6, +7.1]	+84.9 [+78.8, +89.1]
L18	Background	translation	−7.4 [−38.2, +18.6]	+0.1 [−25.0, +23.5]	−48.8 [−103.0, −10.1]
L18	Background	rotation	+2.6 [−22.8, +25.1]	+22.3 [+4.6, +38.5]	−77.1 [−141.5, −22.3]
L18	Background	gripper	−33.5 [−69.1, −3.9]	−0.0 [−17.2, +14.4]	−71.8 [−156.0, −6.0]
L18	Camera	translation	−8.1 [−32.2, +15.5]	+13.4 [+7.1, +20.5]	−72.2 [−114.3, −38.5]
L18	Camera	rotation	−3.3 [−69.4, +34.9]	+21.2 [+12.6, +28.6]	−83.5 [−146.8, −35.7]
L18	Camera	gripper	−39.7 [−66.6, −16.0]	−19.3 [−40.1, −3.0]	−125.9 [−219.4, −58.6]
L18	Light	translation	+12.7 [−7.9, +31.0]	−4.4 [−46.1, +26.2]	−17.0 [−55.1, +13.1]
L18	Light	rotation	−13.3 [−71.3, +30.6]	+9.7 [−12.3, +29.0]	−36.5 [−75.5, −6.5]
L18	Light	gripper	−10.7 [−35.1, +5.9]	+4.0 [−10.0, +16.9]	−37.1 [−98.9, +5.1]
L18	Sensor Noise	translation	−3.9 [−51.5, +25.8]	+5.0 [−11.2, +16.9]	−28.4 [−67.2, +7.3]
L18	Sensor Noise	rotation	+21.2 [−18.6, +54.1]	+28.9 [+16.5, +45.5]	−28.1 [−53.5, −2.3]
L18	Sensor Noise	gripper	−18.7 [−39.6, +1.1]	+6.8 [−9.4, +20.1]	−75.0 [−155.6, −22.3]

800 coefficients per layer; positionwise magnitudes, full hidden responses and first-flow differences use 50, 51,200 and 1,600 features. First flow plus the shared coordinates uses 2,400 features. Each block is centered on its weighted fitting mean and divided by one fitting RMS scale; affine ridge uses an unpenalized intercept and fixed $\lambda = 0.01$. The combined readout scales its two blocks separately. Layers are fitted separately, with position coordinates kept in each feature vector. We score the regenerated first five commands (after the output adapter) separately for translation, rotation and continuous gripper.

Admission rule for rerun recordings. A rerun was admitted if its actions stayed within twice the largest difference observed when the original π_0 runs were repeated on the same input across machines: raw-action maximum absolute difference ≤ 0.018; first-five translation RMS $\leq 3.1 \times 10^{-3}$, rotation RMS $\leq 2.1 \times 10^{-3}$, gripper maximum $\leq 3.9 \times 10^{-3}$; sign agreement where the recorded value exceeds three times the threshold; no NaN. The rule depends on no outcome. It replaced an exact-match criterion after the discrepancy was observed.

Admitted pairs and exclusions. Of 808 runs, 769 were admitted and 39 excluded without retry: 7 runs of four Light settings were missing for technical reasons and 32 exceeded the tolerance (25 on the gripper threshold, 7 on the raw-action threshold, largest 0.27). This left 370 of the 404

planned clean/perturbed pairs on 28 tasks. One further pair was dropped, not retried, because the recorded state of an admitted run failed to write, so 369 pairs enter the analysis. Background, Camera, Light and Sensor Noise contribute 91/89/83/106 pairs on 28/28/27/28 task components. The exclusions spread over 21 family–task cells and all four families (Background 6 of 192 runs, Camera 15 of 204, Light 14 of 192 including the 7 technical, Sensor Noise 4 of 220); 7 of the 31 affected settings lost more than one run. Every statistic compares clean and perturbed recordings from the same rerun; the original actions serve only as the reference for admission, and admitted reruns lie within the original cross-machine scale rather than matching exactly.

S16. A Reusable Protocol for Responses and Paired Task Outcomes

This section states the protocol of the main paper (Secs. 3.2 and 5) as steps another study can follow, and collects the supplement's implementation and verification details. The protocol separates a perturbation's effect from the variation between two clean runs and judges both by paired task outcomes. Estimates can be recomputed from the recorded case-level data without a policy; producing those data requires the policy and its execution pipeline.

1. Fix the plan before observing outcomes. Declare the model and checkpoint, task and visual variant, matched clean rendering, instruction, initial state, two policy-noise draws (0 and 1), episode budget and source code; run all runs of a setting on one machine. Each setting under one policy uses the five runs below, in this order; record missing or invalid runs against the plan. Policy, execution and setting together identify a comparison.

2. Record plans, commands and motion separately. Record the policy's raw actions, the planned commands after the output adapter, the commands sent to the environment and each call's return. Differences in the first five commands (translation, rotation, gripper) are in adapter command units; their root mean square (RMS) gives the size of a change. Record the end-effector position in world meters and its orientation in radians after returned controls 0, 5, 15, 30 and 60, and state the quaternion convention. Record each run's final pose, its step index and whether the two runs of a pair have the same number of steps, and record early stops, unreturned controls, missing poses and unknown outcomes Y explicitly. Fixed-step and final-pose comparisons use the recorded pose at that step. A returned command records what was sent to the controller; the signal it physically applied requires separate logging.

3. Compare the perturbation with the clean rerun. For each noise draw, compare R_n with T_n. R_0–R_0' checks reproducibility; R_0–R_1 is the clean rerun, conditional on the two recorded draws; T_0–T_1 is its perturbed counterpart. For valid binary outcomes, SF is a lost success ($Y_R = 1, Y_T = 0$), FS a gained success, the success change is $\Pr(\mathrm{FS}) - \Pr(\mathrm{SF})$, and their sum is the rate of changed outcomes. Weight the two noise draws one half each, settings equally within a task, and tasks equally. Missing outcomes enter through bounds over missing outcomes; whole-task bootstrap intervals, adjusted for four families where stated, describe sampling uncertainty.

4. State where each internal quantity is recorded. For optional internal recordings of $\pi_{0.5}$, state for every scalar its layer, token region, generation step and recording path. Recomputing the prefix records the pre-block state H, the normalized state Z and the projected cached values V (fixed within a query); recomputing the attention records the full-key probability P, the output after projection and before the gate, W, and the block-end action-expert state H, at the first query and generation step. Keep L1, L9 and L18 and the main-image, wrist-image and text regions where available, and state whether a measure is raw or relative. Reconstruction coefficients describe an observed response; prediction and interventions need their own recorded inputs and operations. Recorded and recomputed states keep separate identities. The original π_0 paired executions stored no internal states; Sec. S15 regenerates first-query internal states for 369 pairs, and the protocol evidence for π_0 is commands, motion at fixed steps and outcomes Y.

5. Report task cost on its matched support. For every comparison with the clean run, perturbed or clean rerun, state the settings, metric and time step it covers, including missing and undefined measurements, and pair the two means on their common cases. Post hoc SF/SS groups describe lost and kept successes among observed outcomes. Later data collections keep their own identities beside the original results, including negative, null and incomplete comparisons.

S16.1. Recomputing estimates from recorded case data

From 536 policy–setting records, the analysis recomputes task-equal outcome estimates, bounds over missing outcomes and end-effector position and orientation comparisons after 5 and 60 controls. It checks that records are unique and complete, with five outcome cells per case and valid units and measurement status, and keeps the original task order and resampling scheme. The primary population has 202 planned settings per policy; a sensitivity analysis uses all 268.

Outcome estimates keep the planned denominator. For each case, all possible assignments of the unknown outcomes are enumerated before each comparison is formed; each reported bound holds for its comparison alone, and different comparisons can reach their endpoints under different assignments. Motion estimates require both noise-matched comparisons and the clean rerun at the same metric and time step. We average over noise draws, settings and tasks in that

Table S37. **In π_0, the 16 shared directions retain far more of an unseen response than clean-feature or random directions.** Share of the response energy around the fitting-data mean (%) at the first query, task-equal mean on each family's measured pairs with pointwise 95% whole-task bootstrap intervals; 369 pairs on 28 task components overall (family counts in text); directions fitted without the test tasks and family. The recordings come from a rerun of π_0 on different hardware, admitted by the action-difference criterion in the text.

Layer	Family	Shared	Clean-feature	Random
L1	Background	44.4 [41.1, 48.0]	2.0 [1.8, 2.1]	1.4 [1.4, 1.5]
L1	Camera	49.2 [43.9, 54.8]	2.2 [2.1, 2.3]	1.5 [1.4, 1.6]
L1	Light	38.5 [35.9, 41.4]	2.2 [2.1, 2.2]	1.5 [1.5, 1.6]
L1	Sensor Noise	50.1 [46.2, 54.0]	2.0 [1.8, 2.2]	1.4 [1.3, 1.4]
L9	Background	72.2 [69.4, 75.1]	1.0 [0.9, 1.1]	1.8 [1.7, 1.9]
L9	Camera	77.2 [74.1, 80.0]	1.0 [0.9, 1.1]	1.7 [1.6, 1.8]
L9	Light	71.9 [69.6, 74.1]	1.1 [1.0, 1.2]	1.9 [1.8, 2.0]
L9	Sensor Noise	70.3 [66.3, 74.0]	1.7 [1.6, 1.8]	1.9 [1.8, 2.0]
L18	Background	43.7 [39.9, 48.1]	21.5 [18.8, 24.4]	1.5 [1.4, 1.6]
L18	Camera	51.9 [47.8, 55.5]	28.1 [24.0, 32.0]	1.5 [1.4, 1.6]
L18	Light	49.2 [46.2, 52.5]	17.5 [15.3, 20.0]	1.4 [1.3, 1.5]
L18	Sensor Noise	43.0 [39.2, 47.1]	21.5 [17.7, 25.9]	1.5 [1.4, 1.6]

Run	Visual input / noise	Purpose
R_0	matched clean / 0	clean run
T_0	perturbed / 0	perturbed, noise of R_0
R_1	matched clean / 1	clean rerun
T_1	perturbed / 1	perturbed, noise of R_1
R_0'	matched clean / 0	repeat of R_0

order, then divide the two task-equal means; a zero clean-rerun mean leaves the ratio undefined. The success-change intervals use four-family-adjusted 98.75% coverage; other outcome and motion intervals are pointwise 95%; bounds over missing outcomes describe what the missing data allow, not sampling uncertainty.

Implementation and verification. Each check states what was verified and what it showed; the checks verify computations on the stated inputs, not the original executions. Data-collection exceptions follow.

Recomputed attention against the policy. One real input is run through two identical branches at generation steps 0, 5 and 9. The recomputed hidden states agree with the recorded ones; the recomputed velocity differs from the policy's by at most 0.03125 in absolute value, with relative errors 0.00367–0.00457. With identical branches, an error relative to the effect is undefined. The check covers hidden states and velocities only: P and W were not recorded during the policy's own generation, so attention has no tolerance or pass rate (Table S1).

Case-level analysis. Run from its instructions in an empty directory, the analysis reproduces 16 outcome panels and 128 metric/time/reference panels, and passes 1,448 checks against the original values, from the case-level records. The package includes the data dictionary, configuration, support tables, expected outputs and edge-case tests.

Policies. The configurations are `pi05_libero` and `pi0_libero`. Checksums of the checkpoint, normalization and tokenizer files of both policies are recorded in the source package.

Reconstruction. The uncentered 16-direction reconstruction results are reproduced to within 2.88×10^{-14} in absolute task-score difference.

Affine readout. The implementation reproduces 324 recorded full-panel MSE and target-energy entries with the uncentered directions, with maximum absolute discrepancy zero.

Nonlinear readouts. The affine readout inside the nonlinear-readout code (Sec. S14) reproduces the expected affine results with absolute differences below 10^{-15} in task-equal MSE and 2×10^{-13} in unscaled relative error reduction, in all four families.

Corrected readout support. The reported readouts use the protocol's 404 pairs with matched commands. A first run had evaluated Light on all 110 common-task pairs instead of the specified 96 with matched commands, which moved its affine values by up to 0.039 in relative error reduction. That first run is kept as a record; its other three families are unchanged.

Later-observation summaries. The shared, clean-feature and random summaries in Table S18 use the original averaging over noise, initial state and task and its intervals. Applying this averaging to all 940 available L18 episodes reproduces all 480 task means and 30 condition means.

Aggregation. The aggregation estimates and correlation intervals are reproduced from the recorded case-level scalars.

Removal test: edit size. In the removal test (Sec. S12), edit sizes are matched on the training average over the three conditions; per condition, the differences exceed the 2% tolerance.

Removal test: data collection. Data collection was paused at 229 of 288 cases and resumed in the fixed dispatch order. No case was selected by outcome. Two interim summaries (at 120 and 220 cases) had been seen before the decision to resume, as the decision record in the package states.

Removal test: execution exceptions. One case was interrupted after initialization; another was replaced under the prespecified technical-failure rule after a startup synchronization failure. Both are identified in the execution records.

Removal test: numerical checks. An independent calculation reproduced the response means exactly. The definitions in Table S29 are transcribed from the run configuration, not recomputed. The package holds the training mean, directions, random-direction seeds, edit sizes and the post hoc outcome bootstrap on available cases with per-draw values.

π_0 **rerun.** The records of the π_0 rerun on different hardware and of its admission rule are included (Sec. S15).

S17. Interpreting Attention Measurements

How should attention measurements of a perturbation be read? Which layer changes more, and whether the change grows during generation, depend on the image region, the computation site and the reference scale. This section and Sec. S20 are an additional study; the main paper mentions it once, in Related Work, and does not rely on it. Figures S18 and S19 give a spatial example and the complete generation profiles (trends: Fig. S23; regional comparisons: Table S26; definitions and checks: Secs. S11.2 and S16.1).

Image-region interventions estimate visual-action attribution [3], and attention-output patching tests language perturbations [2]. We therefore examine both attention probabilities and the output vectors that combine them with transformed values. For a state H, the relative difference is $\|H^p - H^c\|_F / \|H^c\|_F$, pooled over the selected positions and channels.

Regional attention needs its total mass as context. In a separate 36-task study, rotating only the main camera changes deep wrist features although wrist pixels, physical state and instruction are fixed, and attention redistributes within the unchanged wrist image. The redistribution appears larger at L18 than at L9, but weighting it by the wrist's share of all attention reverses the ranking (Sec. S10.4 and Table S26).

The measurement site changes the generation trend. On the 404 matched $\pi_{0.5}$ pairs at L18, recomputed attention probabilities of the clean and perturbed runs become closer from generation step 0 to 9, while the projected outputs, which combine probabilities with transformed values [1] and are measured before gating, move apart. Block-end states recorded during the policy's own generation show smaller absolute but larger relative differences; they include the residual and feed-forward updates, a different stage (Sec. S11.2).

The recomputation runs the frozen blocks on each run's saved generation state. A check on one input at three steps matches the recorded hidden states, with 0.37–0.46% relative velocity error. Probabilities and projected outputs were not recorded during generation, so they could not be compared directly (Secs. S1 and S16.1).

Probabilities and outputs show no clear difference in their link to commands. The difference between the command associations of P and W has intervals that include or touch zero in all four families (Sec. S4, Fig. S13b and Table S7). The generation finding concerns their different profiles: read together, probabilities, outputs and block-end states show changes a single attention map does not.

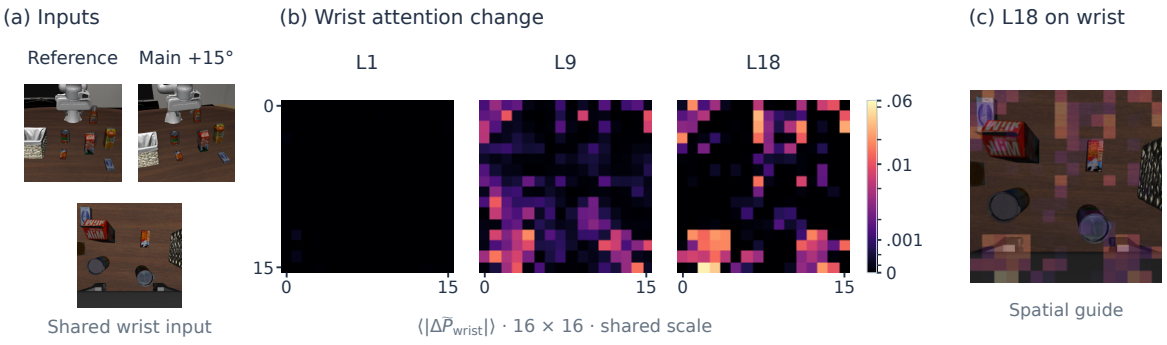


Figure S18. **Attention is redistributed over an unchanged wrist image.** One $\pi_{0.5}$ example, recomputed at generation step 5 with the same plan state, generation time and policy noise in both runs. (a) A 15° main-camera change holds wrist pixels, physical state, and instruction fixed. (b) Wrist probabilities are normalized to sum to one: $\widetilde{P}_{\text{wrist}} = P_{\text{wrist}}/m$. Heatmaps show mean absolute changes over eight heads and ten plan positions, with a shared asinh scale. Brightness indicates change magnitude, regardless of sign. (c) The L18 overlay provides spatial context; the unmixed heatmaps carry quantitative values. Signed maps: Figure S16.

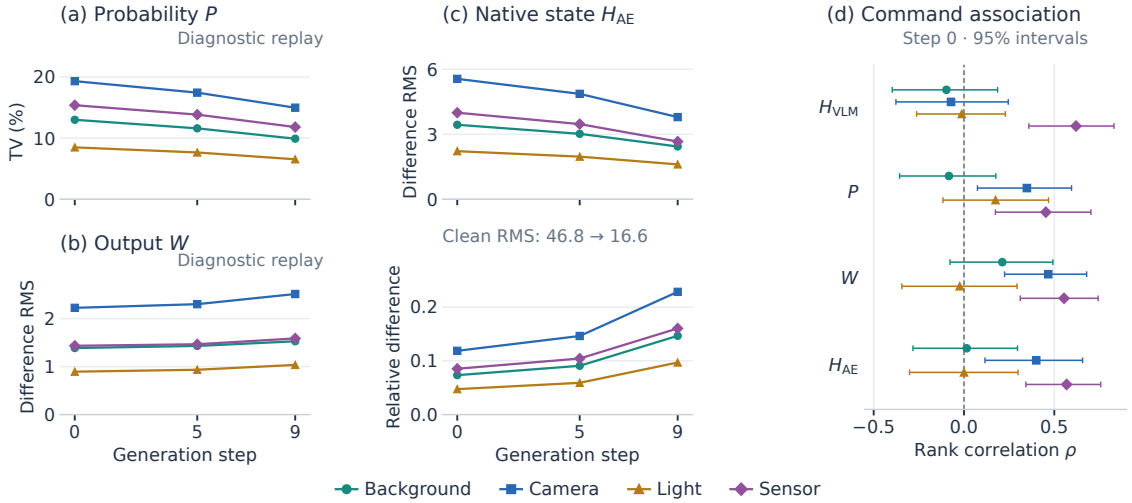


Figure S19. **Attention weights, projected outputs and block-end states follow different trends.** L18 measurements on 404 $\pi_{0.5}$ LIBERO-Plus pairs from 28 tasks. (a,b) Recomputed at each run's saved generation state: full-key probability TV decreases while raw projected-output RMS differences grow. (c) Block-end states H_{AE} recorded during the policy's own generation: differences shrink in raw RMS but grow relative to the clean state. Generation profiles use steps $0/5/9$ of the first query; relative responses are formed per case before task-equal averaging. (d) Post hoc, step-0 task-centered rank correlations with the RMS change of the first five commands (translation), with pointwise 95% task intervals for each correlation. Here P uses full-key TV and the other measures use clean-relative norms; H_{VLM} uses main-camera tokens.

S18. Complete Comparison Panels

This section gives complete versions of the main figures and adds no analysis. Figure S20 combines what the main paper splits across Figs. 1b, 3 and 4, plus the gain of shared over clean-feature directions in predicting commands, $1-\mathrm{MSE}_{\mathrm{shared}}/\mathrm{MSE}_{\mathrm{clean}}$ (positive favors the shared directions). Figure S21 backs Sec. 5.1; Fig. S22 gives all four outcome panels behind Fig. 5, including changed outcomes under a perturbation and under a clean rerun.

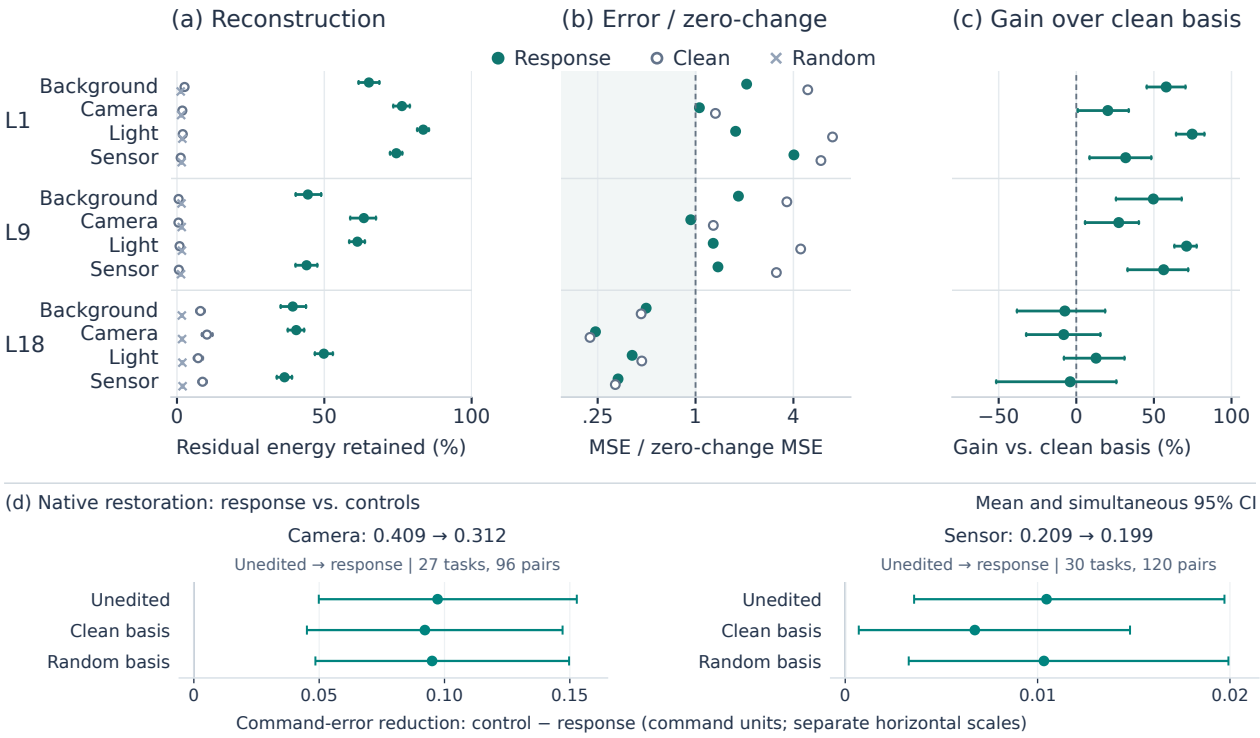
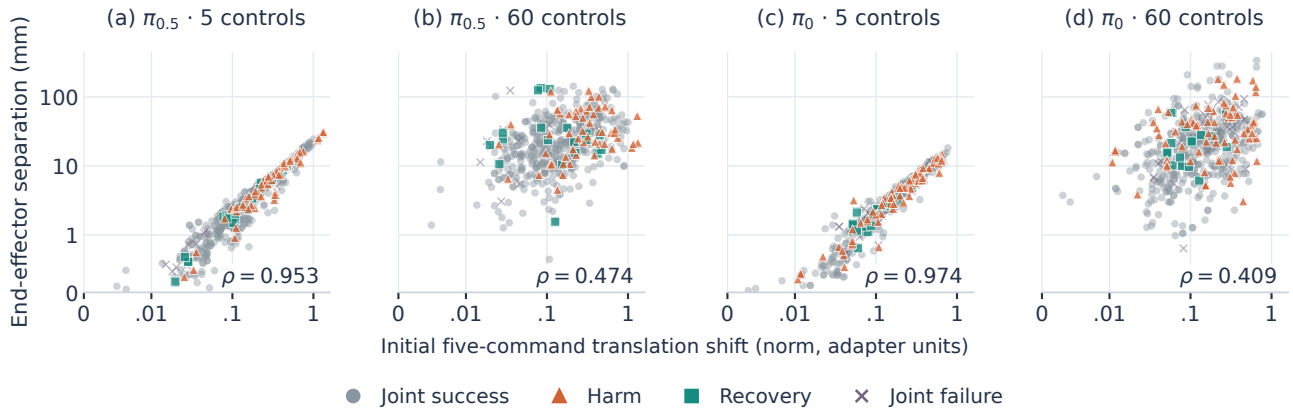


Figure S20. **Shared directions retain unseen responses, and restoring them at L9 reduces command error.** (a–c) The same 404 LIBERO-Plus pairs on 28 tasks; each fit leaves out the test tasks and family. (a) Share of the centered response energy at the first generation step retained by 16 shared directions. (b) Readout error relative to predicting no change; below 1 beats no change (log axis). (c) Gain over clean-feature directions in predicting commands: positive favors the shared directions; panel (b) gives the no-change reference. (a,c) Pointwise 95% task intervals. (d) Restoration at L9, a separate study: 96 Camera and 120 Sensor Noise cases on 27/30 tasks. Command error is the distance between the first five commands and those of the clean run. Points show error reductions (other run minus restoration); intervals are simultaneous 95% over six comparisons. Headers give unedited → restoration error. Separate horizontal scales show the smaller Sensor Noise effect. All controls: Sec. S13.



2858

Figure S21. **The change in the first five commands tracks motion closely after five controls and more weakly after 60.** Paired executions: both times use the same 516/536 pairs for $\pi_{0.5}$ and 526/536 for π_0, on all 40 tasks. A pair enters if its commands and its motion at both times passed validation, without the cross-family restriction of Fig. 5. Color gives the observed paired outcome; all outcome classes are included. ρ: task-weighted rank correlation across the plotted pairs. Shared symmetric-log axes are linear below 0.01 command units and 1 mm. Pointwise 95% intervals and paired early-minus-late differences: Sec. S5.1.

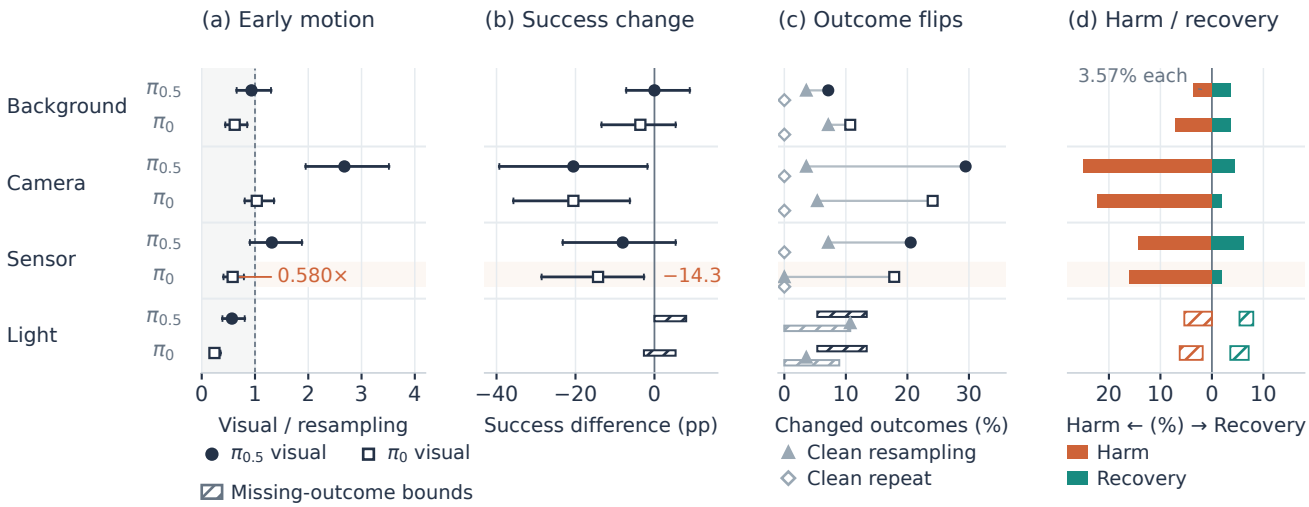


Figure S22. **Motion, net success and paired outcomes show different effects of a perturbation.** 202 settings with a LIBERO-Plus difficulty level, 28 tasks per policy. (a) Motion ratio: end-effector separation after five controls relative to the separation between two clean runs with different policy noise (clean rerun). Ratios divide task-equal means, with pointwise 95% task intervals; shading marks values below the clean rerun. Light has 43/48 settings on 26 tasks ($\pi_{0.5}$), 44/48 on 27 (π_0). (b) Success change with four-family-adjusted 98.75% intervals. (c) Lines connect the rates of changed outcomes under the perturbation and under a clean rerun (descriptive). (d) Lost successes extend left and gained successes right; lengths are positive percentages. Hatching marks bounds over missing outcomes, distinct from sampling intervals. The highlighted row is π_0 Sensor Noise. Secs. S5 and S6.

S19. Sensitivity to the Aggregation of Response Energy

Does the C1 result depend on how response energy is averaged over tasks? No. When task energies are summed before taking the ratio, the shared directions retain 38.4–84.0% of the response energy, against 0.46–10.91% for clean-feature and 1.35–1.98% for random directions, and they exceed both controls in all 12 layer–family cells (Table S38). This supports Sec. 4.1 of the main paper. It is a descriptive check with no new intervals; the task-equal score of the main paper remains the primary measure.

Two ways of aggregating. Let E_t be task t's mean squared response magnitude around the fitting-data position mean, and $S_{t,m}$ its mean squared reconstruction error for directions m. The main score gives tasks equal weight, $|T|^{-1}\sum_t(1-S_{t,m}/E_t)$. The alternative sums errors and energies first,

$$R_m^{\text{energy}} = 1 - \frac{\sum_t S_{t,m}}{\sum_t E_t}, \tag{S28}$$

which gives more weight to tasks with larger responses. Both use the same saved predictions, fitting means and joint task/family holdouts, with nothing refitted. The inputs reproduce all 36 main layer–family–direction means and use the same tasks and energy denominators for all directions.

The result. The shared-direction values differ by at most a few percentage points between the two aggregations (Table S38). Both scores keep the fitting-data position mean, so their response energy includes offsets between fitting and held-out families.

		Main	Energy summed over tasks		
Layer	Family	Shared	Shared	Clean-feature	Random
L1	Background	65.14	66.39	2.27	1.43
L1	Camera	76.35	77.45	1.70	1.38
L1	Light	83.58	84.00	1.89	1.84
L1	Sensor Noise	74.44	76.98	1.10	1.60
L9	Background	44.42	42.62	0.52	1.51
L9	Camera	63.42	67.30	0.46	1.68
L9	Light	61.23	61.01	0.88	1.69
L9	Sensor Noise	43.98	47.22	0.64	1.35
L18	Background	39.29	38.40	8.73	1.66
L18	Camera	40.47	41.80	10.91	1.75
L18	Light	49.84	49.36	8.01	1.75
L18	Sensor Noise	36.50	38.90	8.93	1.98

Table S38. **The shared directions retain far more response energy than the controls under either aggregation.** Share of the response energy around the fitting-data mean (%) on the same 404 held-out pairs and 28 tasks. Main: the main paper's mean of task ratios. The last three columns take the ratio after summing task-level errors and energies. Each cell includes all 28 tasks, with 96/102/96/110 pairs for Background/Camera/Light/Sensor Noise. Descriptive values without intervals.

S20. Attention and Feature Responses during Action Generation

How do visual perturbations show up in the attention computations of $\pi_{0.5}$ during action generation? Changing the main camera alters deep features at the positions of the unchanged wrist image. At L18, clean–perturbed differences in attention weights shrink from the first to the last sampled generation step, while differences in attention output grow, in all four family means. This is an additional study, beside the restoration test of main Sec. 4.2; the main paper mentions it once, in Related Work, and does not rely on it.

Attention outputs depend on both the attention weights and the norms of the transformed value vectors [1], so a change in the weights must be read together with the features being combined. We first examine the content available to attention, then how it is weighted and transformed as the plan is refined.

S20.1. Visual Changes Reach beyond the Altered View

Changing the main camera changes deep features at unchanged wrist-image positions. In a 36-task LIBERO study, we rotate the main camera by 15° while keeping the physical state, wrist image, and language instruction fixed. We examine the wrist-image *value vectors* (V): the feature content that attention reads and combines. These vectors are unchanged at L1 but change at L9 and L18 (Sec. S10.4). A deep feature at a wrist-image position thus carries context from the main view, even when the wrist pixels are identical.

Which layer redistributes wrist attention more depends on whether the wrist's share of attention is counted. In the same study, we normalize the wrist probabilities to sum to one and measure their change by conditional total variation (TV), half the sum of absolute probability differences. Weighting each comparison by the mean total wrist attention of the clean and perturbed runs puts this redistribution on the scale of all attention. At generation step 0 under baseline lighting, L18 has the larger conditional TV, but L9 has the larger mass-weighted TV (Table S39a), in all 36 tasks.

The matched LIBERO-Plus recordings show the same depth pattern under Camera and Sensor Noise: mean wrist-value responses are zero at L1 and nonzero at L9 and L18 (Table S39b), on the 28 common tasks. These values are cached once per observation, and the action expert attends to them at every generation step.

S20.2. Attention-Weight Differences Shrink while Output Differences Grow

What is measured. At each generation step, attention probabilities P combine the cached values with values from the current plan features, and an output projection turns this mixture into features W [1]. We compare the probability TV over all visible token positions with the root-mean-square (RMS) difference in W, measured after the projection and before the attention gate. The first tracks the weighting, the second the resulting features.

To obtain P and W, we rerun the frozen blocks on the saved inputs of generation steps 0, 5 and 9. Each clean or perturbed run supplies its own saved plan state x_j, so later comparisons include the separation between their generation paths. All three steps refine one plan before the robot executes it. The recomputation was checked against recorded hidden states and velocities for one input; P and W were not recorded during the policy's own generation, so they could not be compared directly (Secs. S11.2 and S16.1).

(a) LIBERO: wrist-attention redistribution

Wrist-attention change	L9	L18
Within-wrist TV (%)	15.75	**25.18**
Mass-weighted TV (pp)	**5.45**	2.33

(b) LIBERO-Plus: wrist-value responses

Perturbation	L1	L9	L18
Camera	0	0.514	0.275
Sensor Noise	0	0.415	0.232

Table S39. **Counting the wrist's share of attention reverses the layer ranking; wrist features respond at depth.** (a) Fixed-state LIBERO study: main-camera $0° \rightarrow 15°$, baseline lighting, generation step 0, 36 tasks. Each conditional TV is weighted by its mean clean/perturbed wrist-attention mass before task-equal averaging; the layer ordering reverses in every task. (b) First-query wrist-value differences relative to the clean wrist-value norm, using the Camera (102 pairs) and Sensor Noise (110 pairs) subsets of the matched LIBERO-Plus panel, each on 28 tasks. Entries are task-equal means. Normalized probabilities use %; mass-weighted changes use percentage points (pp). Full measures and value-response intervals: Secs. S8.2 and S10.4.

Weights converge while outputs diverge. At L18, probability differences decrease from the first to the last sampled generation step, while raw output differences increase, in all four family means on the 404 matched pairs (Fig. S23a,b). The two runs assign more similar weights but produce more different output features.

Block-end states shrink in absolute size but grow relative to the clean state. The block-end features H_{AE}, recorded during the policy's own generation after the residual and feed-forward computations, follow the response beyond attention. Their absolute clean–perturbed RMS difference decreases from generation step 0 to 9, while their difference relative to the clean state increases (Fig. S23c,d): the clean-state RMS falls from about 46.8 to 16.6, faster than the response.

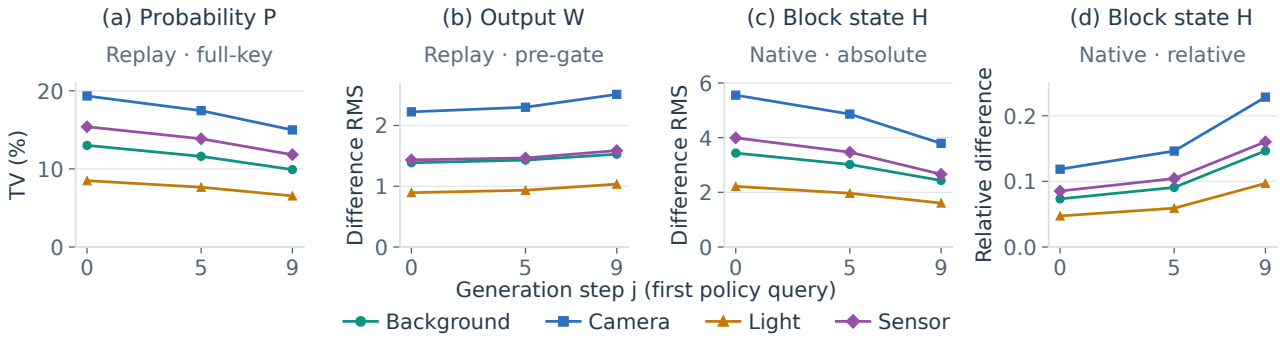


Figure S23. **Attention-weight differences shrink while output-feature differences grow.** At L18, (a) full-key probability TV decreases from the first to the last sampled generation step, while (b) the raw RMS difference in the projected attention output increases. Both are recomputed at each run's saved generation states. The block-end state recorded during the policy's own generation has (c) decreasing absolute differences but (d) increasing differences relative to the shrinking clean state. All panels use the same 404 $\pi_{0.5}$ pairs on 28 tasks; points show task-equal means for each perturbation family at steps 0, 5, and 9, without intervals. Relative responses are formed per case before averaging. Complete profiles and recording scope: Secs. S8.2, S11.2, S16.1, and S17.

Link to commands. At the first generation step, both probability differences and relative output responses are positively associated with the change in the first five commands (translation) under Camera and Sensor Noise (post hoc; Sec. S4).

In short, an unchanged camera image can supply changed deep features, and during generation more similar attention weights can accompany more different outputs. Reading feature content and outputs alongside the weights shows these changes; an attention map alone does not.

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